ENHANCING FORAGING SAFETY: A LOW-RESOURCE MACHINE LEARNING MODEL FOR POISONOUS MUSHROOM IDENTIFICATION IN THE PHILIPPINES

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APPROVAL SHEET

This thesis entitled "**Enhancing Foraging Safety: A Low-Resource Machine Learning Model for Poisonous Mushroom Identification in the Philippines**", prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science, has been examined and is recommended for acceptance and oral examination.

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Table of Contents

Cover Page i
Approval Sheet ii
Acknowledgement iii
Table of Contents iv
List of Tables v
List of Figures vi
Title Page **1**
Abstract **2**

Introduction **3**
Review of Related Literature 6
Conceptual Framework 13
Objectives of the Study 15
Scope and Delimitations 16
Significance of the Study 20

Methods **23**
Research Design 23
Sources of Data 23
Research Instruments 24
Participants 30
Data Collection 31
Data Analysis 33
Research Procedure 34
Ethical Considerations 58

Results **60**

Discussion **83**
Conclusion 91
Recommendations 93

Appendices

Appendix A Cover Letter for End-User System Evaluation
Appendix B API iNaturalist Data Collection Code Snippet
Appendix C Viewing Angle Sample Images
Appendix D Low Fidelity Wireframe
Appendix E Group-Aware Leakage Script
Appendix F Data Transformation Snippet
Appendix G Unvalidated Image Set
Appendix H Validated Image Set
Appendix I ML Expert Consultation Discussion
Appendix J ML Expert Acknowledgement Letter
Appendix K End-User Installation and Testing Guide
Appendix L Test Images for End-Users
Appendix M App Iterations
Appendix N Final Curated Dataset
Appendix O Test Case Table
Appendix P Test Case Outputs
Appendix Q Post-Study Survey Items
Appendix R Participant Consent Summary
Appendix S Experts' Bionotes
Appendix T Editor's Note
Appendix U PlagScan Certificate
Appendix V Docker and Github Repository
Appendix W ACM Formatted Paper
Appendix X Researchers' Resume

List of Tables

Table 1 Distribution of view angle
Table 2 Platform Configuration
Table 3 Model Selection
Table 4 Training configuration
Table 5 Individual Averages and Combined Averages
Table 6 Data Cleaning Results
Table 7 Data Resplit Result
Table 8 Model Comparison
Table 9 Model False Positive and False Negative trends
Table 10 Latency Testing on Selected Low-End device
Table 11 PSSUQ Results by Participant Group per Subscale
Table 12 PSSUQ Results per Subscales
Table 13 Expert Validation Results by Model and Species

List of Figures

Figure 1 Conceptual Framework
Figure 2 Prototype Development Model
Figure 3 Low Resource CRISP-DM Methodology Model
Figure 4 Invalid Mushroom Sample
Figure 5 Data Augmentation Preview
Figure 6 MobileNetV3 Performance Metrics
Figure 7 MobileNetV3 Confusion Matrix
Figure 8 MobileNetV4 Performance Metrics
Figure 9 MobileNetV4 Confusion Matrix
Figure 10 ConvNeXt Base Performance Metrics
Figure 11 ConvNeXt Confusion Matrix
Figure 12 Swin Transformer Base Performance Metrics
Figure 13 Swin Transformer Base Confusion Matrix

Enhancing Foraging Safety: A Low-Resource Machine Learning Model for Poisonous Mushroom Identification in the Philippines

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Abstract

The identification of toxic species of wild mushrooms in the Philippines remains a persistent public health risk, often resulting in avoidable poisoning cases. The study developed a mobile prototype based on low-resource machine learning, capable of differentiating between three toxic species or presumed toxic species, including *Chlorophyllum molybdites*, *Lepiota cristata*, and *Agaricus xanthodermus*, from their edible look-alikes *Chlorophyllum rhacodes, Macrolepiota procera*, and *Agaricus campestris*. Using a curated dataset of 58,184 images across six classes sourced from iNaturalist, transfer learning was applied to fine-tune lightweight convolutional neural network architectures and a Vision Transformer architecture for a hybrid offline-online application. Model evaluation prioritized recall, F-1 score, and confusion matrix analysis over standard accuracy to minimize false negatives, wherein toxic species could be misclassified as edible. MobileNetV4-Large was selected as the offline model, which achieved an average latency of 5.05 seconds on the selected low-end Android device. Meanwhile, Swin Transformer Base was chosen for online use, and it attained a macro F1-score of 93.24% alongside the lowest false-negative rate among toxic classes. Independent machine learning experts reviewed the methodology and model development process. End-user testing with 20 participants was conducted through purposive sampling and a modified PSSUQ-based instrument, which yielded an average score of 4.72 out of 5. These findings suggested that low-resource machine learning models can improve mushroom foraging safety in resource-constrained settings.

Keywords: Mushroom poisoning identification, transfer learning, Vision Transformer, lightweight CNN, mobile deep learning application

ENHANCING FORAGING SAFETY: A LOW-RESOURCE MACHINE LEARNING MODEL FOR POISONOUS MUSHROOM IDENTIFICATION IN THE PHILIPPINES

Introduction

The tropical archipelago of the Philippines is a global biodiversity hotspot with diverse rainforests, highlands, and coastal ecosystems where wild mushrooms thrive abundantly during the rainy season. These fungi play essential ecological roles, from nutrient cycling to symbiotic relationships with plants, and hold cultural significance in foraging practices among rural and indigenous communities. The Philippines is known for its mega-biodiversity, with recent extensive reviews cataloging at least 447 distinct species of wild mushrooms across the archipelago from 2001 up to 2021 (Dulay et al., 2023). The physical characteristics of a mushroom, namely its cap shape, cap color, gill size, gill color, bruises, stalk, and environment, can be utilized to identify the species and toxicity (Ortiz-Letechipia et al., 2024). Yet, the misidentification of wild mushrooms frequently leads to severe poisoning incidents. Recent reports from the Department of Health (DOH) in the Cordillera Administrative Region document multiple outbreaks, including 11 cases involving 34 people hospitalized between May and June 2024 in Benguet and Mountain Province alone (Philippines News Agency, 2024), and earlier incidents where groups of up to 14 individuals fell ill after consuming foraged mushrooms mistaken for edible varieties (Rendon, 2017). Another incident occurred on March 14, 2026, where ten residents, including seven children, were hospitalized after accidentally consuming unspecified poisonous mushrooms in Barangay Rizal, Talibon Town (Inojales, 2026). Such events, often involving gastrointestinal distress, vomiting, diarrhea, and in rare cases more severe outcomes, highlight a persistent public health challenge during wet months when mushrooms fruit prolifically.

Globally, mushroom poisoning remains a concern, with toxic species like those in the Amanita genus responsible for the majority of fatalities worldwide. Poisonous mushrooms in the Philippines often spawn during rainy seasons, where humidity levels are high, and the atmosphere is damp with residual rain. In tropical regions like the Philippines, however, species such as *Chlorophyllum molybdites* (commonly known as the false parasol or green-spored parasol) emerge as a frequent culprit. False parasols are widely distributed in grassy areas, lawns, and open woods across Southeast Asia and the Philippines, closely resembling edible parasol mushrooms (e.g., *Macrolepiota procera* or local look-alikes), leading to accidental ingestion. Its consumption typically causes severe but non-fatal gastrointestinal symptoms, yet it ranks among the most common causes of mushroom-related illnesses in similar climates. With an estimated 447 documented wild mushroom species across 72 families in the Philippines (based on reviews of studies from 2021), accurate identification remains challenging for non-experts, exacerbated by limited access to mycological expertise in remote areas.

Similar to the false parasols, another poisonous mushroom species is the *Lepiota cristata* (Stinking Dapperling), which shares a similar appearance to young edible parasols. *Lepiota cristata* is generally regarded as inedible and possibly poisonous. Its toxicity has not been firmly established: an analysis of Turkish specimens found no amatoxins or phallotoxins, although the authors cautioned that this does not make the species edible (Yılmaz et al., 2018). Because other small *Lepiota* species are poisonous, including *L. brunneoincarnata*, which has caused severe liver injury requiring transplantation (Anbardar et al., 2024), and *L. spiculata*, which contains lethal α-amanitin (Angelini et al., 2020), *L.*

cristata was treated in this study as a precautionary toxic class. As documented on iNaturalist, an online social network dedicated to expert community biodiversity observations, these mushrooms are commonly found growing in woodlands, shady and damp gardens, garden waste, short grass, ditches, and other areas of disturbed ground.

Another recognized poisonous mushroom species is the *Agaricus xanthodermus* (commonly known as the Yellow Stainer), which is frequently mistaken for its edible lookalikes, the *Agaricus arvensis* (Horse Mushroom) and the *Agaricus campestris* (Field Mushroom). Similar to false parasols, this species is often located in woods, lawns, gardens, and hedgerows, and consumption causes the same non-fatal gastrointestinal symptoms. It is stated on iNaturalist that of those who gather Agaricus-style mushrooms, 50% of the poisoning cases stem from the Yellow Stainer alone. Other poisonous species found within the Philippines also include the *Panaeolus cyanescens* (Blue Meanies), *Coprinopsis atramentaria* (Common Ink Cap), and *Hypholoma fasciculare* (Sulphur Tuft).

The broader field of this research lies at the intersection of mycology, public health, environmental science, and artificial intelligence (AI). Machine learning (ML), particularly computer vision techniques using convolutional neural networks (CNNs), has revolutionized species identification in biodiversity-rich but data-scarce regions. Tools like mobile apps have successfully identified plants and fungi in temperate zones, but applications tailored to tropical ecosystems, where lighting, humidity, and morphological variability pose unique challenges, are limited. Existing global datasets and models often underrepresent species available in Southeast Asian countries, creating a gap for localized, low-resource solutions suitable for smartphone deployment in low-connectivity Philippine settings. Traditional

methods rely on field guides, spore prints, or expert consultation, which are often unavailable to expeditioners, curious cooks, indigenous foragers, or environmentalists during outdoor activities. This leads to preventable poisonings, as evidenced by recurring DOH alerts and cases where toxic look-alikes like False Parasol, Stinking Dapperling, and Yellow Stainer are often mistaken for safe edibles.

Existing literature on Philippine mycology focuses primarily on morphological or molecular documentation (e.g., rDNA-ITS sequencing in regional studies), with few explorations of AI applications. While ML models for mushroom identification exist globally, they rarely address tropical variability or optimize for low-resource CNN architectures that run efficiently on affordable smartphones without constant internet or high-end hardware. This study examined whether a low-resource ML-based system could effectively distinguish between poisonous mushrooms and their edible lookalikes that are found within Philippine contexts, thereby enhancing user safety.

Review of Related Literature

Recent analysis of Philippine biodiversity cataloged over 400 distinct species of wild macrofungi across the archipelago (Dulay et al., 2023). While the vast majority of the existing local mycological research prioritizes edible mushrooms and their medicinal uses, these extensive checklists also confirm the widespread presence of internationally recognized toxic species. Notably, two checklists curated by Dulay et al. (2023) and Jacob et al. (2022) documented the occurrence of severe gastrointestinal symptoms from mushroom species within the Philippines. Specifically, these checklists confirm the local presence of the False

Parasol, Yellow Stainer, Stinking Dapperling, and many other toxic species (Blue Meanies, Common Ink Cap, and Sulphur Tuft). The confirmed presence of the highly mistaken and toxic species highlights a critical gap in current literature: while their existence in the Philippines is now established and documented, the lack of dedicated profiling and public awareness regarding these specific hazards remains an issue.

Traditional identification relies on field guides, spore prints, or expert consultation, which are methods considered inaccessible in remote areas or to non-specialists like expeditioners, cooks, or indigenous foragers. This underscores the need for automated, user-friendly systems to reduce preventable poisonings while supporting ethnomycological and conservation efforts.

A study conducted by Ozsari et al. (2024) investigated the application of deep learning models in the classification of macrofungi species, specifically in the identification of mushroom species such as *Amanita pantherina, Boletus edulis,* and *Pleurotus ostreatus*, using deep learning architectures. The study demonstrated that deep learning can significantly improve the accuracy of fungal classification by learning visual patterns and morphological characteristics from image datasets, which highlights the potential of artificial intelligence in supporting biological research and species identification.

MobileNetV3-Large was developed using hardware-aware Neural Architecture Search (NAS) and the NetAdapt Algorithm specifically to optimize its performance on mobile hardware (Howard et al., 2019). When compared to MobileNetV2, it achieved a 3.2-percentage-point improvement in ImageNet top-1 accuracy while also reducing latency by up to 20%. This evidence makes it relevant to the present study, as the proposed application is

intended to perform inference fully offline on smartphones with potentially limited computational and memory resources. Zhu et al. (2023) further demonstrated its applicability to mushroom image classification by using an improved MobileNetV3-Large-based model to classify shiitake mushroom quality in images with complex, unclean backgrounds, achieving 99.91% accuracy with a relatively small model size. While their task differs from poisonous and edible mushroom classification, their findings support the architecture's suitability for mushroom-related image classification. MobileNetV3-Large was therefore included as the baseline lightweight architecture for comparison with the other candidate models in the offline component.

On the other hand, MobileNetV4 represents a later generation of the MobileNet architecture family designed for efficient performance across mobile and edge hardware. Qin et al. (2024) introduced the Universal Inverted Bottleneck and Mobile MQA, a lightweight attention mechanism that provided a reported 39% speedup under their evaluated conditions. The authors evaluated MobileNetV4 on mobile CPUs, GPUs, DSPs, as well as specialized accelerators, including the Apple Neural Engine and Google Pixel EdgeTPU, demonstrating competitive efficiency across different hardware devices and platforms. This is particularly relevant because the proposed application was intended to operate offline on low-end Android smartphones, although it was evaluated on one selected device in this study. Cai et al. (2022) also emphasized the importance of computationally and memory-efficient models for resource-constrained or limited mobile and edge devices. MobileNetV4-Large was therefore included as a second lightweight candidate to determine whether its newer architecture could provide advantages over the MobileNetV3-Large baseline.

ConvNeXt-Base is a modern Convolutional Neural Network (CNN) that incorporates design principles inspired by Vision Transformers while retaining a purely convolutional architecture (Liu et al., 2022). On ImageNet-1K, ConvNeXt-Base achieved an 83.8% top-1 accuracy compared with 83.5% for Swin-Base, while larger-scale training produced results of 85.8% and 85.2%, respectively. Liu et al. (2022) also reported higher inference throughput for ConvNeXt than comparable Swin Transformer models under their GPU evaluation settings. These results show that modern CNNs can provide strong classification performance with efficient inference. ConvNeXt-Base was therefore included as a CNN-based candidate for comparison with the Transformer-based Swin Transformer-Base in the online classification component.

Swin Transformer is a hierarchical Vision Transformer that performs self-attention within local windows and shifts these windows between layers, allowing information to be exchanged across image regions while also keeping computation manageable (Liu et al., 2021). Swin-Base achieved an 83.5% top-1 accuracy score on ImageNet. Its attention-based architecture provides a different or contrasting approach to ConvNeXt-Base, which allows the study to evaluate whether modelling relationships between different image regions can benefit the classification of visually similar mushroom species. However, self-attention does not explicitly guarantee better performance, as effectiveness depends on the dataset and classification task. Ozsari et al. (2024) showed differences in performance among architectures for macrofungi classification. Swin Transformer-Base was therefore included as a Transformer-based candidate for the online classification component, providing a basis for comparison with the convolutional approach of ConvNeXt-Base.

Globally, deep learning, particularly convolutional neural networks (CNNs), has advanced automated species identification, outperforming manual methods in accuracy and speed. Reviews highlight CNNs' success in fine-grained classification of fungi, with models like ResNet, EfficientNet, and Vision Transformers (ViTs) achieving high Top-1 accuracies on datasets like FungiCLEF or MO106. For instance, hybrid approaches combining CNNs with attention mechanisms improve handling of morphological variability, while metadata integration (e.g., habitat, season) boosts performance in real-world scenarios.

In tropical/Southeast Asian contexts, studies emerge but remain limited. A 2024 Malaysian study by Lim et al. (2024) curated a local dataset and compared Vision Transformers (ViTs) and ResNet variants for mushroom ID, addressing poisoning risks from back-to-nature activities in humid climates similar to Philippine challenges. In the Philippines, recent work (Algarme & Villegas, 2025) explores ML for conservation and health awareness, but focuses on broader profiling rather than poisonous look-alikes or mobile deployment. Global fungi recognition systems (e.g., citizen-science apps with CNN ensembles) demonstrate real- time potential, yet they underrepresent tropical species. This study addressed the gap by developing a hybrid offline-online low-resource CNN optimized for both low-connectivity and connected environments in the Philippines.

These solutions address the need for automated identification to mitigate poisoning and aid biodiversity monitoring, but differ in scope: temperate/global models excel in controlled datasets, while tropical applications remained scarce, with gaps in Philippine-specific data and real-field robustness.

To enable deployment on resource-constrained devices (e.g., smartphones in rural Philippines), lightweight CNNs such as MobileNet variations prioritize efficiency without sacrificing accuracy. Studies show MobileNet-based models achieve 90–99% accuracy on plant/fungi tasks with few parameters (e.g., <4M), fast inference, and offline capability ideal for edge devices.

Applications include mushroom classification using improved MobileNetV3 for wild species, which improves performance on complex backgrounds. In related plant disease detection, lightweight models reach 97% to 99% accuracy on tropical datasets while running on low-end hardware. These benefits include reduced computational cost and practical deployment, which bridges academic models to field use.

Existing literature establishes the need for accurate wild mushroom identification in the Philippines to prevent poisoning and support conservation via citizen science. Solutions range from traditional mycology to CNN-based ML, with tropical/regional studies showing promise but limited localization. Lightweight models offer differentiation by enabling mobile/offline use, yet gaps persist: few Philippine-focused ML tools target toxic look-alikes, tropical variability, or low-resource CNNs for edge deployment.

Beyond model performance, the usability and quality of artificial intelligence and machine learning-based applications require validation to ensure reliability and functionality in real-world settings. The Post- Study System Usability Questionnaire (PSSUQ) is a widely accepted validation instrument that supplements this requirement. In a study conducted by Vlachogianni and Tselios (2023), mobile applications validated using PSSUQ scored higher on user satisfaction and system usability compared to other technological categories, with

mobile applications scoring the highest mean, ahead of internet-based and multimedia platforms. This finding supports the instrument's suitability in evaluating mobile-based systems such as the one this study developed.

The use of a confidence-based rejection mechanism to improve the reliability in safety-critical classification systems finds support in existing literature. Corbière et al. (2021) noted that quantifying model confidence is a requirement for deploying deep-learning classifiers in safety-critical applications, and described classifications with a reject option as an approach, where a classifier withholds a prediction rather than risk an unreliable output. This principle informed the study's design of a heuristic three-tier confidence-based warning system intended to reduce the risk of presenting unreliable toxicity classifications to users.

This study picked up where others left off by developing a low-resource CNN architecture focused on False Parasol, Stinking Dapperling, Yellow Stainer, and their edible look-alikes, Shaggy Parasol, Parasol Mushroom, and Field Mushroom, using curated Philippine-relevant data for a mobile tool, and validated through PSSUQ and expert consultations to ensure usability, model performance, and software quality. Significance includes enhanced safety for expeditioners, environmentalists, and the general public, and advancement of equitable AI in tropical hotspots, contributing to public health, United Nations Sustainable Development Goal 15 (Life on Land), and sustainable foraging in the Philippines.

Conceptual Framework

This study was guided by an Input-Process-Output (IPO) conceptual framework that illustrates how a low-resource machine learning system could transform raw mushroom observations into a practical identification tool to support safety in the Philippines. The framework covered the full research pipeline, from dataset curation to model development, evaluation, and deployment, as shown in Figure 1.

The input phase consisted of the raw materials used to build the system. This included the curated image dataset of Philippine found wild mushrooms sourced from iNaturalist, covering three toxic species and their edible look-alikes. This stage also included the pre-trained lightweight CNN and Vision Transformer architectures selected for transfer learning, and the evaluation instruments used later in the study, namely a modified PSSUQ-based questionnaire for usability testing and expert review for model validation.

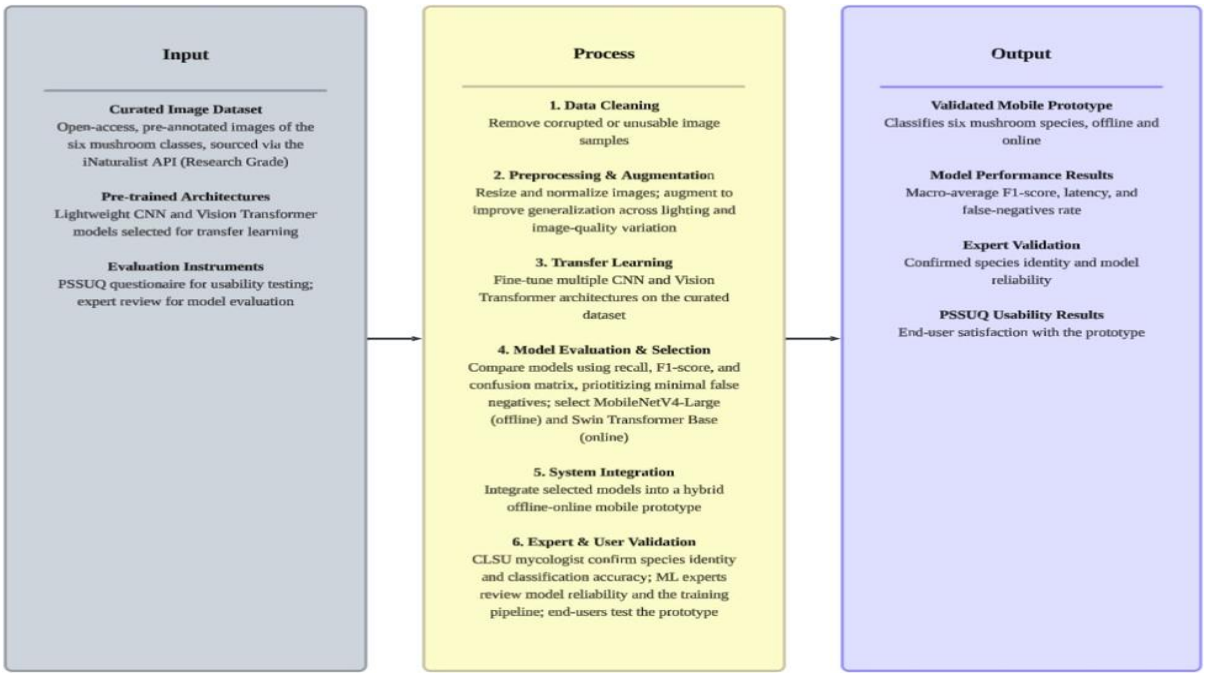
The process phase encompassed all research activities performed on the input materials to produce a working system. Raw images underwent data cleaning to remove corrupted or unusable samples, followed by preprocessing (resizing and normalization), and data augmentation to improve model generalization across lighting and image-quality variation. Transfer learning was then applied to fine-tune multiple CNN and Vision Transformer architectures on the curated dataset. The models were evaluated and compared using recall, F1-score, and confusion matrix analysis, and prioritized the minimization of false negatives given the safety-critical nature of the identification. This produced two selected models: MobileNetV4-Large for offline inference and Swin Transformer Base for online inference. The models were integrated into a hybrid offline-online mobile prototype.

Finally, the system underwent expert validation in which independent mycologists from Central Luzon State University (CLSU) confirmed species identity and were used to evaluate classification accuracy, and the machine learning experts reviewed model reliability and the training pipeline, followed by end-user testing with participants.

The output phase represented the research's final deliverables: a validated, low-resource hybrid mobile prototype capable of classifying the three toxic mushrooms from their respective edible look-alikes. This included the model performance results (macro-average F1-score, latency, and false-negative rates), as well as the expert validation confirming species and model reliability, and the modified PSSUQ-based instrument usability results reflecting end-user satisfaction. Together, these outputs support the study's broader goal of reducing preventable mushroom poisoning and promoting safer foraging practices in the Philippines.

Figure 1

Conceptual Framework



Objectives of the Study

This study aimed to test and evaluate a machine learning prototype for the identification of select poisonous wild mushrooms and their edible look-alikes, using species documented to occur in the Philippines. The general objective was to construct a model relevant to addressing poisoning risks and aligned with the overall aim of creating an accessible hybrid offline-online tool suitable for both remote low-connectivity areas and connected settings. Specifically, it aimed:

- To transform a dataset of images of wild mushroom documented to occur in the Philippines, centered on the toxic False Parasol, the presumed-toxic Stinking Dapperling, Yellow Stainer, and the edible Shaggy Parasol, Parasol Mushroom, and Field Mushroom through appropriate preprocessing techniques.

- To train and test models using transfer learning, including a low-resource Convolutional Neural Network (CNN) architecture and larger candidate models for online use.
- To evaluate and compare the models using standard statistical metrics such as recall, F1-score, and confusion matrix.
- To conduct expert review and end-user feedback collection through modified PSSUQ-based instrument using purposive sampling, with mushroom enthusiasts, environmental groups, ML/AI experts, and the general population.

Scope and Delimitations

This research investigated the application of machine learning, specifically low-resource convolutional neural networks, to the identification of wild mushrooms in the Philippines. The main subject centered on developing a prototype system for distinguishing poisonous species, addressing the persistent challenge of misidentification that contributes to poisoning incidents.

The general purpose of the study was to create an accessible and offline-capable prototype that enhanced safety for non-expert users, such as expeditioners, environmentalists, and the general public, all within a constrained five-month timeline.

The geographical scope was limited to the Philippine archipelago, with emphasis on species and observations commonly reported in regions such as Luzon (e.g., Cordillera highlands, Central Luzon) where the False Parasol, Stinking Dapperling, Yellow Stainer, and other poisonous types were prevalent, as documented in local health advisories and

biodiversity records. In this study, “toxic class” referred to species known or presumed to be poisonous that should not be eaten. *Chlorophyllum molybdites* and *Agaricus xanthodermus* are documented as poisonous; *L. cristata* was included as a presumed-toxic species on a precautionary basis.

Data collection included publicly available images of mushrooms species documented to occur in the Philippines, sourced primarily from open platforms such as iNaturalist. Only images of healthy, non-dried mushrooms were included in the dataset for training. Due to the geographical accessibility constraints of the mushrooms (high-altitude areas and long distances) and recent financial events (surge in gas prices) in 2026, the researchers limited data collection to secondary sources obtained through publicly available datasets, specifically iNaturalist. As a result, image quality, lighting conditions, and angles were not controlled by researchers and may not fully represent the morphological variability of mushrooms found under actual Philippines field conditions.

The dataset does not consist of images verified to have been captured within the Philippines and instead relies on globally sourced images of species confirmed to occur in the country. While the physical characteristics used for identification (e.g., cap, gill, and stem) were not expected to vary significantly by region, environmental factors specific to Philippine conditions, such as humidity, lighting, and regional morphological variation within a species, were not directly captured in the training data. This may affect the model’s ability to generalize to mushrooms photographed under authentic Philippine field conditions, and future work should prioritize collecting Philippine-sourced specimens to validate this assumption.

The dataset size and class distribution were determined by what was publicly available rather than a predetermined sample size, which may affect the model's generalizability to real-world scenarios. Additionally, the model was trained on individual images without labeling by anatomical view (cap, gill, stem). The three-view submission and confidence-averaging mechanism was used at inference and was a design choice rather than a behavior validated by the model itself. The system did not verify that the three submitted images belong to the same specimen, and inconsistent or mismatched inputs could silently produce an unreliable averaged result.

In this study, a model was considered low-resource if it could run on-device on a low-end Android smartphone without reliance on constant internet access, met an inference latency target suitable for real-time use, and remained deployable using standard mobile ML runtimes (e.g., ExecuTorch) without requiring high-end GPU hardware. These criteria aligned with the study's MobileNetV4-Large offline model, which achieved an average inference latency of 5.05 seconds on a selected low-end test device.

For validation, the study involved a convenience sample of participants, including mushroom enthusiasts (e.g., mycologists, foragers, breeders), environmental groups (e.g., farmers and gardeners), future researchers and technology practitioners, and the general public. The participants provided feedback on model performance and system usability through a modified PSSUQ-based instrument using purposive sampling.

The testing strategy included split validation (train/test sets), robustness checks via data augmentation, and standard metrics (accuracy, precision, recall, F1-score, and confusion matrix). Similarly, comparisons between the model architecture were based on a single

evaluation split without inferential statistical testing, and should be interpreted as observed rather than statistically confirmed performance differences.

The separation between training and expert-validation data relied on observation date for observation-level separation, without independent confirmation that no physical specimen was captured across both time windows.

Deployment was limited to a basic Android prototype that supported hybrid offline-online functionality, demonstrated on one selected low-end Android smartphone, using MobileNetV4-Large for offline on- device classification and Swin Transformer Base for online cloud-based classification. While the core identification worked fully offline, cloud integration was included when the internet was available. Full enterprise scalability lay outside the current scope.

The cut-offs of the 3-tier confidence system (85% and 50%) served as heuristic warning thresholds. They were set from research judgment and a review of related literature, not from an empirical analysis of the relationship between model confidence and accuracy, and they should not be read as validated safety boundaries. The displayed confidence values represented raw, uncalibrated softmax outputs, meaning a higher value indicated higher relative confidence rather than a calibrated probability of being correct. Future work should evaluate and calibrate these thresholds and incorporate out-of-scope detection.

Additionally, the system provided no distinct response or alert for inputs outside of the six trained classes. An out-of-scope species would still be classified under one of the six known species with a confidence score using a best-match mechanic, rather than being flagged as unrecognized. No out-of-scope evaluation was designed to measure rejection

performance and the informal tests reported in the results are illustrative only. Future work should validate these thresholds per-prediction confidence and data to confirm they corresponded to the increased misclassification risk, as well as incorporate an out-of-scope detection mechanism. Latency and on-device functionality were evaluated on a single low-end Android device; performance on other devices, operating-system versions, and hardware configurations was not tested.

The toxic status of *L. cristata* was presumed rather than toxicologically confirmed. The model did not assess toxicity; it identified species, and the toxic or edible label was assigned from published sources.

The administered instrument was a modified Post-Study System Usability Questionnaire-based questionnaire. Item 3 asked users whether the app was simple to use "to check whether a mushroom was safe to eat." This wording exceeded what a 6-class identification model could establish, since the app identified species and assigned a toxic or edible class label and did not certify safety. This wording was acknowledged as an instrument limitation, the item was retained as administered, and its ratings were not used as evidence of edibility or safety. A future instrument should read "assist in identifying toxicity or species."

Significance of the Study

This study contributed to the growing field of research at the convergence of artificial intelligence, biodiversity monitoring, and environmental safety by investigating the application of lightweight machine learning models for mushroom identification in tropical

environments. This research also focused on the development of low-resource CNNs capable of operating efficiently on mobile devices and is particularly relevant in developing regions such as the Philippines, where technological infrastructure and connectivity may vary. The hybrid offline-online design ensured the tool remained functional even in areas with poor or no internet while taking advantage of connectivity when available.

The development of a low-resource-based ML model for the identification of poisonous wild mushrooms in the Philippines holds significant importance in the context of public health, environmental awareness, and technological innovation. This study provided a practical and accessible solution to a pressing safety concern. The findings, methodology, and resulting technological tool benefited the following group:

Mushroom Enthusiasts. This study primarily benefits mycologists, Filipino foragers, outdoor enthusiasts, and individuals whose expertise and hobbies involve wild mushrooms. By providing a hybrid offline-online mobile tool capable of distinguishing poisonous species from look-alikes, the research promotes safety, reduces the risk of poisoning, and raises awareness about the proper identification of wild mushrooms. Users can make informed decisions, avoiding health risks associated with accidental consumption of poisonous fungi, such as gastrointestinal distress or more severe complications.

Environmental Groups. Environmental organizations that engage in biodiversity studies, citizen science, and field activities could use the research outcomes to educate students and volunteers about mushroom identification and safe foraging practices.

Future Researchers and Technology Practitioners. This study provides a reference for scholars, computer scientists, and technology practitioners interested in applying machine

learning to biological identification in tropical ecosystems. By demonstrating the effectiveness of low-resource convolutional neural networks in real-world conditions, the study encourages further development of ML optimized for mobile deployment in resource-constrained environments.

General Public. Beyond direct foragers or experts, the general public benefits from reduced incidents of accidental mushroom poisoning, lessening pressure on local healthcare organizations and emergency response services. This study develops scientific literacy and environmental curiosity by making local mycological tools accessible to non-experts. The system's offline and online functionality ensures that those in rural communities and low-connectivity regions have access to health and safety tools.

Methods

Research Design

This study utilized a quantitative research design to develop and evaluate a low-resource machine learning model focused on the identification of poisonous wild mushrooms in the Philippines. Quantitative research was utilized for the collection and analysis of model data, allowing the researchers to objectively measure the model's performance through metrics with a specific target F1-score and recall of 85% or greater on test sets, as well as to summarize the usability ratings from the modified PSSUQ-based instrument. The qualitative strand consisted of the structured consultation with the ML experts, whose feedback on the training pipeline, augmentation techniques, model selection, and deployment strategy was used to refine the model and validate the appropriateness of the overall methodology.

The study adopted the Cross-Industry Standard Process for Data Mining (CRISP-DM) as its primary framework to guide the systematic development of the model, from data understanding and preparation to modeling, evaluation, and deployment. This was complemented by a prototyping methodology, which allowed iterative cycles of testing and refinement to optimize the CNN architecture and the mobile application for efficient performance on mobile devices in resource-constrained or low-connectivity environments.

Sources of Data

In this study, the researchers utilized secondary data sources to develop a low-resource machine learning model capable of identifying poisonous wild mushrooms in the

Philippines. Primary data from end-user surveys and expert consultation were utilized for the evaluation.

Mushroom images were obtained from open-access sources, namely iNaturalist. All images obtained were pre-annotated based on species identification made by experts to ensure reliability in mushroom classification. The dataset was then structured for supervised learning, categorizing samples into six classes based on the selected mushroom species.

Research Instruments

The study employed a set of computational and mobile-based instruments to support the development and evaluation of a low-resource machine learning model for identifying select poisonous mushroom species from their edible look-alikes. These instruments served as the means through which the raw data was captured, transformed into model-ready inputs, and used to produce classification outputs.

Image Dataset Instruments

The primary instrument was a curated image dataset composed of open-source and free-use licensed images from iNaturalist. This instrument served as the foundational input data source from which the machine learning models learned to distinguish toxic mushrooms from edible look-alikes. The dataset consisted of mushrooms in the RGB color model, representing the six target classes and served as a basis for the following model training, validation, and testing.

Model Instruments

The study employed a lightweight convolutional neural network architecture, namely MobileNet variants, due to their computational efficiency and suitability for mobile deployment. The pre-trained models underwent full fine-tuning on the curated dataset using Layer-wise Learning Rate Decay (LLRD) to adjust learning rates across network depth, and a learning rate warmup followed by cosine decay to stabilize training. To improve generalization and reduce overfitting, Mixup and CutMix augmentation, stochastic depth (drop_path_rate), and gradient clipping were applied during training. An Exponential Moving Average (EMA) of model weights was used to improve stability at convergence, while class-weighted loss with label smoothing was used to address class imbalance and reduce the model's overconfidence. These are critical for minimizing the risk of misclassifying toxic species as edible. Transfer learning was applied to fine-tune the pre-trained model for tropical mushroom classification tasks, enabling effective feature extraction despite limited data.

The model was trained on individual mushroom images without explicit labeling by view type (cap, gill, or stem). Each training image was treated as an independent input-label pair regardless of the viewing angle. Confidence scores displayed to users corresponded to the raw softmax output probabilities of the model and were not subjected to post-hoc calibration methods like temperature scaling or Platt scaling. As such, these values should be interpreted as relative confidence indicators rather than calibrated probability estimates of correctness. Models were trained in Python using PyTorch. On-device inference in the final offline application used ExecuTorch within a Flutter application. Post-Training Quantization

and half-precision (FP16) conversion were attempted through ExecuTorch to reduce model size; however, FP16 was not retained due to its on-device latency, and the deployed model ultimately used FP32.

Image Preprocessing Instruments

Image preprocessing instruments included resizing images to 384×384 pixels to match the input requirements of the model. The researchers utilized the Albumentations module to apply normalization, Gaussian blur, rotations and flips, coarse dropouts, and color jitter. These preprocessing steps helped standardize the dataset and improved the quality of input images for model training and evaluation.

System Instruments

A basic Android prototype was developed using Flutter and Dart to demonstrate hybrid offline-online capability. The application performed core image classification fully offline using ExecuTorch with MobileNetV4-Large as the base model, enabling functionality even in low-connectivity environments. It switched to online mode to connect with Swin Transformer Base, a higher-performance model. In both modes, the system outputted toxicity classification results alongside confidence scores and safety warnings. At inference, when a user submitted the 3 required views, each image was processed independently through the same model to produce 3 separate class-probability vectors. These 3 vectors were then fused by averaging the corresponding class probabilities across all views to produce a single fused probability distribution. The final predicted species corresponded to the class with the highest averaged probability, and the displayed confidence score reflected that class's averaged probability value.

To translate model confidence into actionable safety guidance, the application applied a heuristic 3-tier confidence threshold system. Predictions with a confidence score between 85% and 100% were displayed without additional warning, indicating higher relative model confidence without guaranteeing correctness. Predictions falling between 50% and 84% were displayed with a cautionary warning, signaling reduced confidence and advising users to seek expert verification. Predictions under 50% confidence were not displayed at all; instead, the application returned a message indicating that a safe classification could not be produced rather than risking the display of an unreliable species identification. These cut-offs served as heuristic warning thresholds rather than validated safety boundaries.

Classifications were color-coded for readability, with toxic species displayed in red and edible species displayed in green. This approach was designed to minimize the risk of presenting low-confidence predictions and improved readability, particularly given the life-threatening consequences of misidentifying a toxic species as safe.

Evaluation and Validation Instruments

Quantitative evaluation used statistical performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix to measure the effectiveness of the machine learning model in classifying mushroom images.

Qualitative evaluation involved structured feedback surveys administered to ML experts and end-users, including mycologists. These surveys collected qualitative insights regarding the usability, reliability, and practical relevance of the prototype system.

Expert Validation Instruments

To assess the model's reliability, an expert validation study was conducted in which an independent team of mycologists from Central Luzon State University (CLSU) confirmed the species identity of a curated set of mushroom images obtained from iNaturalist under the CC-BY-NC license. To prevent specimen-level data leakage, these validation images were collected exclusively from observations uploaded in August, distinct from the training dataset which was sourced from observations that were uploaded at the latest of June and early July.

This separation was verified at the observation and image level, where no duplicates or observation IDs were shared between the training and expert-validation sets. This approach assumed that a given physical mushroom specimen was not independently photographed and uploaded as a separate observation, which represented a noted limitation of the observation-level leakage control in the study.

This process focused on all six classes, where 12 mushroom samples each were documented for Chlorophyllum and Agaricus pairs, and 11 samples for the Lepiota/Macrolepiota pairs. These sets used three images per sample, namely top (cap), bottom (gill), and side view (stem), for a total of 105 sub-images across all three pairs. Mycologists reviewed each sample and confirmed the species identity prior to model evaluation. These images were then compared against actual model classifications. A prediction by the model was considered misclassified if the model assigned an incorrect species label or if the confidence level fell below 50%, in which case the application withheld the prediction due to unreliable results based on the established heuristic 3-tier

threshold. This supplemented quantitative test-set evaluations by verifying model performance against expert-confirmed identifications.

To assess the developed model pipeline, consultation and validation from 2 machine learning experts were conducted. A structured discussion was held to review the appropriateness of the training pipeline, augmentation techniques, model selections, and deployment strategies. Initial consultations provided feedback to further improve model performance and reliability. Upon model completion, the experts reviewed the finalized system and results, issuing a letter of acknowledgment confirming that the model passed evaluation and that the reported findings proved accurate.

User Validation Instruments

To evaluate the system's overall usability and gauge user satisfaction, a modified PSSUQ-based instrument was used following the prototype finalization. The instrument was adapted from the standard PSSUQ format, with the items rephrased to be more specific to the context of the study and using a 5-point scale where higher scores indicate greater usability and satisfaction. The modified instrument aimed to measure how well the system helped users distinguish toxic mushrooms from their edible counterparts, evaluate the clarity of information and guidance provided by the system, and assess how pleasant and easy the user interface was to navigate.

The questionnaire was distributed to a total of 20 participants, which included mycologists, mushroom foragers and breeders, gardeners and farmers, future research and technology practitioners, and members of the general public (non-experts).

Participants

The primary participants of this study were selected using purposive sampling by predefined inclusion criteria and targeted recruitment channels. Participants were required to have relevant experience related to the study's focus, such as involvement in wild mushroom foraging, cultivation, breeding, background in mycology, environmental science, or technical expertise in machine learning development. To reach individuals meeting these criteria, the researchers sourced the participants through community forums and groups on Facebook such as the Philippine Wild Mushroom, Mushroom Identification, Programmers PH, dedicated to mushroom foraging, mycology, and technological communities, where participants were invited to take part in the study's usability testing and end-user validation.

The participants included the general public, mushroom enthusiasts (e.g., mycologists, foragers, breeders), environmental groups (e.g., farmers and gardeners), and researchers and technology practitioners. The researchers utilized purposive sampling to select individuals who possess expertise in the respective areas: mycology and ML, for expert validation, recruited primarily through accessing expert community groups and the researchers' existing networks. The qualifications and professional backgrounds of the mycologists and machine learning experts were provided in Appendix S, and the curricula vitae of the researchers were provided in Appendix X. Participants received a cover letter stating the purpose of the study, the affiliation of the researchers, and the data privacy terms under which responses were collected (see Appendix A).

Mycologists were selected for their specialized knowledge in mushroom identification, cultivation, and classification, which can support validation of the model's

outputs. Environmental enthusiasts and the general public within rural or indigenous communities were selected because they were more likely to encounter wild mushrooms.

Machine learning experts were selected for their specialized knowledge in artificial intelligence development, as they provided thorough discussions regarding model evaluation and the training pipeline.

All participants were informed of the study's purpose beforehand and provided explicit consent prior to completing their respective survey instruments, in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173).

Data Collection

The data collection for this study involved the selection and preparation of the dataset over a period of five weeks. The researchers utilized publicly available datasets that include False Parasol, Stinking Dapperling, Yellow Stainer, and their edible look-alikes. Images contained various conditions of lighting and quality. These include variations in brightness, blur, noise, and image quality dependent on the user's camera. Each image was reviewed for clarity, relevance, and quality, then separated according to the type of mushroom.

Training and expert validation data were collected in two distinct time batches to reduce the risk of specimen-level data leakage. Training images were sourced from observations uploaded to iNaturalist in late June and early July, while the set reserved for expert validation was drawn from observations uploaded in August.

Data collection was automated using a custom Python pipeline interfacing with the official iNaturalist API (v1/observations) available in Appendix B. The parse argument used in the code "--max_per_species" used a count of 8000 for the dataset.

As image retrieval was performed at the observation level, collecting all photographs attached to a qualifying iNaturalist observation without filtering by viewing angle, the dataset was expected to contain a mix of cap (top), gill (underside), and stem (side) perspectives. To characterize this, a random sample of 60 images (10 per species) was manually reviewed and categorized by dominant viewing angle, full results available in Appendix C. This sample indicated approximately 40% top/cap views, 25% underside/gill views, and 35% side/stem or mixed views, suggesting a reasonable, though top-view-favoring, distribution across viewing angles at the aggregate level. Lepiota cristata (70% top-view) and Macrolepiota procera (70% side- view) each showed strong skew toward a single dominant viewing angle, while Agaricus campestris showed a more even split between gill and side views (40% each).

Notably, this distribution was not uniform across species, as shown in Table 1:

Table 1

Distribution of view angle

Species	Gill	Mixed	Side	Top
Agaricus campestris	40%	10%	40%	10%
Agaricus xanthodermus	20%	10%	20%	50%
Chlorophyllum molybdites	40%	10%	0%	50%
Chlorophyllum rhacodes	20%	0%	30%	50%
Lepiota cristata	10%	0%	20%	70%
Macrolepiota procera	20%	0%	70%	10%

This per-species imbalance was noted as a limitation, as it introduced the possibility that a model could partially associate a species with its dominant viewing angle in the training data rather than purely its morphological features—a potential confound not directly measured in this study.

A total of 59,315 raw images were collected across the six species prior to manual quality-assurance review (see Data Preparation).

Data Analysis

Quantitative measures such as training and validation loss, recall, F-1 score, and confusion matrix were evaluated to assess the effectiveness and quality of the finished models in identifying the species classification of the mushroom. The analysis was performed using the Python programming language, and utilized libraries such as NumPy,

Pandas, Matplotlib, and Scikit-Learn for data manipulation, visualization, and evaluation of the model's output.

Latency measures were analyzed to assess latency performance on the selected low-end Android device. The modified PSSUQ-based instrument responses were analyzed quantitatively to determine the average ratings per item and subscale, providing a numerical basis for evaluating end-user satisfaction and overall system usability. Feedback from the ML expert consultation was analyzed qualitatively through the summarization of the expert's comments on the model training pipeline, model selection, and deployment strategies, which helped support the interpretation of the model's overall performance and validity.

Research Procedure

This study employed the prototyping model as the primary framework for the software development process of the hybrid mobile system. The low-resource Cross-Industry Standard Process for Data Mining model was adopted as a structured subprocess specifically within the quick design and build prototype stages. This combination allowed the prototyping model to manage the full system lifecycle (as illustrated in Figure 2), while the adapted low-resource Cross-Industry Standard Process for Data Mining conformed to the study's constraints in connectivity, computing resources, and device capability (as shown in Figure 3).

Figure 2

Prototype Development Model

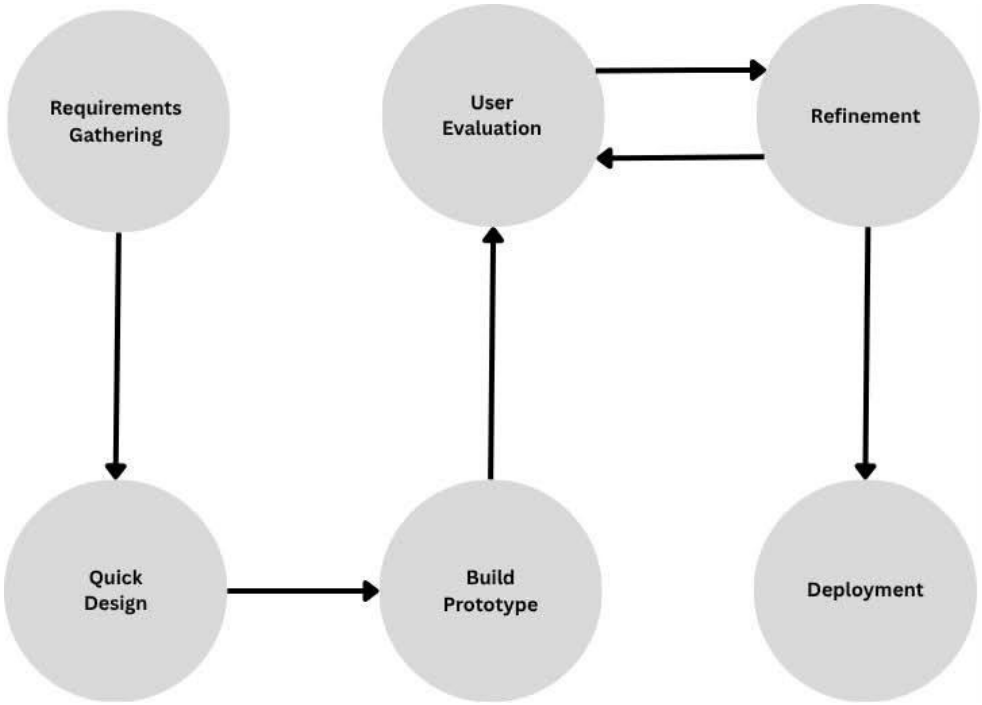
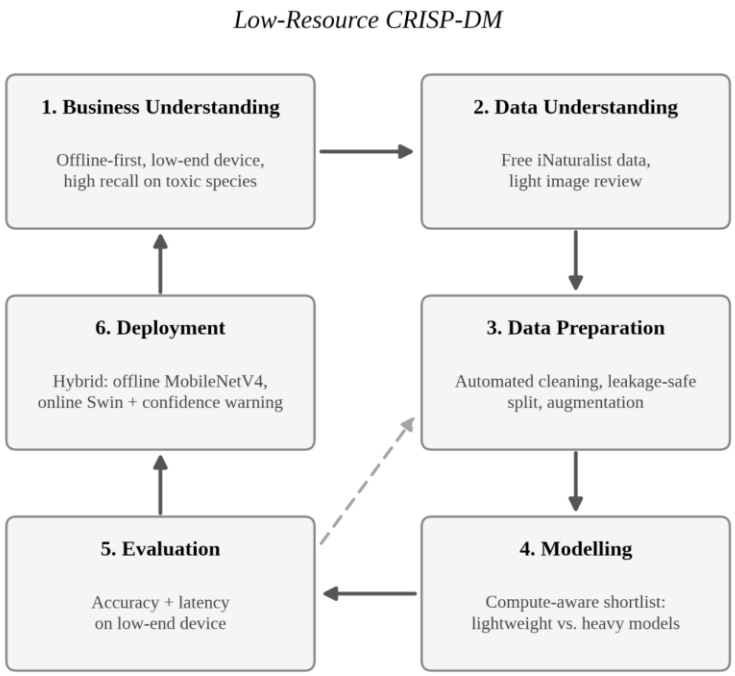


Figure 3

Low-Resource CRISP-DM Methodology Model



The prototyping model was executed by proceeding through 5 main stages: requirements gathering, quick design, build prototype, user evaluation, and refinement.

In the requirements gathering stage, the project foundation was established with clear objectives and priorities:

- Curating a dataset composed of Philippine-relevant wild mushrooms, namely the False Parasol, Stinking Dapperling, Yellow Stainer, and their edible look-alikes.
- Building a low-resource architecture identification model.
- Outlining and developing a hybrid online-offline mobile application.
- Evaluating system performance and model reliability.

The design component was included within the concept stage to establish how the proposed system would function before the actual prototype was developed. A wireframe was prepared to represent the planned user interface, element arrangement, navigation, and user flow of the application. The wireframe served as the visual guide for the development of the working prototype during the build stage, as shown in Appendix D. The system design included:

- User interface flows for the planned mobile application.
- Offline inference logic for mushroom identification when internet connection was unavailable.
- Online synchronization for accessing the online model when internet connectivity was available.
- Data flow between local and cloud components to define how information moved between the mobile application and online classification features.

For the identification results, green was used to indicate a non-poisonous classification, while red was used to indicate a poisonous classification. These visual indicators were incorporated into the wireframe to provide a clear distinction between the safety classifications presented by the application.

The development tools and success metrics were also identified during this stage:

Model development tools:

- Google Colab & Google One
- Python
- Success metrics (F1-score, accuracy, and recall)

- Model size

System development tools:

- Flutter and Dart for mobile application development
- ExecuTorch for on-device model execution
- Google Cloud, Docker, and GitHub (Appendix V)

For the machine learning component, the low-resource CRISP-DM process was applied as a structured sub-process. In the Business Understanding sub-phase, the specific modeling objectives were refined, including the need to:

- Identify the mushroom species from the six representative classes.
- Maintain low computational cost and model size.

Data understanding involved exploratory analysis of the compiled dataset, comprising the following elements:

Dataset Provenance

The dataset used in this study was collected from iNaturalist, a citizen-science biodiversity observation platform. Images were retrieved for six target species, namely *Lepiota cristata*, *Agaricus xanthodermus*, *Chlorophyllum molybdites*, *Macrolepiota procera*, *Agaricus campestris*, and *Chlorophyllum rhacodes*, selected for their inclusion of visually similar edible/toxic species pairs with documented real- world misidentification risk.

Only Research Grade observations were retrieved, representing iNaturalist's highest data-quality tier. This tier required independent community consensus on species identification from multiple contributors, alongside confirmation of wild (non-cultivated) status and the presence of at least 1 photograph. This provided a meaningful degree of label

reliability beyond a single uploader's self-reported identification. Images were further filtered to those released under Creative Commons licensing (CC-BY, CC-BY-NC, or CC0), consistent with permissible reuse for research purposes. It is noted that CC-BY-NC-licensed images restrict commercial use, a consideration relevant to any future commercial deployment of the system.

Observation retrieval was not restricted by geographic location; consequently, while all six target species had documented occurrence in the Philippines, the collected images represented a global sample of each species' iNaturalist observations rather than a Philippines-specific specimen collection. This was noted as a limitation, as regional morphological variation was not fully represented in the training data. Observations were retrieved in order of most recent observation date, rather than through random sampling, a minor collection-methodology detail noted for completeness.

Data preparation planning defined the preprocessing pipeline, which included:

Data Cleaning

The images underwent a manual quality review by the researchers. Each class was manually inspected, and selected images were removed based on the following criteria. A sample of the removed image is shown in Figure 4.

- Inconclusive,
- Unrelated,
- Microscopic Images,
- Unidentifiable mushrooms (mushroom too small),
- Images containing text,

- and dried or dead mushrooms.

Figure 4

Invalid Mushroom Sample



The dataset went through further processing, where it underwent a cross-split leakage check. This was done to verify that no image appeared in more than 1 train, validation, or test split. Another check was run separately, and it concluded that there was a 0.28% (~170 images) cross-split overlap among the subsets, with the script as shown in Appendix E. The researchers checked for this by grouping the images using the observation ID (filename) provided by iNaturalist during the API call within the train-test split. This was largely due to the aggregation of the images within the 80/10/10 constraint, which forced some images to be duplicated.

Data Preprocessing

To improve the model's ability to generalize species features, data augmentation techniques were applied during model training. RandomResizedCrop

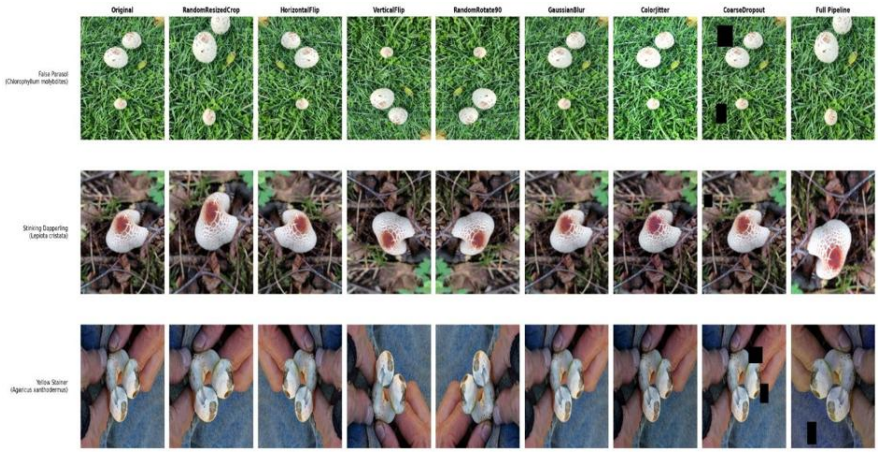
applied scale and framing variation by extracting randomized crops from each base image, while Horizontal Flip, Vertical Flip, and random rotation exposed the model to a broader range of angles and rotations that naturally occurred within user-obtained data. Gaussian blur was applied to mimic the soft focus or blur commonly found in mobile-captured images, while color jitter introduced variety in brightness, contrast, saturation, and hue to simulate natural environments. Coarse dropout added randomly placed rectangular patches to encourage the model to generalize from the overall characteristics of the mushroom rather than focusing on a single feature. Lastly, to make model training more efficient, normalization was applied. The specific configuration of each augmentation is available in Appendix F.

To address the objective of transforming the mushroom image dataset through appropriate preprocessing techniques, a visual demonstration of the augmentation pipeline was conducted on representative samples of the three species central to this study: False Parasol, Stinking Dapperling, and Yellow Stainer. Since data augmentation is applied dynamically during training, meaning the exact transformed images used by the model were never directly observed. This demonstration isolated each transformation step individually, applying it in full to a single sample image per species to make its effect clearly visible.

Figure 5 illustrates the comparison between the original, unmodified image and its transformation under each augmentation technique, including the final transformation pipeline applied during model training.

Figure 5

Data Augmentation preview



The modeling phase defined the model architecture and training environment used during development.

Model Training Procedure

Transfer learning was utilized during the development of the model. Pre- existing model weights were fine-tuned specifically for the study's objectives and curated dataset. The platform's specification is as seen in Table 2.

Table 2

Platform Configuration

Tool	Configuration
CPU	Intel(R) Xeon(R) CPU @ 2.20GHz
RAM	52.96 GB
GPU	NVIDIA L4
CUDA	12.8
OS	Google Colab Cloud Runtime
Python	3.13.15
PyTorch	2.11.0+cu128

Four pre-trained architectures, which span lightweight mobile-oriented models and higher-capacity alternatives, were selected for comparison and selection for the hybrid online-offline system. These are shown in Table 3, along with their corresponding timm identifiers and ImageNet pre-training configurations.

Table 3

Model Selection

Models (timm / ImageNet)
mobilenetv3_large_100.miil_in21k_ft_in1k
mobilenetv4_conv_aa_large.e230_r384_in12k_ft_in1k
convnext_base.fb_in22k_ft_in1k_384
swin_base_patch4_window12_384.ms_in22k_ft_in1k

Model Development

Phase 1: Initial Baseline (MobileNetV3-Large)

Model development began with MobileNetV3-Large, selected as the initial baseline for its lightweight, edge-deployable architecture. An early training configuration, partial-layer unfreezing (only the final blocks and classifier trainable), 320×320 resolution, and 15 training epochs, produced an initial test performance of 87.89% weighted F1-score.

During this phase, a significant infrastructure issue was identified: the training pipeline was reading images directly from a network-mounted cloud storage location, causing severe I/O-bound slowdown unrelated to model computation. This was resolved by copying the dataset to local compute-instance storage prior to training, reducing per-epoch training time substantially and establishing the dataset-handling procedure used for all subsequent training runs.

Following this fix, the training configuration was iteratively refined: input resolution was increased to 384×384, augmentation intensity was reduced (removing redundant color-space transformations suspected of obscuring diagnostically relevant color and bruising cues in visually similar species), the layer-unfreezing method was corrected from an arbitrary parameter-count slice to a named-module-based selection, and a discriminative learning rate schedule was introduced. A validation-loss computation error (a precision-casting mismatch under mixed-precision training) was also identified and corrected during this phase.

Subsequent attempts to further improve MobileNetV3-Large's performance by introducing a richer pretrained checkpoint, learning-rate warmup, and Exponential Moving Average (EMA) weight smoothing produced a performance regression rather than improvement. Diagnostic investigation identified two compounding causes: an Exponential Moving Average decay rate miscalibrated for the training step count of the model, and an input-normalization mismatch between the newly introduced pretrained checkpoint and the preprocessing pipeline. Both issues were corrected, with normalization statistics subsequently resolved programmatically per checkpoint rather than assumed, and Exponential Moving Average weights evaluated in parallel with raw weights each epoch, retaining whichever performed better on the validation set.

Phase 2: ConvNeXt-Base and the Full-Rigor Training Pipeline

To evaluate whether classification accuracy could be substantially improved without deployment-size constraints, ConvNeXt-Base (approximately 88.6 million

parameters) was introduced as an accuracy-focused counterpart to the lightweight MobileNetV3-Large baseline. This phase established a more comprehensive training methodology, subsequently applied to all remaining models in this study: full fine-tuning of all model layers, layer-wise learning rate decay, Mixup and CutMix batch-level augmentation, stochastic depth regularization, gradient clipping, learning-rate warmup with cosine annealing decay, and dual raw/EMA weight evaluation. ConvNeXt-Base trained under this configuration substantially outperformed the MobileNetV3-Large baseline.

Phase 3: Dataset Leakage Correction

A post-hoc audit of the dataset's train/validation/test split, prompted by the observation that source images originated from an observation-based collection method (permitting multiple photographs per physical specimen), identified that 161 images (0.28% of the dataset) belonged to observation groups spanning more than one split. Although the estimated maximum impact on reported accuracy was bounded at approximately 1.4 percentage points, the dataset was re-split using a group-aware method, assigning all images of a given specimen to a single partition, to eliminate this leakage entirely. This corrected split (dataset_resplit) was adopted as the standard dataset for all subsequent training and evaluation in this study, and the leakage-free property was independently re-verified following the re-split.

Phase 4: Unifying the Training Methodology Across Architectures

To ensure a fair, controlled comparison in which any observed accuracy difference between architectures reflects genuine architectural capability rather than

differing training effort, MobileNetV3-Large was retrained from its original partial-freeze configuration into the full-rigor training pipeline established in Phase 2, using the corrected dataset. Both MobileNetV3-Large and ConvNeXt-Base were retrained under this identical methodology, with a reduced 15-epoch training budget adopted after observing diminishing returns beyond this point in preliminary runs.

Phase 5: Architecture-Family Comparison: Swin Transformer

Swin Transformer-Base was introduced as a fourth architecture, deliberately size-matched to ConvNeXt-Base (approximately 87.9 million parameters, versus ConvNeXt-Base's 88.6 million) and trained under the identical full-rigor pipeline and dataset. This pairing was designed specifically to isolate architectural mechanism (convolution versus windowed self-attention) as the variable under study, holding parameter count and training methodology constant. Both models achieved closely comparable performance, differing by less than 0.2 percentage points, suggesting that at this parameter scale, architectural family exerted limited influence relative to model capacity and training rigor.

Phase 6: Generational Comparison Within the MobileNet Family (MobileNetV4)

MobileNetV4-Large (approximately 32.6 million parameters) was subsequently introduced to evaluate whether a newer generation of the MobileNet architecture family would yield accuracy improvements under the same training methodology, dataset, and epoch budget. An architectural augmentation experiment was also conducted, integrating Convolutional Block Attention Module (CBAM) components at each of the model's five stage boundaries; this modification did not

produce a measurable accuracy improvement over the unmodified MobileNetV4-Large baseline within the training budget evaluated, and was not adopted for the final deployed model.

Phase 7: Precision and Deployment Evaluation

Following architecture selection, post-training precision reduction was evaluated across all trained models as a candidate compression strategy. INT8 quantization was tested via two approaches: dynamic quantization and ONNX Runtime static post-training quantization. It was ultimately not adopted: dynamic quantization provided negligible size reduction for convolution-dominant architectures, while static quantization caused severe accuracy degradation on attention-containing architectures, a known limitation of naive post-training quantization applied to attention mechanisms. FP16 (and mixed-precision, retaining FP32 for batch-normalization layers) was tested as an alternative and found to preserve accuracy reliably; however, latency evaluation revealed FP16 inference to be consistently slower than FP32 across every architecture tested, and unsuitable for mobile deployment when evaluated in an on-device inference runtime. Consequently, FP32 was retained as the deployment precision for the final application.

Based on the cumulative results of this iterative process, MobileNetV4- Large was selected as the deployed model for offline, on-device operation, and Swin Transformer-Base was selected as the deployed model for online operation where computational resources and connectivity permit higher-accuracy inference.

Training Configuration:

- Full fine-tuning (no frozen backbone layers)
- Layer-wise Learning Rate Decay (LLRD)
- Mixup + CutMix
- Stochastic depth (drop_path_rate)
- Gradient clipping
- Learning rate warmup + cosine decay
- Exponential Moving Average (EMA)
- Class-weighted loss + label smoothing

The same techniques were applied to each model, with the following per-architecture values. The specifications of each training configuration are illustrated in Table 4.

Table 4

Training Configuration

Item	MobileNetv3	MobileNetv4	ConvNeXt	Swin
Optimizer	AdamW	AdamW	AdamW	AdamW
Initial LR	5e-5	5e-5	5e-5	5e-5
LLRD decay factor	0.85	0.85	0.85	0.85
LLRD depth	9	7	6	6
Weight decay	0.05	0.05	0.05	0.05
Batch Size	128	64	32	32
Mixup / Cutmix	0.2 / 1.0 / 0.5	0.2 / 1.0 / 0.5	0.2 / 1.0 / 0.5	0.2 / 1.0 / 0.5
Label smoothing	0.1	0.1	0.1	0.1
Stochastic depth	0.2	0.2	0.2	0.2
Grad clip norm	1.0	1.0	1.0	1.0
Seed	42	42	42	42
Epochs (full run)	15	15	15	15
Warmup epoch	AdamW	AdamW	AdamW	AdamW
Early-stop patience	5e-5	5e-5	5e-5	5e-5
EMA window target	0.85	0.85	0.85	0.85
Checkpoint selection rule	9	7	6	6

Model Evaluation

The evaluation phase included the collaboration of selected individuals in relation to the research objectives.

Consultations were conducted with mycologists from CLSU and two independent ML experts in order to validate the species identification of the expert- model comparison set and review the appropriateness of the model selection and training pipeline, as detailed in Expert Validation. The qualifications and professional backgrounds of the mycologists and ML experts who participated in these consultations were provided in Appendix S. All consultations were conducted digitally through private messaging (e.g., Facebook Messenger) due to prevailing geographical and scheduling constraints.

Image samples for the validation set were compiled and uploaded to Google Drive and sent to the mycologists and their team to identify. These images were separated by genus (e.g., Chlorophyllum, Agaricus, Lepiota/Macrolepiota); however, the identification of the specific species was left unlabelled, and the images were labelled as set 1, set 2, up to set 12. The unvalidated images are stored within a Google Drive link, available in Appendix G.

The validation made by the mycologists were completed within a week of being given the Google Drive link, and they confirmed the specific identifications for the validation image set through file renaming, following the review of the images shared via Google Drive. The validated images are stored in Google Drive link, available in Appendix H.

The comparison results of the model's performance against the expert- validated image sets are presented in Table 13 under Results, which reports the model's classification accuracy per species relative to the mycologists' confirmed identifications.

Separately, a structured discussion was held with two independent ML experts, conducted digitally via private messaging, to review the selected training pipeline, augmentation strategy, model selection, and deployment approach. Both ML experts were given a detailed discussion of the model's architecture and training techniques, which include the augmentation strategy along with performance metrics and results generated from the training process, provided in full in Appendix I.

Based on their review, the ML experts recommended shifting the primary focus from accuracy alone to both recall and the confusion matrix analysis, given the safety-critical nature of misclassifying toxic species as edible. This recommendation was adopted in the model evaluation presented in the results of the study.

Upon completion of the model evaluation, both the ML experts issued a signed letter of acknowledgement confirming the model's validity, provided in Appendix J.

Initial Prototyping

During the build prototype stage, both the machine learning model and the full hybrid application were constructed based on the design and requirements established earlier. The model base was fine-tuned using transfer learning on the curated dataset, with initial model variants trained and debugged iteratively. Parallel to model development, the Android prototype was coded to integrate the model, enabling fully offline classification with confidence scores and safety warnings while implementing online features to access a higher-performing model.

Post-Training Quantization (PTQ) was attempted to substantially reduce offline model size and latency while preserving accurate and confident scores for on- device use, but

was unsuccessful in system integration. Each build delivered a working increment of the prototype, such as smoother offline-online transitions and better user interfaces.

During prototype development, the single-image design was flagged during this iteration because a single mushroom photo, such as one showing only the cap, is often not diagnostically sufficient in mycology; toxic species like *Chlorophyllum molybdites* and its edible look-alike *Chlorophyllum rhacodes* are frequently distinguished by gill or stalk features rather than cap appearance alone. A one-shot classification risked a confidently wrong answer, which is a direct patient-safety concern given the app's poison-identification purpose.

The revision required the system to create and integrate evidence from multiple angles (top, bottom, side) into one combined result, rather than relying on a single photo's prediction. Each of the three photos was run through the model independently, producing its own full probability distribution across all six species (via softmax, so each image's scores sum to 1.0). The system then averaged these three probability scores per class; the final score for a given species was the mean of that species' probability from the top photo, the bottom photo, and the side photo. The species with the highest averaged score became the final prediction, and its averaged probability was displayed as the final confidence percentage.

This approach differed from simply picking the best single photograph or performing a majority vote of labels, as it allowed weaker or ambiguous evidence from 1 angle to be balanced out by stronger, more distinctive evidence from another (e.g., an inconclusive cap photograph could be outweighed by a highly distinctive gill photograph), while still requiring consistent agreement across angles for a high final confidence score to emerge.

To validate this approach, a controlled test was conducted where all three angle photos (cap, gills, stalk) were taken of a single known toxic mushroom species, *Chlorophyllum molybdites* (False Parasol), and run through both classifiers. The results are presented in Table 5.

Table 5

Individual Averages and Combined Averages

Metric	Offline (MobileNetV4-Large)	Online (Swin Transformer Base)
Final Label	*Chlorophyllum Molybdites*	*Chlorophyllum Molybdites*
Final Averaged Confidence Score	90.25%	90.80%
Top Confidence Score	89.81%	89.70%
Bottom Confidence Score	90.50%	91.08%
Side Confidence Score	90.44%	91.85%
Confidence Spread	0.7	2.1
Threshold bucket Score	High >85%	High >85%

Both models returned the correct classification with high confidence and full inter-image agreement; all three individual photos independently ranked the correct species as their own top prediction, with a confidence spread of under 3 percentage points across angles for both models. This indicated that the 3 capture angles were consistent with one another rather than contradictory, which illustrated the operation and inter-view consistency of the averaging approach. The online model returned slightly higher confidence (90.88% vs. 90.25%). In the on-device latency test on the selected low-end device (Table 10), online mode was also faster on average (4.08 seconds vs. 5.05 seconds) because inference was

performed on the server. Online latency depended on network conditions and was not tested across different connection speed.

The system also introduced the confidence threshold feature to manage uncertainty in the identification results: predictions with confidence scores of 85% to 100% were presented as the identified result without additional restrictions or warnings, predictions between 50% and 84% were accompanied by a cautionary warning, and predictions below 50% were withheld because model confidence was considered insufficient for presenting the classification as a reliable result. This proved particularly important because the application distinguished between poisonous and non-poisonous mushrooms, where an uncertain prediction could not be presented with the same certainty as a high-confidence result. At this stage, results were color-coded, with green representing non-poisonous and red representing poisonous classifications.

End User Evaluation

The User Evaluation stage focused on testing and validating the constructed prototype with mycologists, ML experts, and end-users to assess whether the system met its intended requirements and proved usable.

Mycologists confirmed species identifications against model classifications using an expert-validation image set, machine learning experts reviewed the appropriateness of model selection and the training pipeline, and end-users evaluated system usability and satisfaction through the modified Post-Study System Usability Questionnaire-based instrument, as detailed in Expert Validation and User Validation. Feedback gathered from the evaluation stage determined whether the prototype satisfied study requirements or required further refinement in later stages.

All communication with the 20 participants was conducted through digital private messaging via Facebook Messenger, where they received a short description of study objectives, basic instructions on how to install and run the application on their Android devices, and links to installation files and sample images for testing, available in Appendix K.

The sample images were deliberately gathered outside the time-scope of the training image set to ensure that the participants evaluated the system on genuinely unseen mushroom images rather than the samples the model may have been exposed to during training (stored in Google Drive and available in Appendix L).

After reaching the target of 20 participants, recruitment through Facebook Messenger was closed, and the collected responses were compiled and analyzed. Average ratings per



item and subscale, separated by user groups, are presented in Table 11 under Results, while general averages per subscale are presented in Table 12.

Prototype Refinement

In the refinement stage, feedback collected from user evaluation was used to revise the model and application as needed, looping back to quick design or build prototype stages where necessary until the prototype satisfied study requirements. Once refinement was complete, the finalized mobile application underwent final integration testing to verify end-to-end functionality, including smooth transitions between offline core identification and online enhancements.

Across the study, the prototype passed through three iterations (see Appendix M), with each cycle incorporating feedback from User Evaluation to refine the model and application, looping back to Design or Build Prototype stage where necessary.

First Iteration

The initial prototype established the first working version of the mushroom identification system, implementing single-image classification as its core mechanism. The initial developed prototype design showcasedthe following:

- Built an offline-only classification pipeline that returned a species prediction from a submitted photograph.
- Implemented single-image input, where a user submitted 1 photograph of a mushroom and received 1 classification result, without multi-angle capture or confidence tiering.

Second Iteration

In Iteration 2, three-angle capture and the confidence threshold system were introduced to address the reliability limits of single-image classification, along with the online classification pipeline for the Swin Transformer Base model. This iteration illustrates the new features and revisions made to the system.

- Introduced three-angle image capture, requiring top, bottom, and side photos of the mushroom instead of a single photo.
- Developed the averaging mechanism that combines the three photos' predictions into one final classification and confidence score.
- Added online classification, connecting to a higher-performing cloud-based model when an internet connection was available, alongside the existing offline classification.
- Implemented the confidence threshold system, sorting predictions into tiers based on the model's certainty.
- Added color-coded results, using green and red to visually flag non-poisonous versus poisonous classifications.

Third Iteration

Following recommendations from the thesis defense panel, the final prototype refined the presentation of results without changing the underlying averaging or threshold logic established in Iteration 2:

- Updated to display information on how the specific predicted mushroom species was classified, based on the mushroom features that informed the result.
- Updated to provide clearer terms and messaging for invalid images at the results section.
- Changed the cautionary warning color from its prior treatment to yellow, to visually distinguish the 50–84% precautionary tier from the red used for confident poisonous classifications.
- Retained green, unchanged, for safe and edible classifications.

Ethical Considerations

Researchers were bound by the rules of ethics that govern standards of professional conduct expected of everyone in the field of computer science and research. Research misconducts such as the fabrication of research data, plagiarism, academic dishonesty, intellectual property violation, and non-adherence to known ethical standards were strictly prohibited. To ensure the originality of the manuscript, the researchers subjected the paper to a plagiarism check; the resulting certificate is provided in Appendix U. The manuscript was also reviewed by a language editor for grammar, clarity, and flow, as certified in the Editor's Note in Appendix T.

In the context of mushroom foraging, the researchers prioritized the use of verified fungal datasets and strict validation protocols to ensure the model's reliability, as misidentification poses severe health risks or fatalities to users. Informed consent was obtained from all participants, and any personal information was treated with strict

confidentiality and used solely for academic purposes to maintain the integrity and safety of the study in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173).

To maintain transparency regarding system limitations, the researchers explicitly stated that the machine learning model could not achieve 100% accuracy in identifying species. Because severe repercussions exist for misidentifying a poisonous mushroom as edible, the software was designed to aid decision-making—not to serve as the ultimate authority or as a substitute for consulting a professional mycologist. As part of the measures implemented to manage this operational risk, warning banners were incorporated into the software interface, warning users of the inherent risks associated with consuming wild mushrooms identified by the software.

Results

Data Transformation Results

The data cleaning process, as shown in Table 6, removed a total of 1,131 raw images, which produced a final dataset of images suitable for model training and evaluation.

Table 6

Data Cleaning Results

Class	Raw Images	Removed Images	Final Images
Agaricus_Campestris	8,213	187	8,026
Agaricus_Xanthodermus	9,791	284	9,507
Chlorophyllum_Molybdites	10,067	169	9,898
Chlorophyllum_Rhacodes	10,338	134	10,204
Lepiota_Cristata	10,971	281	10,690
Macrolepiota_Procera	9,935	76	9,859

This correction, as stated in the Research Procedure, shown in Table 7, preserved the same approximately 80/10/10 split while eliminating cross-split contamination. The dataset is available in Appendix N.

Table 7

Data Resplit Result

Split	Original	New	% of total
Train	46,454	46,474	79.87%
Val	5,815	5,897	10.14%
Test	5,828	5,813	9.99%

Model Training Results

Model training followed the transfer-learning setup, platform configuration, and model architectures previously mentioned in the Research Procedure (Data Preparation and Model Training Procedure), which included the training configuration applied uniformly across to the four candidate models. The resulting baseline and comparison performance of each trained model is presented below.

Model Evaluation

Baseline performance metrics used in evaluating the model included accuracy and, for model generalization, the macro average for recall. All six classes roughly had a similar number of test images (approximately 700 to 1,000); thus, the macro-average F1-score was used. This served to gather the metric independently from each class and reduce bias toward larger classes, as it takes only the unweighted arithmetic value. Test-time augmentation was not used in any reported evaluation or in the deployed application. All metrics in Tables 8 and 9 and Figures 6 through 13 were obtained from single-pass inference on the fixed test set, matching the deployed configuration.

MobileNetV3-Large

MobileNetV3-Large achieved a baseline F1-score of 87.17% at the 15th epoch. With a batch size of 128, training time resulted in approximately 50 minutes (where each epoch took approximately three minutes and 20 seconds). Performance was lowest among the Agaricus and Chlorophyllum classes, specifically with the recall of the two toxic species: *Agaricus xanthodermus* (Yellow Stainer) at 83.33% and *Chlorophyllum molybdites* (False Parasol) at 83.37%, falling below the minimum 85% threshold set by the study, as illustrated in Figure 6. This indicated that the model produced a greater-than-average number of false negatives, carrying the risk of classifying these two toxic mushroom species as their edible counterparts. The confusion matrix shown in Figure 7 illustrated this risk directly. Approximately 102 images of the Yellow Stainer were misclassified as *Agaricus campestris* (Field Mushroom), and 75 images of False Parasols were misclassified as *Chlorophyllum rhacodes* (Shaggy Parasol).

Figure 6

MobileNetV3 Performance Metrics

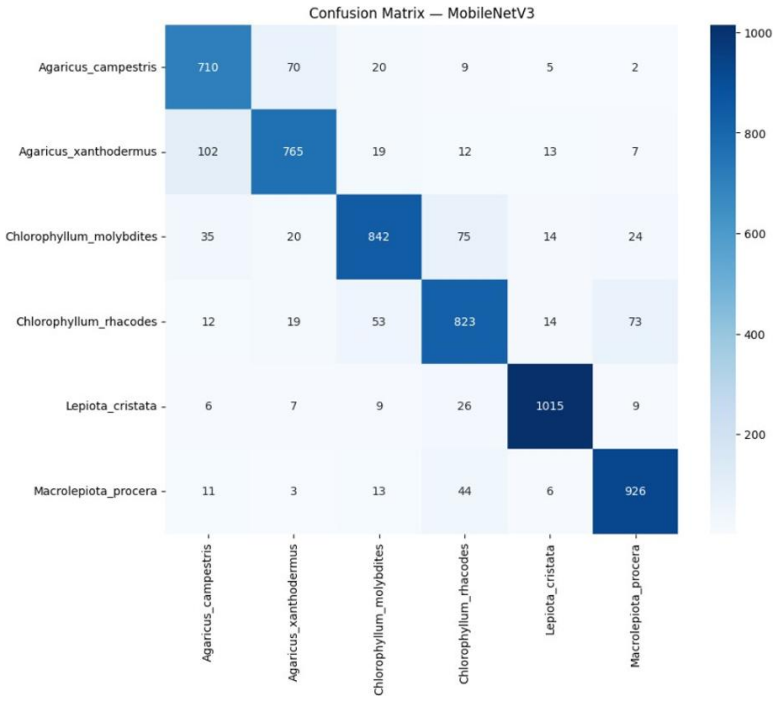
```
--- MobileNetV3 Classification Report ---
                        precision    recall  f1-score   support

     Agaricus_campestris     0.8105    0.8701    0.8392       816
   Agaricus_xanthodermus     0.8654    0.8333    0.8491       918
Chlorophyllum_molybdites     0.8808    0.8337    0.8566      1010
  Chlorophyllum_rhacodes     0.8322    0.8280    0.8301       994
       Lepiota_cristata     0.9513    0.9468    0.9490      1072
   Macrolepiota_procera     0.8895    0.9232    0.9061      1003

                accuracy                         0.8741      5813
               macro avg     0.8716    0.8725    0.8717      5813
            weighted avg     0.8747    0.8741    0.8740      5813
```

Figure 7

MobileNetV3 Confusion Matrix



MobileNetV4-Large

The MobileNetV4 model achieved a baseline F1-score of 90.56% at the 15th epoch. With a batch size of 64, training time was approximately 1 hour and 42 minutes, with each epoch taking approximately 6 minutes and 50 seconds. Compared to MobileNetV3-Large, MobileNetV4-Large offered more parameters while still remaining hardware-efficient on mobile devices. The Agaricus and Chlorophyllum classes showed a notable improvement, with the recall of the Yellow Stainer at 88.47% and the False Parasol at 89.60%, successfully meeting the minimum threshold of 85% as illustrated in Figure 8. The confusion matrix in Figure 9 showed improvement compared to MobileNetV3-Large, with false-negative misclassifications for the Yellow Stainer dropping from 102 down to 68, and for the False Parasol dropping from 75 down to 39, while introducing a trend of an increased number of false positives alongside reduced false negatives.

Figure 8

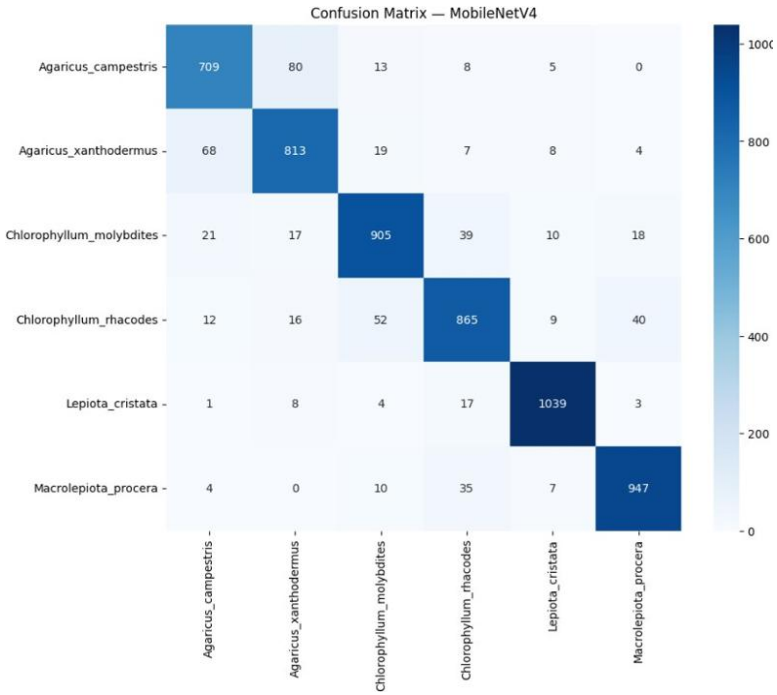
MobileNetV4 Performance Metrics

--- MobileNetV4 Classification Report ---

	precision	recall	f1-score	support
Agaricus_campestris	0.8699	0.8699	0.8699	815
Agaricus_xanthodermus	0.8704	0.8847	0.8775	919
Chlorophyllum_molybdites	0.9023	0.8960	0.8992	1010
Chlorophyllum_rhacodes	0.8908	0.8702	0.8804	994
Lepiota_cristata	0.9638	0.9692	0.9665	1072
Macrolepiota_procera	0.9358	0.9442	0.9400	1003
accuracy			0.9080	5813
macro avg	0.9055	0.9057	0.9056	5813
weighted avg	0.9079	0.9080	0.9079	5813

Figure 9

MobileNetV4 Confusion Matrix



ConvNeXt-Base

The researchers wanted to select a model for the online mode of the system, where the model would be hosted in a cloud service. This provided flexibility in selecting models that would be heavier in terms of computational resources, such as storage requirements and latency.

ConvNeXt-Base achieved a baseline F1-score of 92.98% at the 15th epoch. With a batch size of 32, training time was approximately 3 hours and 50 minutes, with each epoch taking approximately 15 minutes and 18 seconds. While the *Agaricus* and *Chlorophyllum* classes remained the lowest, they showed further improvement compared to previous models, with the recall of the False Parasol at

93.47% and the Yellow Stainer at 91.94%, as shown in Figure 10. The trend of increasing false positives and fewer false negatives continued from MobileNetV4-Large. The confusion matrix in Figure 11 supported this trend, as false positives for *Agaricus campestris* (Field Mushroom), specifically those misclassified as the Yellow Stainer, increased from 73 in MobileNetV4-Large to 80, while false negatives decreased to 54 from 68.

Shaggy Parasol misclassified as False Parasols decreased from 52 to 50, while False Parasol misclassified as Shaggy Parasol decreased to 30 from 39.

Figure 10

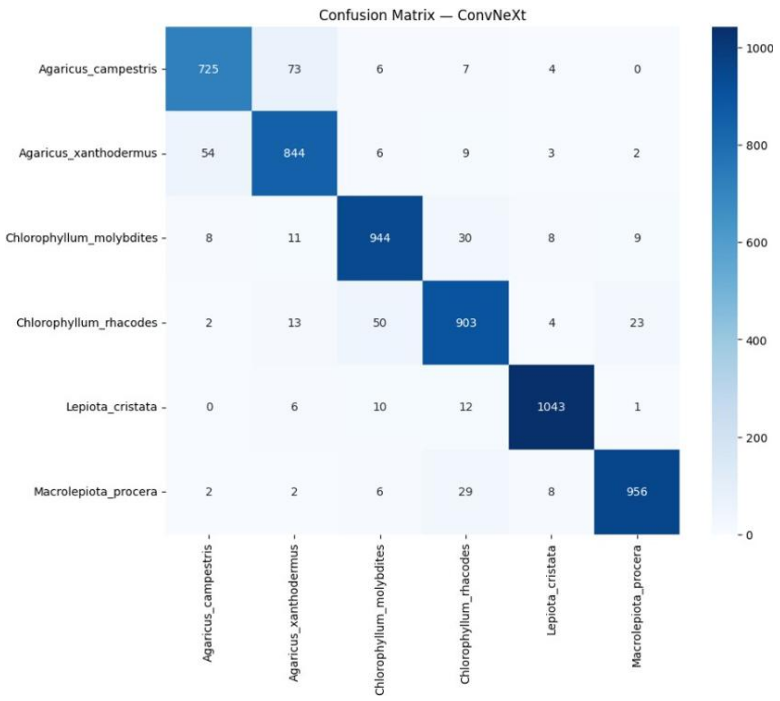
ConvNeXt Base Performance Metrics

--- ConvNeXt Classification Report ---

	precision	recall	f1-score	support
Agaricus_campestris	0.9166	0.8896	0.9029	815
Agaricus_xanthodermus	0.8894	0.9194	0.9041	918
Chlorophyllum_molybdites	0.9237	0.9347	0.9291	1010
Chlorophyllum_rhacodes	0.9121	0.9075	0.9098	995
Lepiota_cristata	0.9748	0.9729	0.9739	1072
Macrolepiota_procera	0.9647	0.9531	0.9589	1003
accuracy			0.9315	5813
macro avg	0.9302	0.9295	0.9298	5813
weighted avg	0.9318	0.9315	0.9316	5813

Figure 11

ConvNeXt Confusion Matrix



Swin Transformer Base

Swin Transformer-Base achieved the highest F1-score of 93.24% at the 12th epoch. With a batch size of 32, training time resulted in approximately 5 hours and 12 minutes, with each epoch taking approximately 26 minutes. Compared to ConvNeXt-Base, the model achieved the best balance in recall among all models within the *Agaricus* and *Chlorophyllum* classes, with the recall of the Yellow Stainer at 93.46% and the False Parasol at 93.27%. The model produced consistently high recall and F1-scores across all six classes, all within the approximately 90% range as illustrated in Figure 12. While the number of overall misclassifications decreased, the model still followed the trend of exhibiting higher false positives than false negatives: Field

Mushrooms misclassified as the Yellow Stainer dropped from 73 to 68, and vice versa (44 out of 54), as shown in Figure 13. Instances of Shaggy Parasols misclassified as False Parasols and vice versa remained at 50 and 30, respectively.

Figure 12

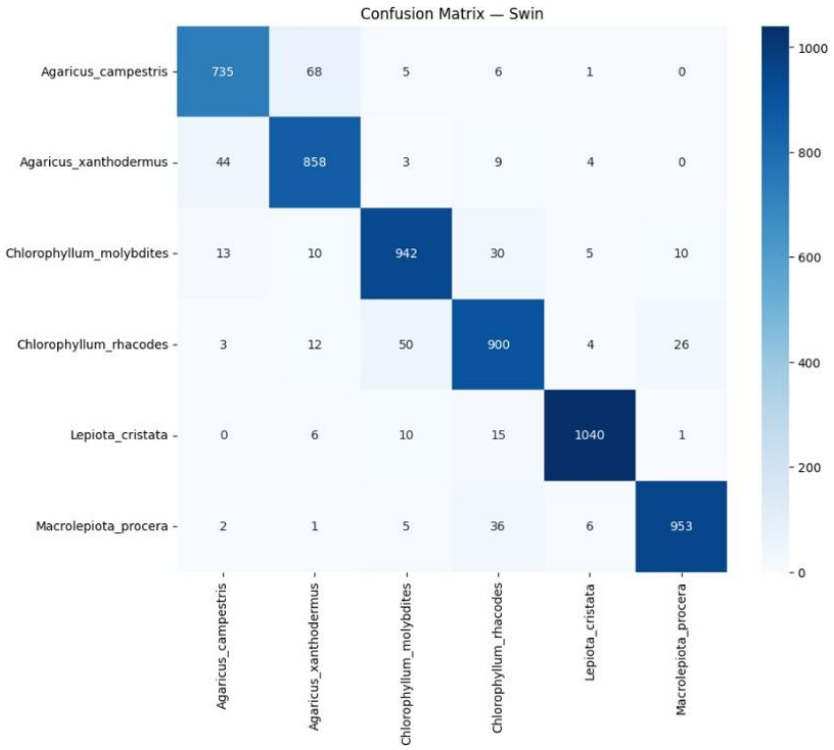
Swin Transformer Base Performance Metrics

--- Swin Classification Report ---

	precision	recall	f1-score	support
Agaricus_campestris	0.9222	0.9018	0.9119	815
Agaricus_xanthodermus	0.8984	0.9346	0.9162	918
Chlorophyllum_molybdites	0.9281	0.9327	0.9304	1010
Chlorophyllum_rhacodes	0.9036	0.9045	0.9041	995
Lepiota_cristata	0.9811	0.9701	0.9756	1072
Macrolepiota_procera	0.9626	0.9501	0.9563	1003
accuracy			0.9338	5813
macro avg	0.9327	0.9323	0.9324	5813
weighted avg	0.9341	0.9338	0.9339	5813

Figure 13

Swin Transformer Base Confusion Matrix



Model Summary

Across all four models, the accuracy and macro-average recall improved consistently from MobileNetV3-Large's 87.17% to Swin Transformer Base's 93.24%. This improvement was accompanied by a consistent decline in false negatives for both of the weakest toxic classes: Yellow Stainer misclassified as Field Mushroom dropped from 102 to 44, and False Parasol misclassified as Shaggy Parasol dropped from 75 to 30. Aligning with the safety-first priority outlined by the study, the number of false positives remained stable and preferable to false negatives, and hovered around the 30-80 range. This indicated that the model improved at avoiding higher-risk errors of misclassifying a toxic mushroom as edible, with a

stable increase in the lower-risk error of misclassifying edible species as toxic. Based on this trend, MobileNetV4-Large and Swin Transformer Base offered the strongest balance of overall accuracy and recall, with the reduced false-negative risk for toxic species. Metric comparison is as shown in Table 8.

Table 8

Model Comparison

Model	F1-score	Macro Avg. Recall
MobileNetV3-Large	87.17%	87.25%
MobileNetV4-Large	90.56%	90.47%
ConvNeXt-Base	92.98%	92.95%
Swin Transformer Base	93.24%	93.23%

The interpretation of this table includes False Positive (FP) of Field Mushroom -> Yellow Stainer and Shaggy Parasol -> False Parasol, and False Negative (FN) of Yellow Stainer -> Field Mushroom and False Parasol -> Shaggy Parasol is illustrated in Table 9.

Table 9

Model False Positive and False Negative trends

Model	FP (Field Mushroom -> Yellow Stainer)	FN (Yellow Stainer -> Field Mushroom)	FP (Shaggy Parasols -> False Parasols)	FN (False Parasol -> Shaggy Parasol)
MobileNetV3-Large	70	102	53	75
MobileNetV4-Large*	80	68	52	39
ConvNeXt-Base	73	54	50	30
Swin Transformer Base*	68	44	50	30

**Chosen Models*

System Testing

Prior to expert and user validation, the completed mobile application underwent system testing to confirm that it remained functional in both online and offline modes, reliable in handling varied image quality inputs (e.g., blurry, low-light) and demonstrated input validation for incomplete or irrelevant images. The complete test case documentation including the steps, inputs, expected outputs, actual outputs, and result is provided in Appendix O, while screenshots showcasing each result are available in Appendix P. System functionality was tested in both the online and offline mode of the system. Top, bottom, and side image uploads were accepted by both modes, and the system successfully returned a classification result. Input validation testing included uploading non-mushroom photos and incomplete image input. For non-mushroom photos, the system responded with



low-confidence predictions along with a message that notified the users that the uploaded images may be too blurry or may contain irrelevant objects. This was observed on a small set of test images and is not a general rejection mechanism for non-mushroom or out-of-scope inputs. For incomplete image input, the system disabled the submit button, which prevented returning results that were made with incomplete data. Reliability was tested by submitting images of varied quality, namely blurry and low-light images. Rather than returning a confident but potentially inaccurate result, the app applied the designed heuristic three-tier confidence threshold and displayed a warning and message for the low-confidence results.

For out-of-scope mushrooms, the researchers randomly selected 2 species each for toxic and edible categories from the mushroom checklist compiled by Dulay et al. (2023): *Amanita muscaria* (Fly Agaric), *Leucocoprinus birnbaumii* (Plantpot Dapperling) for toxic, *Pleurotus ostreatus* (Oyster Mushroom), and *Schizophyllum commune* (Splitgill Mushroom) for edible. The Fly Agaric sample was flagged as 'Image Unclear' and withheld results in both the offline and online modes. Offline, Plantpot Dapperling was misclassified as *Lepiota cristata* with an 'Inconclusive Match' warning, and results remain the same online but without the warning. The Oyster Mushroom got misclassified as *Agaricus xanthodermus* in both offline and online modes, both having an 'Inconclusive Match' warning. For Splitgill Mushroom, it was misclassified as *Lepiota cristata* in offline mode with an 'Inconclusive Match' warning while in online mode, it withheld results and was flagged as 'Image Unclear'.

Across the four out-of-scope species, withholding occurred in three of eight runs (Fly Agaric in both modes; Splitgill Mushroom online). The other five runs assigned the sample to a trained toxic class, four with an “Inconclusive Match” warning and one (Plantpot Dapperling, online) without any warning. The withheld results were labeled “Image Unclear,” not as an unknown species. None of the assigned samples was classified as an edible trained class, but with n = 4 and three of the six classes being toxic, this may be chance and should not be treated as a property of the model. This informal test illustrated behavior on unfamiliar inputs and did not measure out-of-distribution detection.

While this pattern was based on a small, informal test set (n=4) and should not be treated as a statistically validated property of the model, it remained consistent with the broader false-negative and false-positive trends observed across the 4 primary evaluated architectures in this study, where larger models similarly demonstrated a skew toward the safer failure direction. Overall, system testing showed that the core features functioned in both modes under the tested conditions, though it did not evaluate reliability on unknown species. This testing successfully showcased the readiness of the system for expert and end-user validation.

On-Device Performance Testing

To objectively evaluate the real-world latency of the developed system, the researchers used a low-end Android device chosen as a representative example of a low-specification smartphone. The device specifications were as follows:

- Octa-core 1.8 GHz Cortex-A53 Processor
- Qualcomm Adreno 506 GPU
- Qualcomm SDM450 Snapdragon 450
- 2GB RAM
- 32 or 64GB Storage

Six offline and online latency testings were conducted on the selected low-end device, with the results presented in Table 10. The offline model, MobileNetV4-Large, recorded an average latency of 5.05 seconds, while the online model, Swin Transformer Base recorded 4.085 seconds. These covered the full three-view submission and classification result. The online mode's faster latency reflects the advantage of offloading inference to server-side, where it remained unconstrained by the device's limited capacity aside from internet strength. Despite being tested on a low-end device and its hardware, both modes consistently returned classifications within 3 to 6 seconds, a suitable timeframe for real-world mushroom identification. Results were from six runs per mode on this single device and may differ on other hardware. Table 10 presents the latency of each of the six trials per mode. Offline mode (MobileNetV4-Large) averaged 5,050.33 ms (SD = 522.30), ranging from 4,414 to 5,724 ms, while online mode (Swin Transformer Base) averaged 4,084.50

ms (SD = 764.44), ranging from 3,030 to 5,033 milliseconds. On average, online mode was about 966 ms (19.1%) faster. Online latency also varied more across trials, which proved consistent with its dependence on network strength. All results derived from six runs per mode on a single device and might differ on other hardware configurations.

Table 10

Latency Testing on Selected Low-End Device

Trial	Offline Mode (ms)	Online Mode (ms)
1	5,133	5,033
2	4,881	3,808
3	4,591	4,552
4	4,414	3,489
5	5,724	3,030
6	5,559	4,595
Mean	5,050.33	4,084.50
SD	522.30	764.44

User Feedback

The modified Post-Study System Usability Questionnaire-based instrument, provided in Appendix Q, was completed by a total of 20 participants selected through purposive sampling, as demonstrated by the number of respondents providing consent to participate in the end-user survey (Appendix R). Respondents were classified into 4 distinct user groups based on their self-reported backgrounds: General Users (n=10), mycologists, foragers, and

outdoor enthusiasts representing mushroom enthusiasts (n=3), Farmers/Breeders/Gardeners representing environmental groups (n=3), and Future Researchers and Technology Practitioners (n=4). To examine whether usability perceptions varied by participant background, responses were first grouped by participant type, and their subscale scores were computed individually before being summarized across the entire sample.

Interpreting results by group, General Users and Environmental Groups rated the system most favorably among the 4 user groups, yielding overall scores of 4.76 and 4.85, respectively. In contrast, Mushroom Enthusiasts and Future Researchers/Technology Practitioners rated the system lower, at 4.58 and 4.52, respectively. System Usefulness scored lowest among these latter 2 groups, suggesting that participants with greater domain expertise or technical familiarity maintained higher expectations regarding system efficiency and reliability compared to participants without such backgrounds. Interface Quality and Information Quality scored consistently high across all 4 user groups, ranging from 4.58 to 4.87, which indicated that the overall visual design, layout, and clarity of information were well received regardless of participant profile or expertise level. Given the small sample sizes of Mushroom Enthusiasts and Environmental Groups (3 participants each), these group-level patterns should be interpreted descriptively rather than as statistically confirmed conclusions. Averages per participant group are illustrated in Table 11.

Table 11

PSSUQ Results by Participant Group per Subscale

Subscale	General Users (n=10)	Mushroom Enthusiasts (n=3)	Environmental Groups (n=3)	Future Researchers & Tech Practitioners (n=4)
Interface Quality	4.80	4.78	4.78	4.58
Information Quality	4.80	4.73	4.87	4.70
System Usefulness	4.71	4.42	4.88	4.38
Total	4.76	4.58	4.85	4.52

Interpreting the results by subscale, Interface Quality scored the highest with a score of 4.75 out of 5, indicating that the system's overall design, layout, and overall interactivity were favorable regardless of user expertise. Information Quality, which covers error handling, recovery, and clarity of system information, was rated high with a score of 4.78, indicating that the system's presentation of relevant information (e.g., toxic status, species label, warnings, and confidence level) was clear and easily understood even to non-experts. The system's error management was designed to limit interactions to what was strictly necessary (uploading photos and selecting model mode), which limited the errors users can encounter while using the app, thus resulting in the high average obtained on the subclass.

System Usefulness, which pertains to efficiency, learnability, and ease of use, showed the lowest score out of the 3 subscales, at an average of 4.63 out of 5. The average score

indicates the system could be further improved, with additional comments requesting on-hand reference material that can be used for comparison and more detailed classification reasoning to improve user trust in the system. Expert users identified areas for improvement to strengthen the system's capabilities beyond its current state.

These results, illustrated in Table 12, suggest that the system was well-received by all the evaluated participants, with a strong liking for the app's interface and information quality, with opportunities for improvement present in system usefulness.

Table 12

PSSUQ Results per Subscales

Subscale	Average Score out of 5
Interface Quality	4.75
Information Quality	4.78
System Usefulness	4.63
Total	4.72

ML Expert Feedback

Following review of the model's training procedure and evaluation results, the ML expert raised the following points, which were addressed as follows:

On data splitting and cross-validation

The experts asked whether the dataset split was performed prior to augmentation, and whether images from the same specimen (or their augmented variants) were prevented from appearing across multiple partitions. It was clarified that the split was performed before augmentation, and that augmentation was applied



exclusively to the training partition. Furthermore, the experts questioned whether cross-validation was utilized; it was confirmed that no cross-validation was employed, and the study maintained a single fixed training, validation, and test partition throughout all evaluations.

On Evaluation Consistency

The experts questioned why support counts and certain metrics in the classification reports appeared to differ from those implied by the confusion matrices. It was confirmed that all reported results were generated from the same fixed test set, using consistent normalization parameters (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]; as applied in the MobileNetV4-Large pipeline) across both training and testing phases.

On Test-Time Augmentation (TTA)

The experts asked which reported results used TTA and whether TTA is active in the deployed application. It was clarified that TTA was used in earlier experimental runs and was subsequently removed. All reported results use single-pass inference without TTA, matching the deployed application.

On the definition of "low-resource"

The experts requested a precise, measurable basis for the term "low-resource," including model size, parameter count, memory requirements, inference time, and performance on the target device. It was clarified that the "low-resource" classification was based on architecture and on-device size: MobileNetV4-Large (32.6M parameters, 120MB) was selected after MobileNetV3-Large (5.5M

parameters) produced unsatisfactory results. Post-Training Quantization (PTQ) in half precision was attempted to further reduce on-device footprint; while experimental results showed improvement, on-device deployment was ultimately unsuccessful (592ms inference at FP32 versus 78,929ms at FP16), a limitation that could not be resolved within the scope of the study.

On misclassification risk between toxic species and their edible look-alikes

The experts inquired how many poisonous specimens were misclassified as their edible look-alikes, and what safeguards mitigate this risk. It was clarified that a prediction confidence cutoff of approximately 85% is used to trigger safety warnings to the user.

On handling of out-of-distribution (unknown) species

The experts asked whether the model could return an "Unknown" or "Unable to Identify Reliably" result for mushrooms outside the six trained classes, rather than forcing a prediction. It was clarified that the underlying neural network always generates a classification prediction across trained classes; however, the application layer applies safety threshold cutoffs to flag low-confidence results to the user.

On the appropriate evaluation metric for a safety-critical, imbalanced-risk classification task

The experts noted that because three of the six classes (*Chlorophyllum molybdites, Agaricus xanthodermus,* and *Lepiota cristata*) are toxic or presumed toxic, and three others (*Chlorophyllum rhacodes, Agaricus campestris,* and *Macrolepiota procera)* are their edible look-alikes, overall accuracy is not an

appropriate primary metric for this study. The expert emphasized that recall on the toxic classes, and the direction of misclassifications shown in the confusion matrix, are the metrics that matter most in this setting; a single toxic specimen (e.g., *Chlorophyllum molybdites*) misclassified as its edible look-alike (e.g., *Chlorophyllum rhacodes*) can be masked by a high overall accuracy figure (e.g., 93%). The expert pointed out that recall for the toxic classes *Agaricus xanthodermus* and *Chlorophyllum molybdites* reached only approximately 88.47% and 89.60%, respectively, in MobileNetV4-Large (Figure 8), and recommended prioritizing false-negative reduction across all model selection decisions.

Mushroom Experts

To evaluate the model's performance outside the simulated test-set, an expert validation study was conducted using the mushroom samples identified by the mycologists from CLSU. Both the offline (MobileNetV4-Large) and online (Swin Transformer-Base) models were evaluated against the same 35 expert-confirmed samples across all six classes, along with one additional out-of-scope sample (*Marasmius* sp.) included as an illustrative check of the model's behavior when it encounters a species outside the trained classes.

As shown in Table 13, the offline model correctly classified 30 of the 35 samples (85.71%), while the online model correctly classified 31 of 35 samples (88.57%), excluding the out-of-scope sample. Both results remained consistent with the accuracy and recall ranges observed during the test set evaluation, indicating that model performance aligned appropriately with independent expert-verified samples.

The *Marasmius* sp. sample yielded low-confidence results from both online and offline models, triggering a withheld classification rather than risking a prediction for an out-of-scope species. However, this single sample served merely as an illustrative test and did not establish general out-of-distribution protection, given that in separate informal tests, most unfamiliar samples were still assigned to one of the six classes.

Table 13

Expert Validation Results by Model and Species

Class	Status	Expert-Confirmed	MobileNetV4 Correct	Swin Transformer Base Correct
C. molybdites	Toxic	8	5	6
C. rhacodes	Edible	4	3	3
A. xanthodermus	Toxic	5	4	5
A. campestris	Edible	7	7	6
L. cristata	Toxic	5	5	5
M. procera	Edible	6	6	6
Total	-	35	30	31

Discussion

This study aimed to address accidental poisoning from misidentifying toxic mushrooms by developing a low-resource deep learning classification system capable of distinguishing toxic Philippine wild mushrooms from their edible look-alikes. Across the 4 identified objectives that span dataset curation, model training and testing, model evaluation, and expert and end-user validation, the findings demonstrated that low-resource transfer learning-based architectures are capable of achieving strong performance metrics while also emphasizing trade-offs between model accuracy and safety priorities highlighted by the study. These findings extended the gap identified in the Review of Related Literature, where Philippine mycological checklists (Dulay et al., 2023; Jacob et al., 2022) confirmed the local occurrence of toxic species such as the False Parasol, Yellow Stainer, and Stinking Dapperling, but noted the lack of dedicated profiling and public-facing tools to help non-specialists distinguish them from edible look-alikes. These studies established that the species occur in the Philippines; they did not show that a model trained on iNaturalist images generalizes to Philippine field conditions. Because the training images were globally sourced, the present results spoke to species relevance, not to performance under local humidity, lighting, and regional morphological variation. Philippine-sourced validation therefore remained necessary before any claim about field use.

One of the primary findings was the refocus from accuracy as the sole metric to both recall and the confusion matrix. Initial evaluation of the trained models relied primarily on overall accuracy as the main performance indicator, which is common practice in image classification tasks and consistent with how prior mushroom-classification studies such as

Ozsari et al. (2024) report performance. However, upon consultation with a machine learning expert in the field, it became evident that accuracy alone did not represent the model appropriately when its main purpose was to prevent accidental poisoning. Given that the consequence of a false negative in this context posed a significant life-threatening risk, the evaluation approach was revised to prioritize recall and the confusion matrix. Prioritizing recall on toxic classes reflected the asymmetric cost of errors in this task: classifying a toxic mushroom as edible proved far more harmful than the reverse. The confusion matrix made the direction of errors visible, which accuracy alone hid. Corbière et al. (2021) addressed a different issue: they argued that reliable confidence estimates are required for safety-critical classifiers and described withholding predictions when confidence was low. The study's 3-tier warning system was conceptually informed by this reject-option idea, but it differed in an important way: Corbière et al. learned a confidence estimator, whereas this study used raw softmax scores with heuristic thresholds. The warning system therefore should be read as a practical safeguard, not a validated implementation of that approach. This shift shaped the interpretation and evaluation of model performance across the four selected model architectures and influenced final model selection for both offline and online modes of the system.

Among the four models, MobileNetV4-Large was selected as the model for offline, edge mobile deployment. The MobileNet family is built for low-resource environments and optimized for low latency, small model size, and on-device inference without network connectivity, consistent with Cai et al. (2022)'s emphasis on the importance of computationally and memory-efficient models for resource-constrained devices. This aligned

broadly with project objectives, providing classification offline (outside reachable services) where computational overhead, memory, and latency mattered more than accuracy gains. Findings by Howard et al. (2019) and Zhu et al. (2023) noted that the MobileNet family proved well-suited to mushroom-image classification tasks. MobileNetV4-Large reduced toxic-to-edible misclassifications relative to MobileNetV3-Large (for example, Yellow Stainers misclassified as Field Mushrooms fell from 102 to 68 images) at the cost of increased false positives. Its accuracy (90.80%) lay well below the 99.91% that Zhu et al. (2023) reported, but the 2 could not be compared directly: Zhu et al. classified shiitake mushroom quality, a task involving different classes, images, and error costs. Separating visually similar toxic and edible species from uncontrolled citizen-science photographs proved a harder problem, which might explain much of the gap, although this study did not test that explanation. Its ability to run offline on the selected low-end Android device supported the efficiency rationale of Cai et al. (2022) and Howard et al. (2019); however, because latency was measured on only one device, this support remained limited to that specific device.

Swin Transformer-Base was selected as the model for online mode, where server-side and cloud inference removed hardware constraints present in offline deployment. The model achieved the highest metrics among all evaluated architectures and recorded the lowest false-negative rates for both critical misclassification pairs (Yellow Stainer and False Parasol), where careful consideration was applied given the safety implications of misidentifying toxic mushrooms for their edible counterparts. Notably, Swin Transformer-Base represented the only Vision Transformer-based architecture in the comparison, with all other models relying

on convolutional neural networks (CNNs). Swin Transformer-Base achieved the highest macro F1-score (93.24%) and the fewest toxic-to-edible misclassifications among the 4 models. Liu et al. (2021) designed the architecture to model relationships across image regions through shifted-window self-attention, which might have contributed to observed performance on visually similar species; however, this study could not directly attribute the result to self-attention. Swin Transformer-Base also represented the largest and most heavily trained model, and the comparison utilized a single data split. Earlier literature noted that attention mechanisms do not by themselves guarantee superior performance (Ozsari et al., 2024).

The three-view submission and probability averaging were introduced because a single photograph, especially of the cap, often proved non-diagnostic for look-alike species. Aggregating several views of a single object constituted an established concept in computer vision. However, this study did not compare single-view with three-view performance, and Table 5 represented merely a single illustrative case. Averaging could also dilute a confident correct view or allow 1 poor view to lower confidence below a warning threshold. Training directly on grouped views and comparing it with single-view prediction would be necessary to establish a definitive benefit.

Confidence-based warnings depended on model confidence being informative. In this study, displayed confidence values consisted of raw softmax outputs, and the 85% and 50% cutoffs were set by research judgment rather than derived from the mathematical relationship between confidence and accuracy. They represented heuristic warning levels, meaning that a high displayed confidence did not automatically imply a high probability of being correct.

Calibration analysis (such as reliability diagrams and expected calibration error) remained necessary before thresholds could be treated as rigorous safety boundaries.

The model functioned as a closed-set classifier trained on six classes, meaning the application assigned unfamiliar inputs to one of the six classes unless confidence fell below thresholds. In informal out-of-scope tests, only some unfamiliar samples were withheld, and several were assigned to a trained toxic class, including one instance without any warning. The single *Marasmius sp.* sample that was withheld served as an illustrative case and did not establish general out-of-distribution protection. Given that the Philippines harbored hundreds of documented wild mushroom species, field use would regularly encounter species outside the It should be noted that performance comparisons among the 4 candidate model architectures relied on performance metrics (e.g., F1-score, recall, and confusion matrix analysis) evaluated on a single train/validation/test split, without inferential statistical testing. While performance improvements across model generations were considered meaningful for model selection, these comparisons should be interpreted as observed differences rather than statistically confirmed advantages of one architecture over another.

Following the selection of MobileNetV4-Large for offline, on-device deployment, further optimization strategies were explored to improve its suitability for low-resource mobile environments. In the search for low-resource techniques, model quantization was evaluated, specifically Post-Training Quantization (PTQ) in considering an on-device deployment strategy. FP16 (Half-Precision Floating Point) constituted a reduced-precision numerical format representing model weights and computations using 16 bits instead of the standard 32 bits used in FP32. In principle, this halved memory footprints and could

accelerate inference on hardware with native FP16 support. This study evaluated FP16 (alongside mixed precision, which retained FP32 for BatchNorm layers while converting the rest of the model to FP16 (Taklai et al., 2025) as a candidate compression technique and confirmed correct execution: model accuracy was preserved within a fraction of a percentage point of the FP32 baseline across every architecture tested, and file sizes were reliably reduced by approximately 50%.

However, when integrated into an actual mobile deployment pipeline using ExecuTorch within a Flutter application, a single mushroom identification required approximately 80 seconds to complete, compared to the deployed FP32 model's end-to-end latency of approximately 5.05 seconds per classification. This resulted from incomplete hardware and software support for native FP16 execution across all operations within the model, necessitating repeated upcast and downcast conversion overhead. This latency result diverged from findings by Taklai et al. (2025), whose evaluation of FP16 quantization did not report deployment-level latency penalties. This discrepancy might stem from the specific mobile runtime and hardware used in this study rather than an inherent shortcoming of FP16 quantization itself. Given this finding, FP32 was retained as the deployed precision format for the final mobile application.

Expert and end-user validation offered valuable perspective on model and system capabilities and reliability. Experts signed acknowledgement forms clarifying whether model findings and results derived through thorough consultation proved sufficient and validated. This suggested that the selected training pipeline, model choices, and deployment strategy met the objectives set by the study while introducing actionable areas for improvement:

reducing toxic-class misclassifications, expanding testing on out-of-scope species, and providing robust performance evidence across diverse entry-level devices. Notably, concerns regarding model size and quantization constraints (FP16 and INT8) were addressed above.

On the 35 mycologist-confirmed samples, accuracy reached 85.71% for MobileNetV4-Large and 88.57% for Swin Transformer-Base, compared with test-set accuracies of 90.80% and 93.38%, respectively. Both models therefore scored approximately 5 percentage points lower on external samples. This proved consistent with the premise that externally confirmed specimens fused from 3 uncontrolled views are inherently harder than a held-out split from the same image source, though empirical evidence remained limited. Wilson confidence intervals (approximately 71% to 94% and 74% to 96%) encompassed test-set accuracies, meaning the performance gap could not be statistically distinguished from chance. Per-class results also revealed weaker performance on toxic classes than suggested by the test set: *Chlorophyllum molybdites* was correctly identified in 5 of 8 samples by MobileNetV4-Large and 6 of 8 by Swin Transformer-Base. Because error directions were not reported, whether these errors involved toxic-to-edible misclassifications remained unknown. A larger, systematically sampled external dataset tracking error directions is required.

The mean score of 4.72 out of 5 indicates favorable perceived usability on the study's modified 5-point scale. Because the instrument was adapted from the standard PSSUQ (reworded items, a 5-point scale with higher scores indicating greater usability), the score is not directly comparable with results from standard PSSUQ studies. Vlachogianni and Tselios (2023) support the use of PSSUQ-type instruments for usability evaluation; they are cited for

the suitability of the approach, not as a benchmark for this score. The sample of 20 participants, with only three in two of the groups, also limited generalization.

Disaggregating these results by participant group clarified where the lower System Usefulness rating originated. On the System Usefulness subscale, General Users (n = 10) and Environmental Groups (n = 3) gave favorable ratings of 4.71 and 4.88, respectively, while Mushroom Enthusiasts (n = 3) and Future Researchers and Technology Practitioners (n = 4) gave lower ratings of 4.42 and 4.38 (Table 11). The same ordering appears in the total scores: Environmental Groups (4.85) and General Users (4.76) rated the system higher than Mushroom Enthusiasts (4.58) and Future Researchers and Technology Practitioners (4.52). Interface Quality remained consistently high across all 4 groups, suggesting that visual design and presentation did not drive the lower ratings.

The item eliciting the most varied response was "I feel comfortable using this app to check a mushroom's toxicity status," which ranged from 2 to 5 with an average of 4.20 out of 5. This finding aligned with prior literature on trust in AI-driven classification and diagnostic systems, which correlated distrust with the black-box nature of artificial intelligence and the difficulty non-technical users face in interpreting classification processes and results (Li et al., 2021). This insight supported the recommendation to integrate explainability features such as Grad-CAM, enabling users to visualize which morphological features informed specific predictions. Participant requests for reference images and detailed justifications pointed in the same direction, and such explanations could also assist users in evaluating when predictions should not be trusted. This discomfort proved reasonably heightened in health-related applications where incorrect predictions directly threatened user safety.

Conversely, items receiving the highest ratings (averaging 4.90 out of 5) pertained to result delivery speed, information clarity, and general interface satisfaction. Notably, high ratings for result speed supported the core design objective of creating a low-resource model suitable for on-field identification, as empirically demonstrated on the selected low-end Android device.

Because Mushroom Enthusiasts and Environmental Groups comprised only 3 participants each, these group-level differences should be read descriptively rather than as statistically confirmed conclusions. A larger, more balanced sample across user groups would be required to rigorously test whether domain expertise and its relationship with trust affected perceptions toward mobile classification applications. This constituted a notable limitation, as the small sample sizes of the lowest-numbered groups rendered subscale averages extremely sensitive to individual responses and limited cross-group comparisons. Future studies should recruit larger, more evenly distributed participant samples to validate these trends.

Future iterations of the system should consider training models directly on multi-view or group image inputs rather than relying on post-hoc averaging of independent single-image predictions, thereby ensuring proper validation of the fusion mechanism. Additionally, comprehensive confidence calibration analyses should be conducted to validate or adjust thresholds utilized within the confidence-based warning system.

Conclusion

This study developed and evaluated a prototype, low-resource machine learning mobile application designed to support safer identification and screening of wild mushrooms

in low-connectivity settings. The system targets three toxic or presumed-toxic species documented to occur in the Philippines, namely *Chlorophyllum molybdites* (False Parasol), *Agaricus xanthodermus* (Yellow Stainer), and *Lepiota cristata* (Stinking Dapperling, a presumed-toxic species), and distinguishes them from their visually similar edible counterparts, *Chlorophyllum rhacodes* (Shaggy Parasol), *Agaricus campestris* (Field Mushroom), and *Macrolepiota procera* (Parasol Mushroom). To reduce model overestimation, a curated dataset of 58,184 images across six target classes was assembled from iNaturalist and preprocessed using a group-aware split strategy providing observation-level isolation across the training, validation, and test partitions.

Based on evaluation on a single fixed split, two models were selected for a dual-mode architecture. For offline use, MobileNetV4-Large was chosen as the on-device model, achieving a macro F1-score of 90.56% and an average end-to-end latency of 5.05 seconds on the selected low-end Android device. For online use, Swin Transformer Base was selected as the cloud-based model, achieving the highest macro F1-score of the four models at 93.24%. Swin Transformer Base also showed fewer toxic-to-edible misclassifications than MobileNetV3-Large, reducing Yellow Stainer misclassified as Field Mushroom from 102 to 44 and False Parasol misclassified as Shaggy Parasol from 75 to 30. These are observed differences on one split and were not statistically tested.

The system was examined through test-set evaluation, mycologist-confirmed external samples, and end-user testing. On 35 samples confirmed by mycologists from Central Luzon State University (CLSU), accuracy was 85.71% for MobileNetV4-Large and 88.57% for Swin Transformer Base. This small sample is a descriptive check, not a field validation. End-

user testing with 20 participants, using a modified PSSUQ-based instrument, produced a mean of 4.72 out of 5 on the study's scale, with System Usefulness at 4.63, Information Quality at 4.78, and Interface Quality at 4.75.

Taken together, these findings suggested that lightweight on-device models, paired with a heuristic three-tier confidence-warning interface, could support safer mushroom identification and screening without constant connectivity. The work demonstrated potential for broader deployment in rural communities, agricultural environments, and public health contexts. Effects on actual poisoning incidents were not measured. Further work remains necessary regarding Philippine-sourced validation, multi-device latency testing, confidence calibration analysis, out-of-scope detection, and rigorous validation of the three-view fusion mechanism prior to wider deployment. The application functions strictly as an aid to identification and does not replace expert consultation before consuming any wild mushroom.

A condensed version of this study, formatted according to ACM conference paper guidelines, is provided in Appendix W.

Recommendations

Based on the findings and limitations identified in this study, several recommendations are proposed for future research and development:

Future studies should expand the dataset to include additional Philippine wild mushroom species beyond the initial six examined in this study, namely *Chlorophyllum molybdites*, *Agaricus xanthodermus*, *Lepiota cristata*, and their respective edible look-alikes. A broader taxonomic scope would enhance the generalizability of the model and its practical

utility for field identification, given the extensive diversity of wild fungi documented across the Philippines.

The integration of explainable AI (XAI) in the mobile app is also recommended. End-user evaluation in this study revealed discomfort and comparatively lower trust scores due to the vagueness and black-box nature of the classification process. By incorporating explainable methods such as Grad-CAM, users are provided with clearer justification for a given prediction. This is particularly important for health-related applications, where user trust directly influences system adoption and decision-making.

Further refinement of out-of-scope handling is recommended as well. The developed system withholds or flags some low-confidence results, but it is a closed-set classifier and has no dedicated detection of unknown species. Future work should add open-set recognition or out-of-distribution detection and evaluate it on a larger, systematically chosen set of unfamiliar species.

This would reduce the risk of the system confidently misclassifying unfamiliar species as one of the trained classes.

Lastly, future development should implement and evaluate post-training quantization (PTQ), including FP16 and INT8 formats, using deployment frameworks with more complete native low-precision support. This study retained FP32 for the final application as FP16 quantization via ExecuTorch within Flutter required repeated upcasting and downcasting due to incomplete native FP16 support, increasing inference latency to approximately 80 seconds despite reducing file size by 50%. Future work should explore

alternative mobile inference runtimes or hardware with better native FP16 or INT8 support to achieve these storage compression benefits without incurring deployment latency penalties.

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Appendix A

Cover Letter for End-User System Evaluation

Greetings!

We are fourth-year BS Computer Science students from the **School of Computing (SoC) at Holy Angel University (HAU)**, conducting a survey in partial fulfillment of the final requirements for our course **CS Independent Study 2 (6CISTUDY2)**.

Our thesis is entitled **Enhancing Foraging: A Low-Resource Learning Model for Poisonous Mushroom Identification in the Philippines**, focusing on low-resource detection of edible and poisonous mushrooms using computer vision and image recognition.

As end users, we kindly ask you to answer the following questions by **honestly evaluating the finalized system,** wherein your responses will help us **gather relevant data for our thesis**.

In compliance with **Republic Act No. 10173**, otherwise known as the **Data Privacy Act of 2012**, all information provided will be treated with **strict confidentiality and will be used solely for research purposes.** All data will be securely stored and will be properly disposed of after the completion of this study.

By proceeding with this survey, you voluntarily agree to the collection and use of your data for the purposes stated above.

Thank you for your time, and we greatly appreciate your participation!

Appendix B

API iNaturalist Data Collection Code Snippet

```
SPECIES = {
    "Stinking_Dapperling": "Lepiota cristata",
    "Yellow_Stainers": "Agaricus xanthodermus",
    "False_Parasol": "Chlorophyllum molybdites",
    "Parasol": "Macrolepiota procera",
    "Field_Mushroom": "Agaricus campestris",
    "Shaggy_Parasol": "Chlorophyllum rhacodes",
}

def download_image(url, save_path):
    try:
        r = requests.get(url, stream=True, timeout=30)
        if r.status_code == 200:
            with open(save_path, 'wb') as f:
                for chunk in r.iter_content(1024):
                    f.write(chunk)
            return True
    except Exception as e:
        print(f"  Download failed: {e}")
    return False

def scrape_species(taxon_name, output_dir, max_images=8000):
    out_dir = Path(output_dir) / taxon_name.replace(" ", "_")
    out_dir.mkdir(parents=True, exist_ok=True)

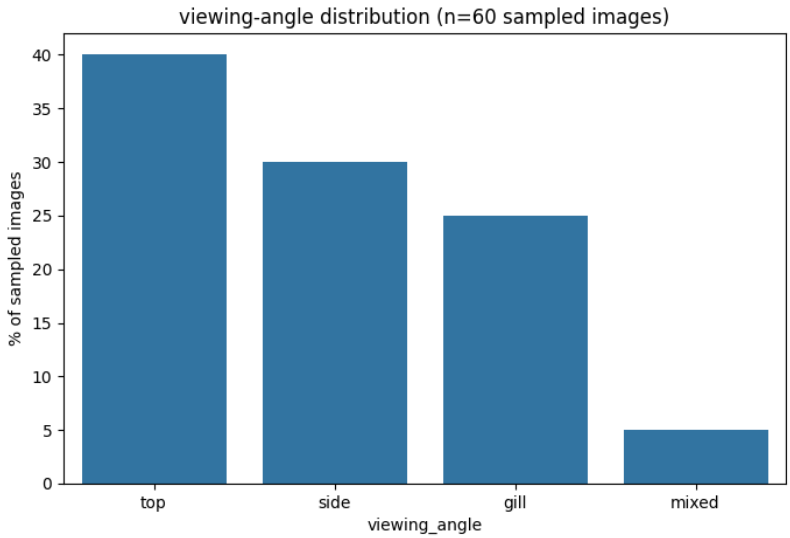
    print(f"\nScraping {taxon_name} → {out_dir}")

    page = 1
    per_page = 100
    downloaded = 0
    base_url = "https://api.inaturalist.org/v1/observations"

    while downloaded < max_images:
        params = {
            "taxon_name": taxon_name,
            "quality_grade": "research",
            "has_photos": "true",
            "order_by": "observed_on",
            "order": "desc",
            "per_page": per_page,
            "page": page,
            "photo_license": "cc-by,cc-by-nc,cc0"
        }
        try:
```

Appendix C

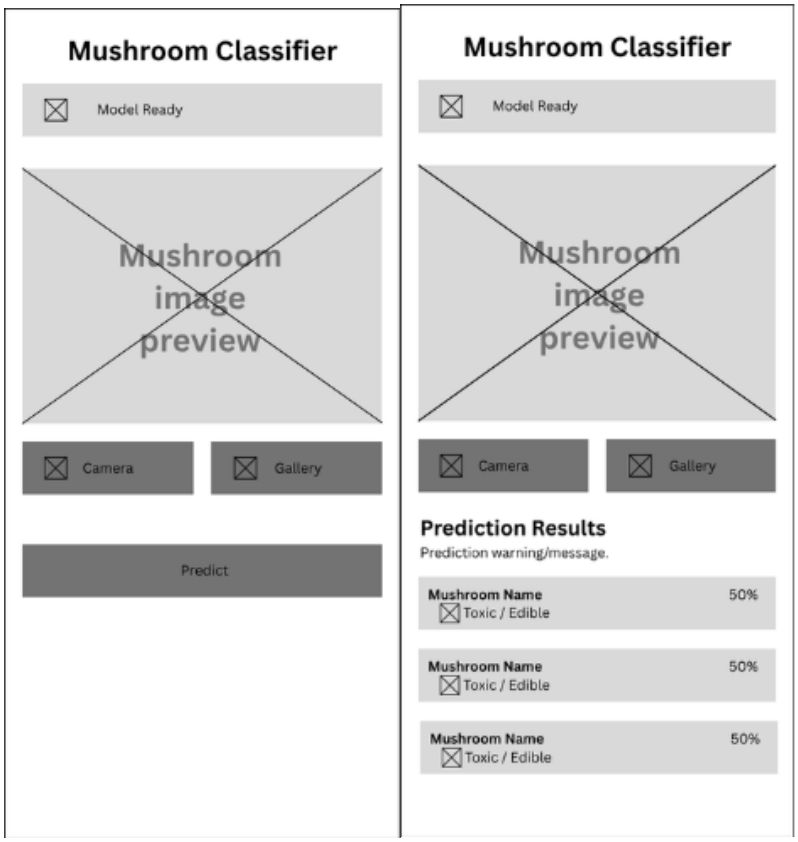
Viewing Angle Sample Images



https://drive.google.com/drive/folders/1k0OB45In2smgJjmDpUWE8grleU-Ih_lN?usp=sharing

Appendix D

Low Fidelity Wireframe



Appendix E

Group-Aware Leakage Script

```
print("\nPerforming group-aware split (random_state fixed for reproducibility)...")

gss1 = GroupShuffleSplit(n_splits=1, train_size=0.8, random_state=seed)
train_idx, rest_idx = next(gss1.split(all_paths, y, groups))

rest_groups = groups[rest_idx]
gss2 = GroupShuffleSplit(n_splits=1, train_size=0.5, random_state=seed)
val_rel_idx, test_rel_idx = next(gss2.split(rest_idx, y[rest_idx], rest_groups))
val_idx = rest_idx[val_rel_idx]
test_idx = rest_idx[test_rel_idx]

split_map = {"Train": train_idx, "Val": val_idx, "Test": test_idx}
```

Appendix F

Data Transformation Snippet

```
IMG_SIZE = 384

train_transform = A.Compose([
    A.RandomResizedCrop(size=(IMG_SIZE, IMG_SIZE), scale=(0.7, 1.0), ratio=(0.85, 1.15), p=1.0),
    A.HorizontalFlip(p=0.5),
    A.VerticalFlip(p=0.3),
    A.RandomRotate90(p=0.3),
    A.GaussianBlur(blur_limit=(3, 7), p=0.15),
    A.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.15, hue=0.05, p=0.5),
    A.CoarseDropout(max_holes=4, max_height=32, max_width=32, p=0.2),
    A.Normalize(mean=NORM_MEAN, std=NORM_STD),
    ToTensorV2(),
])
```

Appendix G

Unvalidated Image Set



https://drive.google.com/drive/folders/1k0OB45In2smgJjmDpUWE8grleU-Ih_lN?usp=sharing

Appendix H

Validated Image Set



https://drive.google.com/drive/folders/1w0Zc6cp-n3CTB7p8qud07_M5TG1kdljJ?usp=sharing

Appendix I

ML Expert Consultation Discussion

How were the training, validation, and test sets separated?
Was the dataset split performed before augmentation, and were images of the same mushroom specimen or augmented versions prevented from appearing across different partitions?

A: Yes, the split was performed before the augmentation. Augmentation was applied exclusively during training, and only the training partition was given the extensive augmentations.

Were all reported results generated from exactly the same final test set?
Why do the support counts and some metrics in the classification reports differ from those shown or implied by the confusion matrices?

A: Since the final tests were done via the fixed dataset, all reports are from the same test set. The Normalization -> mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225] were used on both training and testing.

Which results are without TTA and which results use Test-Time Augmentation (TTA)?
Is TTA also used in the actual deployed application, or only during experimental evaluation?

A: TTA was conducted only for experimental evaluations.

What specifically does "low-resource" mean in this study, and what measurements demonstrate it?
What are the model size, parameter count, memory requirements, inference time, hardware requirements, and performance on the intended target device?

A: Low resources' basis was the model architecture and size on the device. We trained the MobileNetv4-Large with a parameter count of 32.6M and got a model size of 120mb. We had tested on a smaller parameter count before MobileNetv4, the MobileNetv3-Large (5.5M), and we were unsatisfied with the results. To further shrink model size on the device, we attempted to use PTQ in Half Precision; experimental results showed improvement. Unfortunately, implementation on-device came back undesirable with MobileNetV4-Large having 592ms on FP32 and 78,929ms on FP16, and we weren't able to remedy it.

--- MobileNetV3 Classification Report ---

	precision	recall	f1-score	support
Agaricus_campestris	0.8105	0.8701	0.8392	816
Agaricus_xanthodermus	0.8654	0.8333	0.8491	918
Chlorophyllum_molybdites	0.8808	0.8337	0.8566	1010
Chlorophyllum_rhacodes	0.8322	0.8280	0.8301	994
Lepiota_cristata	0.9513	0.9468	0.9490	1072
Macrolepiota_procera	0.8895	0.9232	0.9061	1003
accuracy			0.8741	5813
macro avg	0.8716	0.8725	0.8717	5813
weighted avg	0.8747	0.8741	0.8740	5813

How many poisonous mushrooms were incorrectly classified as their edible look-alikes?
How do the researchers interpret these errors, and what safeguards are implemented to reduce the risk associated with such misclassifications?

https://drive.google.com/drive/folders/1EuCqaW6fHVKl8GAbg38-9XhM63P76Q8B?usp=sharing

Appendix J

ML Expert Acknowledgement Letter

ML Expert Acknowledgement Letter 1

CERTIFICATE OF MACHINE LEARNING MODEL VALIDATION

Enhancing Foraging Safety: A Low-Resource Machine Learning Model For Poisonous Mushroom Identification In The Philippines

Danielle Laurence B. Lenon, Hans Christian S. Ocampo, Bianca Claire L. Ochoa, and Bianca Venice L. Ochoa
Bachelor of Science in Computer Science

This is to certify that the undersigned has acted as an **Expert Evaluator** for the machine learning application developed for the research study mentioned above.

The prototype was subjected to hands-on testing to evaluate its user interface and predictive outputs. The proponents have successfully completed the end-of-development evaluation survey and have satisfactorily answered all queries regarding the model's localized classification performance and usability.

Therefore, the machine learning model prototype is hereby acknowledged as **VALIDATED** for academic presentation.

Evaluated by:

SIGNATURE

PROF. DARYL IVAN HISOLA, MSIT
College of Computing and Information Sciences, Cor Jesu College

Date Signed: _____

ML Expert Acknowledgement Letter 2

CERTIFICATE OF MACHINE LEARNING MODEL VALIDATION

Enhancing Foraging Safety: A Low-Resource Machine Learning Model For Poisonous Mushroom Identification In The Philippines

Danielle Laurence B. Lenon, Hans Christian S. Ocampo, Bianca Claire L. Ochoa, and Bianca Venice L. Ochoa
Bachelor of Science in Computer Science

This is to certify that the undersigned has acted as an **Expert Evaluator** for the machine learning application developed for the research study mentioned above.

The prototype was subjected to hands-on testing to evaluate its user interface and predictive outputs. The proponents have successfully completed the end-of-development evaluation survey and have satisfactorily answered all queries regarding the model's localized classification performance and usability.

Therefore, the machine learning model prototype is hereby acknowledged as **VALIDATED** for academic presentation.

Evaluated by:

Engr. Gerald Tiocson, ECE, 1PHN -- Divine Word College of Calapan
SIGNATURE

[Full Name and Affiliation]

Date Signed: September 17, 2026

Appendix K

End-User Installation and Testing Guide

Good day! We are asking for your participation in evaluating a system prototype developed under the study "Enhancing Foraging Safety: A Low-Resource Machine Learning Model for Poisonous Mushroom Identification in the Philippines". The primary objective of the application is to assist in distinguishing the 3 identified toxic mushrooms species (False Parasol, Yellow Stainer, Stinking Dapperling) from their common edible-look alikes.

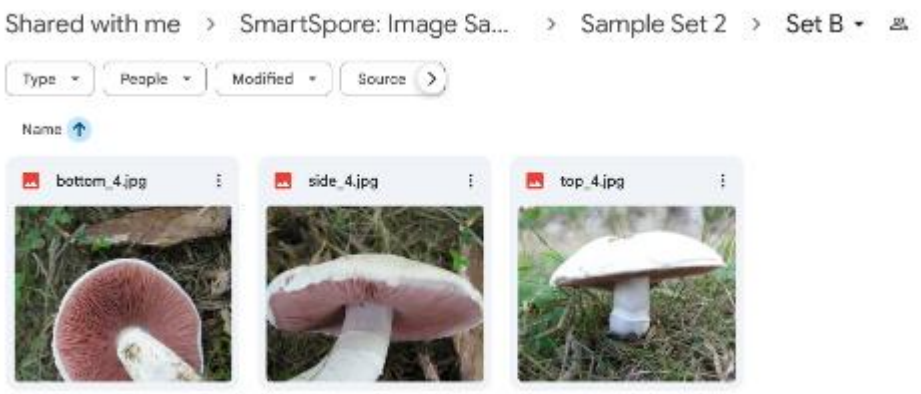
This study is open to the following user groups:
- General Public (Non-experts)
- Mushroom Enthusiasts (Mycologists, Foragers, & Outdoor Enthusiasts)
- Environmental Groups (Farmers, Breeders, & Gardeners)
- Research and Technological Practitioners

App Guide:
https://drive.google.com/drive/u/5/folders/14-YpPzLDMKq_H0SexoLWAMbkV1csP7h1
- Install the APP (APK) from the attached Google Drive link, under a folder named 'App Installation'. Download 'app-arm64-v8a-release.apk' and install it on your device.
- There are sample photos users can use to test the application, located at the attached Google Drive Link, under folders 'Set 1-3'.
- Upload the images to their designated angle (top, bottom, side), then press identify.

After exploring and testing the app, we encourage you to take the usability questionnaire below.
https://forms.gle/mUjMCtjxrc2qtVub7

Appendix L

Test Images for End-Users

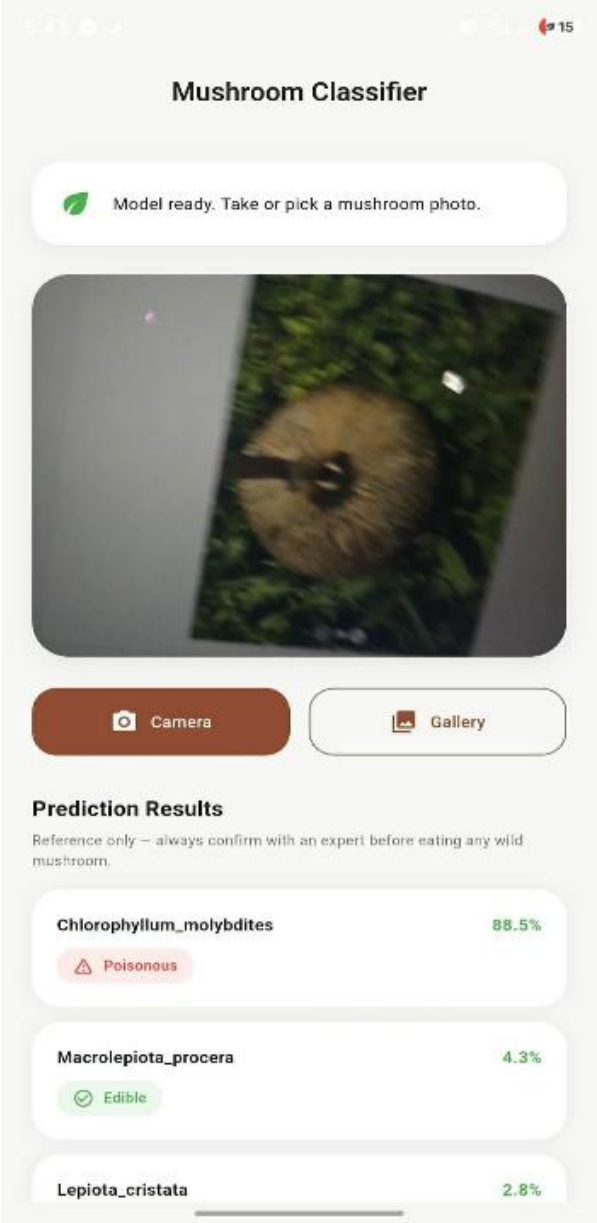


https://drive.google.com/drive/folders/14-YpPzLDMKq_H0SexoLWAMbkV1csP7h1?usp=sharing

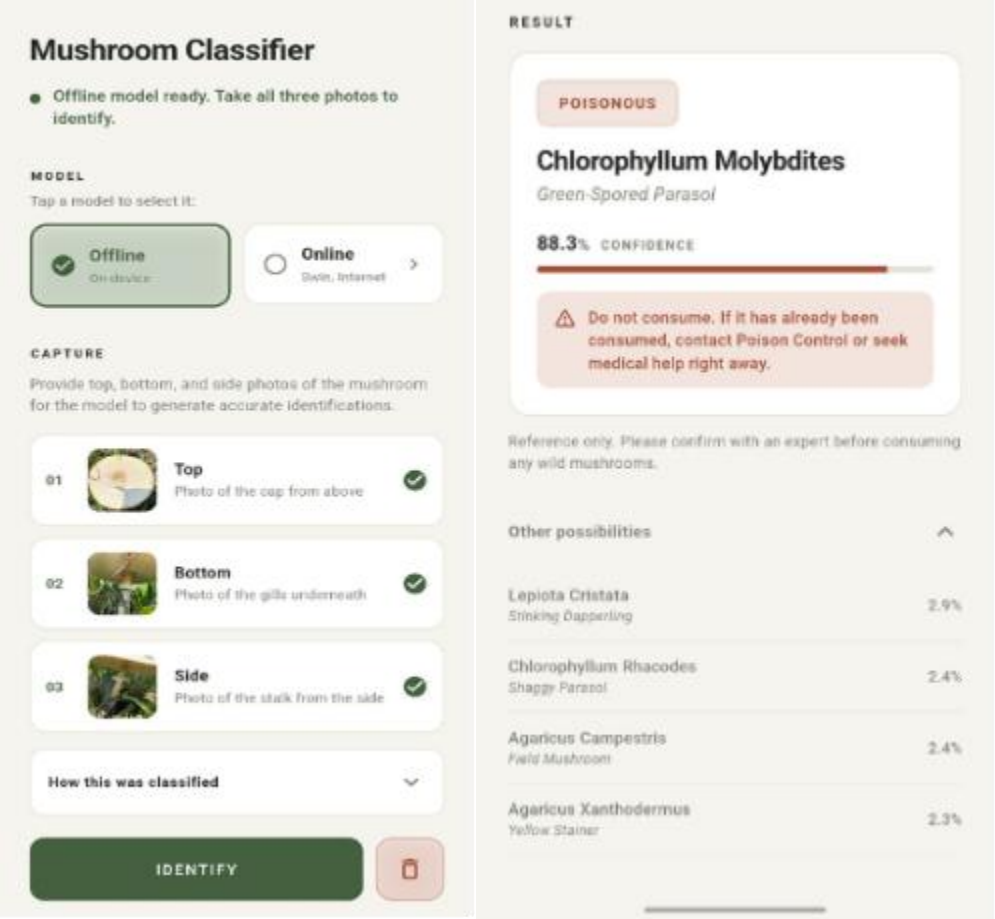
Appendix M

App Iterations

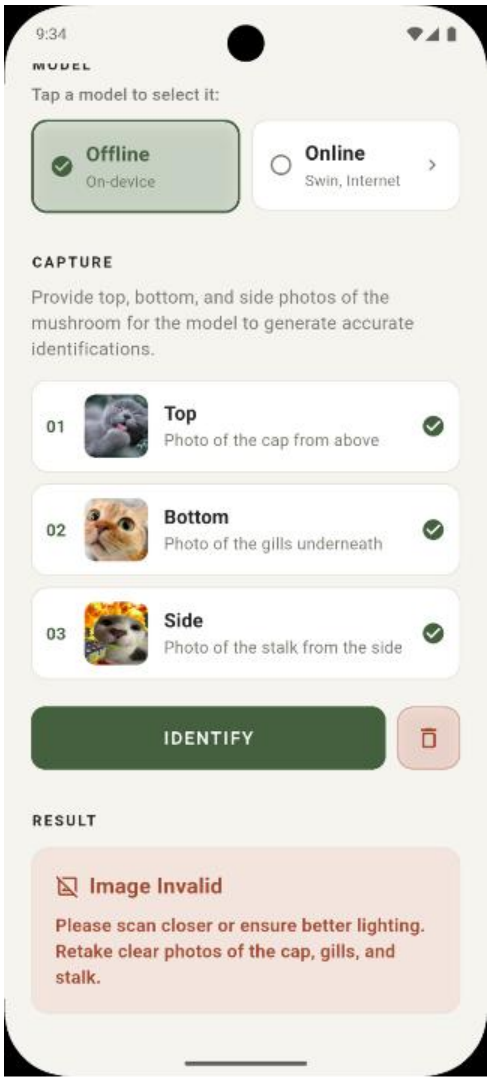
App First Iteration



App Second Iteration



App Third Iteration



App Third Iteration Detailed Results



Appendix N

Final Curated Dataset



https://drive.google.com/file/d/1eIlByDbiHOF-IZFjJXeC8DnzAZrf0CdS/view?usp=sharing

Appendix O

Test Case Table

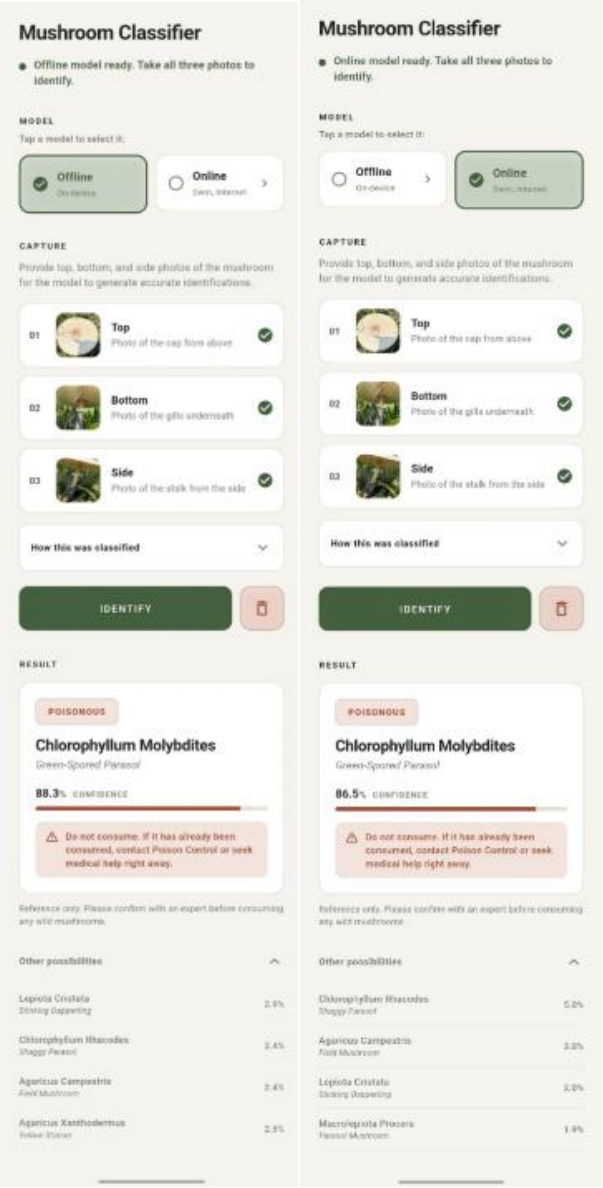
Test Case ID	Test Case Objective	Prerequisite	Steps	Input Data	Expected Output	Actual Output	Status
TC_01	Test User Image Upload (Offline)	Image Upload should be enabled	1. Select Top/Bottom/Lateral Button 2. Select Upload Option (Gallery/Camera) 3. Select Photo 4. Submit	Clear and Valid mushroom photos	Prediction should be displayed	See Appendix P	PASS
TC_02	Test User Image Upload (Online)	Image Upload should be enabled	1. Select Online Mode 2. Select Top/Bottom/Lateral Button 3. Select Upload Option (Gallery/Camera) 4. Select Photo 5. Submit	Clear and Valid mushroom photos	Prediction should be displayed	See Appendix P	PASS
TC_03	Test Non-Mushroom Photo	Image Upload should be enabled	1. Upload top/bottom/side 2. Submit	Non-Mushroom Photo	No prediction available	See Appendix P	PASS
TC_04	Test Incomplete Image Input	Image Upload should be enabled	1. Upload Images but leave 1 empty 2. Submit	Two of three required images	Grey/disabled out the submit button	See Appendix P	PASS
TC_05	Test Blurry/Low-Quality	Image upload should be enabled	1. Upload a blurry or low-light image 2. Submit	Blurry or low-light mushroom photo	App should display results	See Appendix P	PASS

	Image Input						
TC_06	Test out-of-scope or unknown mushroom species	Image upload should be enabled	1. Upload an out-of-scope or unknown mushroom image 2. Submit	Out-of-scope or unknown mushroom images	App should display low-confidence or withhold results	See Appendix P	PASS

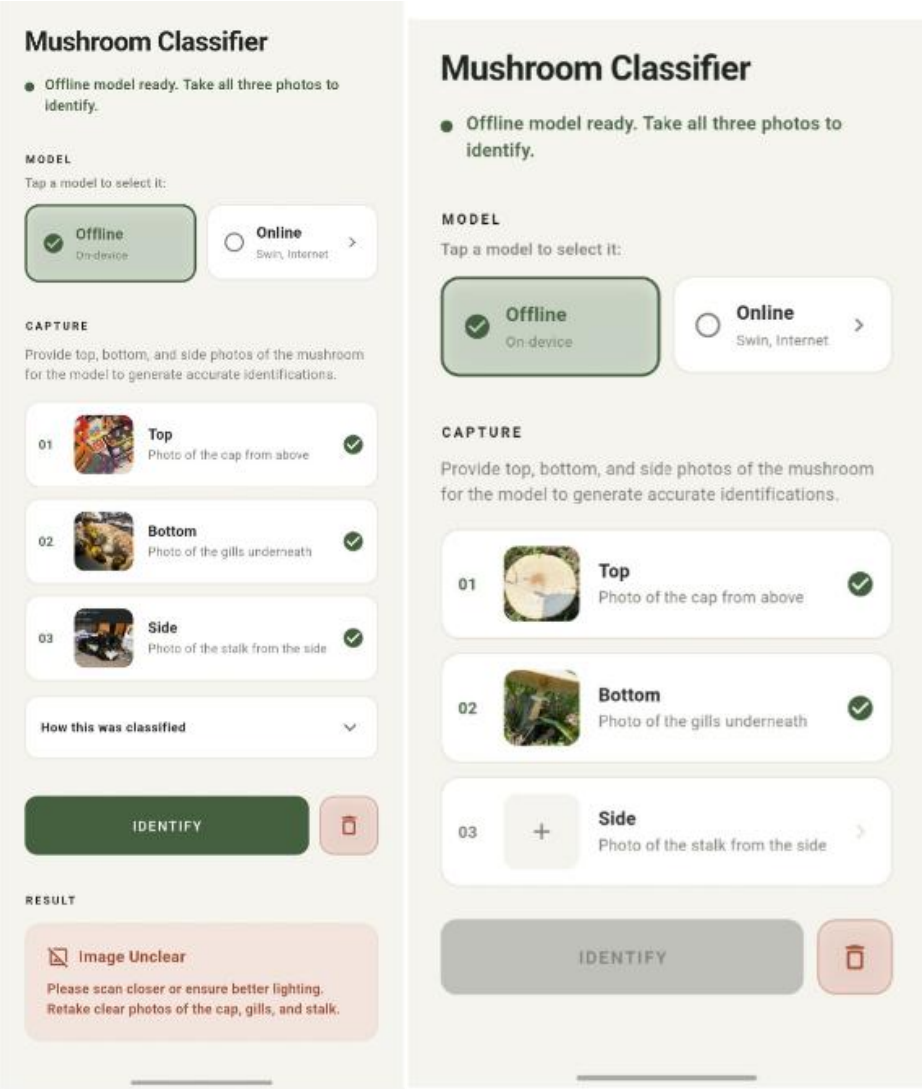
Appendix P

Test Case Outputs

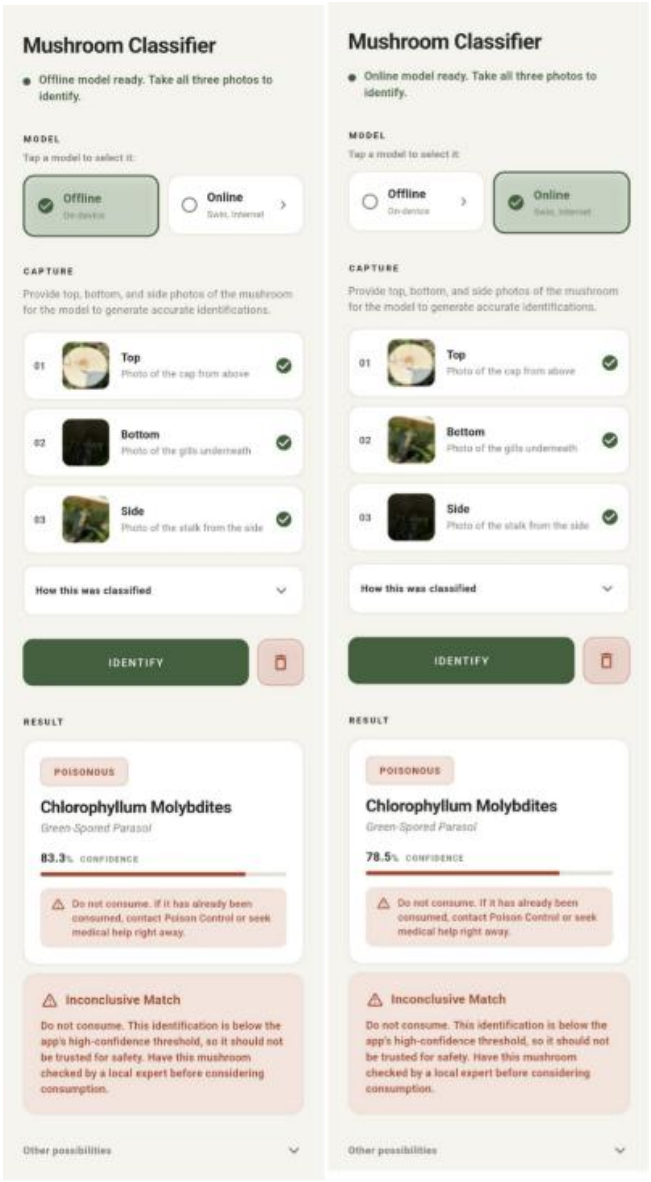
Outputs for Image Uploading Testing for Offline and Online Modes (TC_01-02)



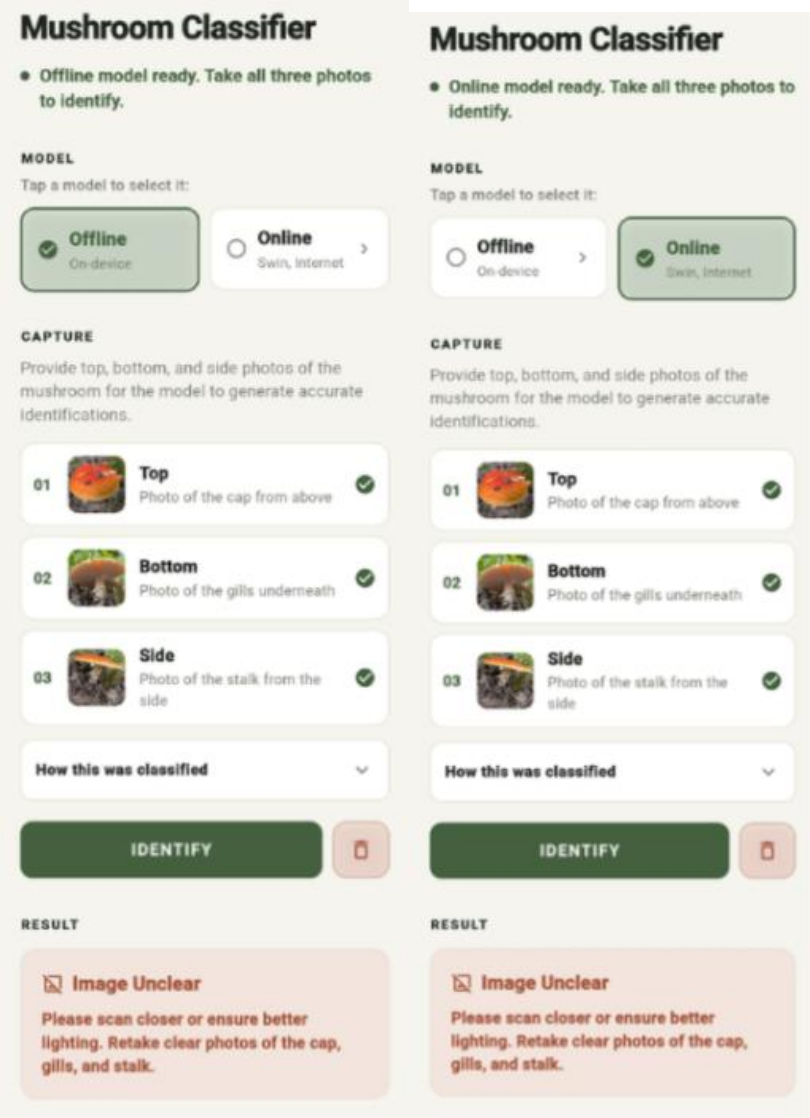
Outputs for Incomplete or Invalid Image Upload Testing (TC_03-04)



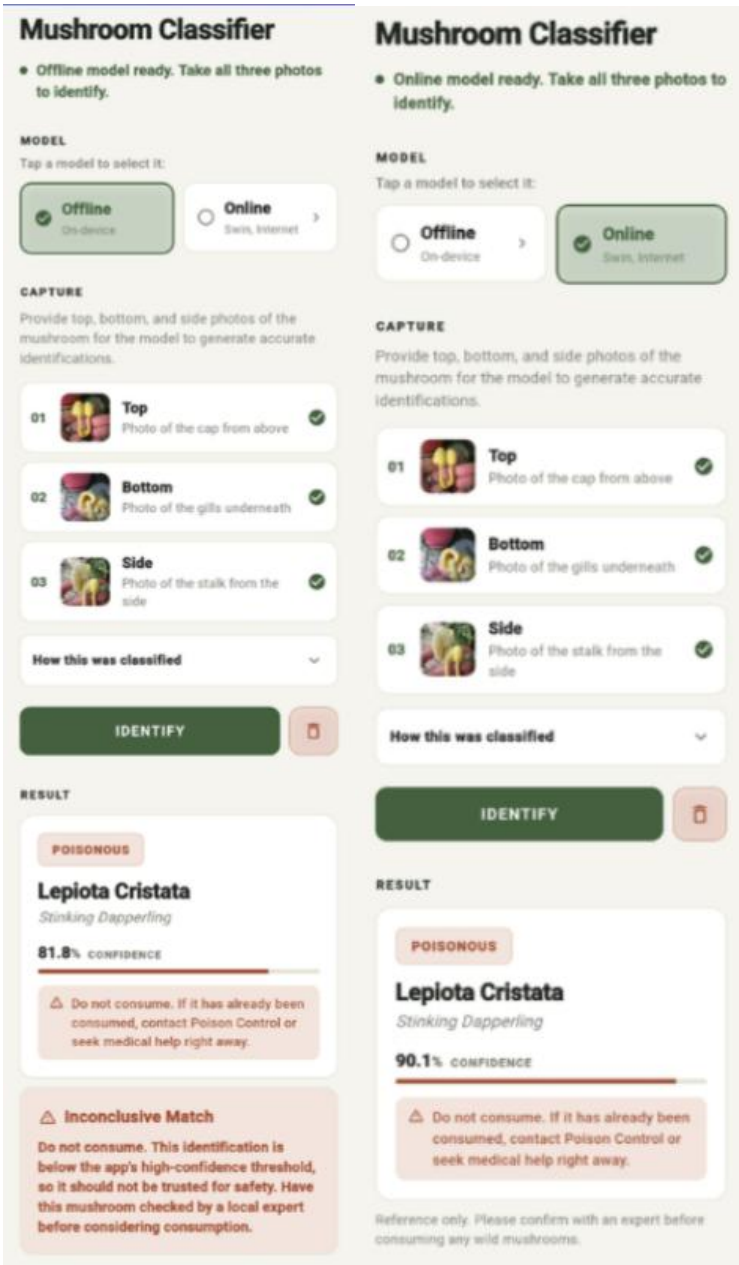
Outputs for Testing Low-Quality Image Uploads (TC_05)



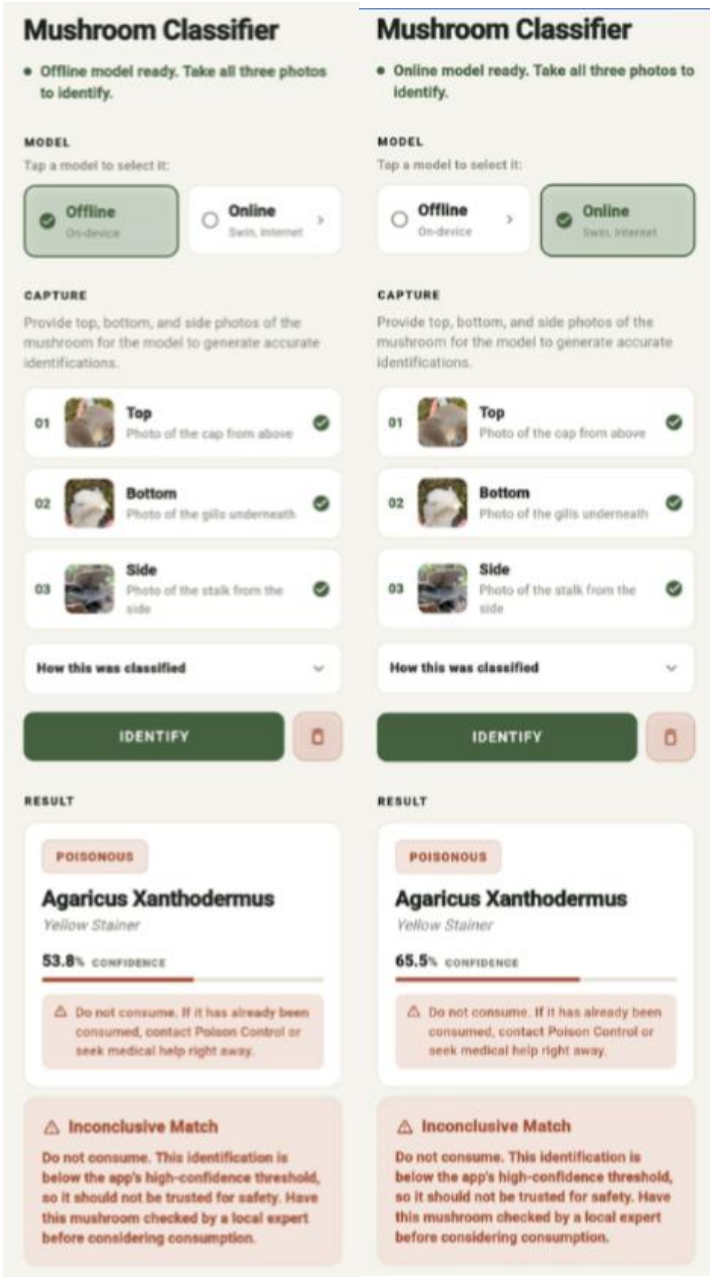
Outputs for Testing Out-of-Scope Mushroom Image Uploads Image 1 (TC_06)



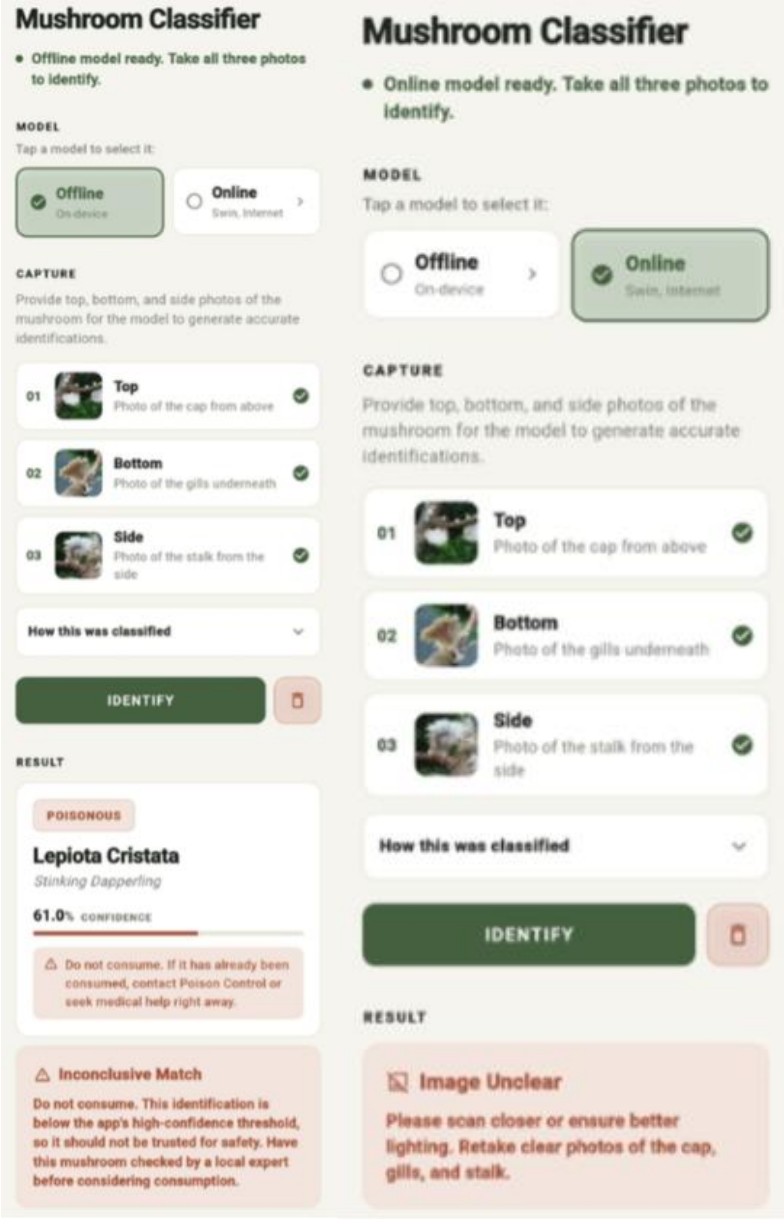
Outputs for Testing Out-of-Scope Mushroom Image Uploads Image 2 (TC_06)



Outputs for Testing Out-of-Scope Mushroom Image Uploads Image 3 (TC_06)



Outputs for Testing Out-of-Scope Mushroom Image Uploads Image 4 (TC_06)



Appendix Q

Post-Study Survey Items

Modified Post-Study System Usability Questionnaire (PSSUQ)-based Instrument

Study: Enhancing Foraging Safety: A Low-Resource Machine Learning Model for Poisonous Mushroom Identification the Philippines

Part 1. Informed Consent

By participating in this survey, we will gather your data anonymously, and you will respond voluntarily.

- ☐ I consent
- ☐ I do not consent

Part 2. Respondent Profile

Which user group do you belong to?

- ☐ General Users
- ☐ Mushroom Enthusiasts (Mycologists, Foragers, Outdoor Enthusiasts)
- ☐ Farmers/Breeders/Gardeners (Environmental Groups)
- ☐ Future Researchers and Technology Practitioners
- ☐ Other:

Part 3. System Usability Rating

No.	Questions	5 (Strongly Agree)	4 (Agree)	3 (Neutral)	2 (Disagree)	1 (Strongly Disagree)
1	By participating in this survey, we will gather your data anonymously, and you will respond voluntarily.					
2	Overall, I am satisfied with how easy it is to photograph a mushroom and get a toxicity classification.					

3	It was simple to use this system to check whether a mushroom was safe to eat. No additional instructions were needed to get a mushroom classification.					
4	I was able to effectively identify the mushroom's species and toxicity using this app.					
5	I was able to quickly get a classification result after uploading or taking mushroom photos.					
6	I was able to use the app efficiently without needing to repeat unnecessary steps.					
7	I feel comfortable using this app to check a mushroom's toxicity status.					
8	It was easy to learn how to use the app to get a mushroom classification.					
9	I believe I could become more productive or efficient in practicing mushroom foraging safety by using this app.					
10	The system gave error messages and instructions that clearly told me how to fix problems or get more accurate results.					
11	Whenever I made a mistake using the system, I could recover easily and quickly.					

12	The information (such as species label, toxicity status, messages) provided by this system was clear.					
13	It was easy to find the information I needed (species label, toxicity status, messages).					
14	The information provided by the system is easy to understand even to non-experts.					
15	The interface of this system was pleasant and readable.					
16	I liked using the interface of this system.					
17	This system has all the functions and capabilities I expect it to have. Overall, I am satisfied with this system.					

Part 4. Comments and Suggestions (optional)

Additional Comments:

_____

_____

Appendix R

Participant Consent Summary

By participating in this survey, we will gather your data anonymously, and you will respond voluntarily.

Copy chart

20 responses



...

Appendix S

Experts' Bionotes

Daryl Ivan Hisola

Role in the study: ML Expert

Daryl Ivan Hisola is a software developer and machine learning practitioner with over 15 years of experience. He builds machine learning and computer vision models for real-world applications, with projects in crop disease detection, fruit defect detection, and food quality grading. His work uses deep learning and machine learning approaches such as YOLOv8, Faster R-CNN with Inception V2, k-Nearest Neighbors, and artificial neural networks. He has co-authored several studies on these systems, many of them published through the IEEE International Conference on Artificial Intelligence in Engineering. His expertise also includes Software Engineering, Information Systems, Cybersecurity, and Data Privacy. He is affiliated with the College of Computing and Information Sciences at Cor Jesu College, Digos City, Davao del Sur, Philippines. He holds a Bachelor of Science in Computer Science from Cor Jesu College and has completed the academic requirements for the Master of Science in Information Technology at Ateneo de Davao University. In this study, he served as an expert evaluator, reviewing the machine learning model architecture, training pipeline, and deployment strategy, and providing technical validation of the prototype.

Gerald Tiocson

Role in the study: ML Expert

Engr. Gerald Tiocson is a Program Chairperson at Divine World College of Calapan and a technical consultant with over 20 years of experience, specializing in electronics engineering and applied machine learning. He designs computer vision systems for real-world automation, with expertise in YOLO-based object detection. He co-authored "Maintaining the Use of Face Masks in an Indoor Establishment Conforming with the COVID-19 Protocols Thru Computer Vision Model: An Approach to Detect Mouth as Region of Interest," published in the International Research Journal on Innovations in Engineering, Science and Technology (2021). He holds a Bachelor of Science in Electronics Engineering and is a member of the Institute of Electronics Engineers of the Philippines (IECEP). In this study, he served as an expert evaluator, reviewing the machine learning model architecture, training pipeline, and deployment strategy, and providing technical validation of the prototype.

Rich Milton R. Dulay

Role in the Study: Mushroom/Mycologist Expert

Dr. Rich Milton R. Dulay is a Professor VI of the Department of Biological Sciences, and Director of the Center for Tropical Mushroom Research and Development of Central Luzon State University. He is a registered microbiologist, a mycologist with expertise in mushroom science and biotechnology, conservation and myko-pharming of wild, edible and useful Philippine mushrooms. He completed his Bachelor of Science in Biology at Central Luzon State University and his Master of Science in Biology and Doctor of Philosophy in Biology at De La Salle University-Manila. He was a Fulbright Advanced Research Fellow at the National Center for Agricultural Utilization Research, US Department of Agriculture, Illinois, USA in 2014. He has published over 180 scientific papers in the Web of Science-, Scopus-indexed and peer-reviewed journals, 3 books on Philippine mushrooms, and 5 Chapter in a Book. Dr. Dulay has 1 Patent, 14 utility models relating to the cultivation of Philippine mushroom and development of mushroom-based food products and registered 3 copyrights relating to the production technology of commercially-cultivated mushrooms. Dr. Dulay also received Best Scientific Poster and Paper Awards in various scientific conferences. He was awarded as one of the 2022 Outstanding Young Scientists by the National Academy of Science and Technology (NAST) and conferred as Diplomate in Microbiology by the Philippine Academy of Microbiology. In this study, he served as an expert validator, reviewing and confirming the specific species identifications for the wild mushroom image validation sets to ensure the accuracy of the model's evaluation.

Appendix T

Editor's Note

HOLY ANGEL UNIVERSITY SCHOOL OF COMPUTING

DR. MARLON I. TAYAG
Dean

This is to certify that the thesis document, entitled "**ENHANCING FORAGING SAFETY: A LOW-RESOURCE MACHINE LEARNING MODEL FOR POISONOUS MUSHROOM IDENTIFICATION IN THE PHILIPPINES**" authored by the following proponents:

Lenon, Danielle Laurence B.
Ocampo, Hans Christian S.
Ochoa, Bianca Claire L.
Ochoa, Bianca Venice L.

has been checked for grammatical and typographical errors.

Yours truly,

Maria Cristina C. Nogoy, LPT, MAEd
Faculty, Comm and Languages Department

October 11, 2026

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Email: hau@hau.edu.ph | www.hau.edu.ph

Appendix U

PlagScan Certificate

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CERTIFICATION

This certifies that the research paper entitled **"Enhancing Foraging Safety: A Low-Resource Machine Learning Model for Poisonous Mushroom Identification in the Philippines"** by Danielle Laurence B. Lenon, Hans Christian S. Ocampo, Bianca Claire L. Ochoa, and Bianca Venice L. Ochoa, is essentially clear of plagiarism, as subjected to Turnitin review. Scanned and reviewed by the University Research Office on October 12, 2026 with the following details:

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Appendix V

Docker and GitHub Repository

V.1 Overview

SmartSpore was containerized using Docker to make its release build reproducible. A Dockerfile in the smartspore_app folder packages the Flutter SDK, the Android SDK, and the application's dependencies into a single image, so the release APK can be built without installing Flutter or Android tooling on each computer. This approach reduces reliance on manually configuring the build environment.

In addition, GitHub provides public access to the project's source code, including the Flutter application, the bundled offline model, the model training notebook, and the research documents. The offline model file (about 125 MB) is stored with Git LFS.

GitHub Public Access: https://github.com/C1AUS-Y/SmartSpore/blob/main/README.md

GitHub Public Access README.md: https://github.com/C1AUS-Y/SmartSpore/blob/f21aa1ceb740c888eeae92ce0e12fb332349f5c8/README.md

Docker Build Files: https://github.com/C1AUS-Y/SmartSpore/tree/f21aa1ceb740c888eeae92ce0e12fb332349f5c8/smartspore_app

V.2 Deployment Environment

The build environment consisted of the following components:

- Mobile application framework: Flutter and Dart
- On-device inference: ExecuTorch with MobileNetV4-Large
- Containerization Platform: Docker Desktop
- Version control: Git with Git LFS
- Build output: release APK for Android devices

The step-by-step build procedure is documented in the repository README under "Build the APK with Docker".

Appendix W

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ACM First Page

Enhancing Foraging Safety: A Low-Resource Machine Learning Model for Poisonous Mushroom Identification in the Philippines

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ABSTRACT

The identification of toxic species of wild mushrooms in the Philippines remains a persistent public health risk, often resulting in avoidable poisoning cases. The study developed a mobile prototype based on low-resource machine learning, capable of differentiating between three toxic species or presumed toxic species, including *Chlorophyllum molybdites*, *Lepiota cristata*, and *Agaricus xanthodermus*, from their edible look-alikes *Chlorophyllum rhacodes, Macrolepiota procera*, and *Agaricus campestris*. Using a curated dataset of 58,184 images across six classes sourced from iNaturalist, transfer learning was applied to fine-tune lightweight convolutional neural network architectures and a Vision Transformer architecture for a hybrid offline-online application. Model evaluation prioritized recall, F-1 score, and confusion matrix analysis over standard accuracy to minimize false negatives, wherein toxic species could be misclassified as edible. MobileNetV4-Large was selected as the offline model, which achieved an average latency of 5.05 seconds on the selected low-end Android device. Meanwhile, Swin Transformer Base was chosen for online use, and it attained a macro F1-score of 93.24% alongside the lowest false-negative rate among toxic classes. Independent machine learning experts reviewed the methodology and model development process. End-user testing with 20 participants was conducted through purposive sampling and a modified PSSUQ-based instrument, which yielded an average score of 4.72 out of 5. These findings suggested that low-resource machine learning models can improve mushroom foraging safety in resource-constrained settings.

CCS CONCEPTS • Computing methodologies → Computer vision; Neural networks; Transfer learning • Human-centered computing → Mobile computing • Applied computing → Health informatics

Keywords: Mushroom poisoning identification, transfer learning, Vision Transformer, lightweight CNN, mobile deep learning application, MobileNetV4, Swin Transformer

1. INTRODUCTION

The tropical archipelago of the Philippines is a global biodiversity hotspot where wild mushrooms thrive abundantly during the rainy season, with recent reviews cataloging at least 447 distinct species from 2001 to 2021 (Dulay et al., 2023). While physical characteristics such as cap shape, gill color, bruising, and stalk structure can be used to identify species and toxicity (Ortiz-Letechipia et al., 2024), misidentification of wild mushrooms contributes to recurring poisoning incidents. News reports, including accounts sourced from the Department of Health (DOH), document 34 hospitalizations across Benguet and Mountain Province between May and June 2024 (Philippines News Agency, 2024) and an earlier outbreak involving up to 14 individuals (Rendon, 2017). More recently, 10 residents, including seven children, were hospitalized in Talibon, Bohol, in March 2026 after eating unspecified poisonous mushrooms (Inojales, 2026).

In tropical regions such as the Philippines, *Chlorophyllum molybdites* (False Parasol or Green-Spored Parasol) is a commonly reported culprit, closely resembling generally edible parasol mushrooms such as *Macrolepiota procera* (Parasol Mushroom) and *Chlorophyllum rhacodes* (Shaggy Parasol). Similarly, *Lepiota cristata* (Stinking Dapperling) shares morphological traits with young edible parasols. Its toxicity is not firmly established, as an analysis of Turkish specimens found no amatoxins or phallotoxins, though the authors cautioned this does not make the species edible (Yılmaz et al., 2018). Because other small Lepiota species are dangerous, including *L. brunneoincarnata*, which has caused severe liver injury requiring transplantation (Anbardar et al., 2024), and *L. spiculata*, which contains lethal α-amanitin (Angelini et al., 2020), L. cristata was treated as a precautionary toxic class. Another widespread hazard is *Agaricus xanthodermus* (Yellow Stainer), reported by iNaturalist contributors to account for roughly half of poisoning cases among Agaricus-type foragers (iNaturalist, n.d.), and frequently mistaken for the edible *Agaricus campestris* (Field Mushroom).

Traditional identification relies on field guides, spore prints, or mycological consultation, which are largely inaccessible to non-specialists, indigenous foragers, and rural communities. While computer vision and deep learning have advanced automated species identification globally, existing applications appear to give limited attention to tropical look-alike hazards or to low-resource smartphones in low-connectivity settings. This study addresses this gap by developing and evaluating a hybrid offline-online mobile prototype built on transfer learning. The system integrates MobileNetV4-Large via ExecuTorch for fully offline on-device inference and Swin Transformer Base for online classification, together with a three-view probability-averaging mechanism and a heuristic three-tier confidence warning system intended to support safer foraging decisions. The remainder of this paper presents the related literature, methods, results, and discussion of the prototype's evaluation.

ACM Second Page

2. REVIEW OF RELATED LITERATURE

2.1 Philippine Mycological Biodiversity and Toxic Look-Alikes

Philippine mycological checklists compiled by Dulay et al. (2023) and Jacob et al. (2022) confirm the local occurrence of toxic species such as *Chlorophyllum molybdites*, *Agaricus xanthodermus*, and *Lepiota cristata*, and document severe gastrointestinal symptoms associated with mushroom species found in the country. However, these checklists note a lack of dedicated profiling and public-facing tools that help non-specialists distinguish toxic species from their edible look-alikes.

2.2 Deep Learning in Macrofungi Classification

Ozsari et al. (2024) evaluated deep learning architectures for classifying macrofungi species, including *Amanita pantherina*, *Boletus edulis*, and *Pleurotus ostreatus*, and reported performance differences among the architectures. In Southeast Asia, Lim et al. (2024) curated a local Malaysian dataset and compared Vision Transformers (ViTs) and ResNet variants for mushroom identification, addressing poisoning risks from outdoor activities in humid climates similar to those in the Philippines. Locally, Algarme and Villegas (2025) explored machine learning for conservation and health awareness in the Philippines, but their work focused on broader mushroom profiling rather than toxic look-alike differentiation or mobile deployment.

2.3 Lightweight CNNs for Edge Deployment

Deploying deep learning on resource-constrained mobile and edge devices requires computationally and memory-efficient models (Cai et al., 2022). MobileNetV3-Large was developed using hardware-aware Neural Architecture Search and the NetAdapt algorithm to optimize performance on mobile hardware, achieving a 3.2-percentage-point ImageNet top-1 accuracy improvement over MobileNetV2 while reducing latency by up to 20% (Howard et al., 2019). Zhu et al. (2023) applied an improved MobileNetV3-Large-based model to classify shiitake mushroom quality against complex backgrounds, achieving 99.91% accuracy with a relatively small model size. Their task differs from toxic-versus-edible classification, but it supports the suitability of the architecture for mushroom imagery. MobileNetV4 is a later generation of the family. Qin et al. (2024) introduced the Universal Inverted Bottleneck and Mobile MQA attention, reporting a 39% speedup under their evaluated conditions across mobile CPUs, GPUs, DSPs, and accelerators such as the Apple Neural Engine and Google Pixel EdgeTPU. This efficiency across hardware makes it relevant for offline deployment on low-end Android devices.

2.4 High-Capacity CNN and Vision Transformer Architectures

Where network connectivity permits server-side inference, higher-capacity models can be used. ConvNeXt-Base modernizes standard convolutional networks using design principles from Vision Transformers while remaining purely convolutional, reaching 83.8% top-1 accuracy on ImageNet-1K with high inference throughput (Liu et al., 2022). Swin Transformer Base applies hierarchical self-attention within shifted local windows, allowing information to be exchanged across image regions at manageable computational cost, and reaches 83.5% top-1 accuracy on ImageNet-1K (Liu et al., 2021). Its attention-based design offers a contrasting approach to ConvNeXt-Base for distinguishing visually similar classes.

2.5 Confidence Estimation and System Usability Validation

Corbière et al. (2021) argued that reliable confidence estimates are required for deploying classifiers in safety-critical applications and described withholding predictions when confidence is low (a reject option) as a way to reduce the risk of unreliable outputs. For usability validation, Vlachogianni and Tselios (2023) found that mobile applications evaluated with the Post-Study System Usability Questionnaire (PSSUQ) scored higher on user satisfaction and system usability than other technology categories, supporting its use for evaluating mobile applications.

3. CONCEPTUAL FRAMEWORK

This study was guided by an Input-Process-Output (IPO) conceptual framework illustrating how raw citizen-science mushroom observations were transformed into an evaluated hybrid mobile identification tool (Figure 1).

The Input phase comprises the curated image dataset of six wild mushroom species (three toxic or presumed-toxic species and their three edible look-alikes) sourced from the iNaturalist API (Research Grade observations only), pre-trained CNN and Vision Transformer checkpoints, and the evaluation instruments (a modified PSSUQ questionnaire and expert validation sets).

The Process phase covers data cleaning and group-aware splitting to prevent specimen-level leakage; image preprocessing (resizing to 384×384 pixels, normalization) and augmentation via Albumentations; full fine-tuning through transfer learning across four candidate architectures; model evaluation prioritizing recall, macro F1-score, and false-negative minimization; integration into a Flutter/ExecuTorch hybrid mobile application with three-view probability averaging and heuristic confidence thresholds; and expert validation with Central Luzon State University (CLSU) mycologists and machine learning specialists, followed by end-user testing.

The Output phase delivers the evaluated hybrid Android prototype, comparative model performance metrics, expert validation results, and PSSUQ usability ratings.



Figure 1: Input-Process-Output conceptual framework for the low-resource mushroom identification system.

ACM Third Page

4. OBJECTIVES OF THE STUDY

This study aimed to develop and evaluate a machine learning prototype for identifying select poisonous wild mushrooms and their edible look-alikes documented to occur in the Philippines, as an accessible hybrid offline-online tool for both low-connectivity and connected settings. Specifically, the study aimed:

1. To curate and transform an image dataset of six wild mushroom species (toxic *Chlorophyllum molybdites* and *Agaricus xanthodermus*, presumed-toxic *Lepiota cristata*, and edible *Chlorophyllum rhacodes*, *Macrolepiota procera*, and *Agaricus campestris*) using group-aware splitting and data augmentation.
2. To train and fine-tune lightweight CNN architectures (MobileNetV3-Large, MobileNetV4-Large) for offline on-device use and higher-capacity architectures (ConvNeXt-Base, Swin Transformer Base) for online inference.
3. To evaluate and compare the models using recall, macro F1-score, and confusion matrix error analysis, with accuracy and on-device latency on a low-end Android smartphone as supporting measures.
4. To assess the system through independent mycological identification, machine learning expert review, and end-user usability testing with a modified PSSUQ-based instrument, using purposive sampling of mushroom enthusiasts, environmental groups, ML/AI experts, and the general population.

5. METHODS

5.1 Research Design

This study used a quantitative design complemented by qualitative expert consultations. System development followed a prototyping approach across three iterations, each incorporating requirements, design, build, user evaluation, and refinement. The machine learning pipeline followed a Low-Resource Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology (Figure 2).

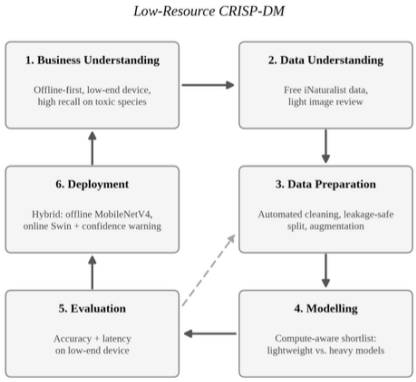


Figure 1: Low-Resource CRISP-DM methodology model.

5.2 Dataset Curation and Preprocessing

Images were collected through a custom Python pipeline using the iNaturalist API (v1/observations), restricted to Research Grade observations released under CC-BY, CC-BY-NC, or CC0 licenses. A total of 59,315 raw images were retrieved across the six classes. Manual quality assurance removed 1,131 images that were inconclusive, microscopic, dried or dead, or contained text, leaving 58,184 images.

To eliminate specimen-level leakage, the dataset was re-split with a group-aware method so that all photographs of one specimen fell in the same partition. The resulting 80/10/10 split was 46,474 training (79.87%), 5,897 validation (10.14%), and 5,813 test (9.99%) images. Images were resized to 384×384 pixels and normalized. Training used Albumentations (RandomResizedCrop, flips, random 90-degree rotations, Gaussian blur, color jitter, coarse dropout) together with batch-level Mixup and CutMix.

5.3 Model Training and Platform Configuration

Models were trained in Python 3.13.15 with PyTorch 2.11.0+cu128 on a Google Colab runtime with an NVIDIA L4 GPU, an Intel Xeon CPU at 2.20 GHz, and 52.96 GB RAM. Four pre-trained timm checkpoints were fully fine-tuned under one 15-epoch pipeline: mobilenetv3_large_100.miil_in21k_ft_in1k, mobilenetv4_conv_aa_large.e230_r384_in12k_ft_in1k, convnext_base.fb_in22k_ft_in1k_384, and swin_base_patch4_window12_384.ms_in22k_ft_in1k. Training used AdamW with an initial learning rate of 5×10^{-5}, weight decay of 0.05, layer-wise learning rate decay (0.85), cosine annealing with warmup, stochastic depth (0.2), class-weighted cross-entropy with label smoothing (0.1), and an Exponential Moving Average (EMA) of weights.

5.4 System Architecture and Inference Pipeline

The hybrid Android prototype was built with Flutter and Dart. Offline mode ran MobileNetV4-Large locally through ExecuTorch in FP32, while online mode sent requests to Swin Transformer Base

hosted on Google Cloud. Because a single view is often non-diagnostic, users submit three photographs per specimen: top (cap), bottom (gills), and side (stalk). Each image yields a six-class softmax vector, and the system averages the three: $P_fused(c) = (1/3)\ \Sigma_i\ P_i(c)$, for views $i \in \{top, bottom, side\}$. The class with the highest fused probability is selected.

A heuristic three-tier confidence system manages uncertainty. Predictions with ≥85% confidence are shown directly (red for toxic, green for edible). Scores of 50% to 84% trigger a yellow cautionary warning advising expert verification. Scores below 50% are withheld as "Image Unclear / Inconclusive."

5.5 Validation and Evaluation Metrics

Models were evaluated on the held-out test set (N = 5,813) with emphasis on toxic-class recall, since a toxic specimen misread as an edible look-alike is the costliest error. The study set a target of at least 85% recall and macro F1-score. Metrics were computed in Python with Scikit-Learn and NumPy.

5.5.1 Accuracy

The proportion of correct classifications across all six classes, Accuracy = (TP + TN) / (TP + TN + FP + FN), reported on the test set (N = 5,813) and the CLSU expert-confirmed set (N = 35).

5.5.2 Precision

The share of predictions for a class that truly belong to it, Precision = TP / (TP + FP), macro-averaged so all six classes count equally.

5.5.3 Recall

The share of actual specimens of a class that the model found, *Recall = TP / (TP + FN). For the toxic classes this directly measures the rate of dangerous false negatives.*

5.5.4 F1-Score

The harmonic mean of precision and recall, F1 = 2 × (Precision × Recall) / (Precision + Recall), macro-averaged across classes and used as the primary metric for comparing architectures. Confusion matrices were used to examine the direction of errors between toxic species and their edible look-alikes.

On-device latency was measured over six trials per mode on a low-end Android smartphone (Qualcomm Snapdragon 450, octa-core 1.8 GHz Cortex-A53, Adreno 506 GPU, 2 GB RAM). Mycologists from Central Luzon State University (CLSU) confirmed 35 temporally separated iNaturalist specimens (105 sub-images uploaded in August, distinct from the June/July training cutoff), plus one out-of-scope specimen (*Marasmius* sp.). Two independent ML experts reviewed the training pipeline and evaluation methodology. Twenty purposively sampled participants (10 general users, 3 mushroom enthusiasts, 3 environmental group members, and 4 future researchers and technology practitioners) rated usability with a modified PSSUQ-based instrument, analyzed descriptively across the System Usefulness, Information Quality, and Interface Quality subscales.

5.6 Ethical Considerations

The study followed the Data Privacy Act of 2012 (Republic Act No. 10173) and obtained informed consent from all survey participants. All iNaturalist images were used under their Creative Commons licenses. Because mushroom misidentification carries serious health risks, the interface displays warning banners stating that the tool is a decision-support aid and not a substitute for professional mycological consultation.

5.7 Research Procedure

Model development proceeded in seven phases alongside three prototype iterations. (1) A MobileNetV3-Large baseline was trained, and a cloud-storage input bottleneck was resolved by moving data to local instance storage. (2) A unified 15-epoch full-fine-tuning pipeline (Section 5.3) was established and first benchmarked on ConvNeXt-Base. (3) A leakage audit found a 0.28% cross-split specimen overlap (161 images), so the full 58,184-image dataset was re-split by observation ID, giving zero specimen overlap. (4) MobileNetV3-Large and ConvNeXt-Base were retrained on the corrected split under the identical pipeline. (5) Swin Transformer-Base (about 87.9M parameters) was trained to compare windowed self-attention with the size-matched ConvNeXt-Base (about 88.6M). (6) MobileNetV4-Large (about 32.6M parameters) was trained to test whether a newer MobileNet generation improved on V3. (7) INT8 and FP16 post-training quantization were evaluated, and FP32 was retained for deployment. MobileNetV4-Large was selected for offline use and Swin Transformer-Base for online use.

The Flutter prototype went through three iterations. Iteration 1 offered single-image offline classification. Iteration 2 added three-view probability averaging, cloud inference, the three-tier confidence system, and color-coded results. Iteration 3, following panel recommendations, added morphological explanations of each result, clearer invalid-image messages, and changed the cautionary warning color to yellow.

5.8 Testing

Testing had four parts. First, all four models were evaluated by single-pass inference (no test-time augmentation) on the held-out test set, and three-view fusion was verified with controlled cap, gill, and stalk photographs of *C. molybdites* in both modes. Second, six functional test cases (TC_01 to TC_06) covered offline and online three-image classification, non-mushroom uploads, incomplete input (two of three views), blurry or low-light images, and out-of-scope species (*Amanita muscaria*, *Leucocoprinus birnbaumii*, *Pleurotus ostreatus*, *Schizophyllum commune*) to observe closed-set behavior. Third, six end-to-end latency trials were timed per mode on the Snapdragon 450 device. Fourth, both deployed models were tested against the 35 CLSU-confirmed specimens plus the out-of-scope *Marasmius sp.*, while 20 end users tested the app on unseen sample sets and completed the modified PSSUQ survey.

6. RESULTS

6.1 Dataset Curation and Splitting Results

Manual cleaning removed 1,131 invalid images from the 59,315 images retrieved, leaving 58,184 curated images across the six classes. Group-aware splitting then partitioned them to eliminate cross-split specimen overlap (Table 1).

Table 1: Dataset Cleaning and Group-Aware Split Distribution

Species Class	Status	Raw Images	Removed	Final
Agaricus campestris	Edible	8,213	187	8,026
Agaricus xanthodermus	Toxic	9,791	284	9,507

ACM Fifth Page

Chlorophyllum molybdites	Toxic	10,067	169	9,898
Chlorophyllum rhacodes	Edible	10,338	134	10,204
Lepiota cristata	Toxic*	10,971	281	10,690
Macrolepiota procera	Edible	9,935	76	9,859
Total	n/a	59,315	1,131	58,184

*Note. *Presumed toxic on a precautionary basis. The group-aware split yielded 46,474 training (79.87%), 5,897 validation (10.14%), and 5,813 test (9.99%) images.*

6.2 Model Classification Performance and Error Direction

Accuracy, macro-average recall, and macro F1-score rose steadily from the lightweight baseline to the Vision Transformer (Table 2).

Table 2: Classification Performance and Critical Misclassification Counts on the Test Set (N = 5,813)

Metric	MobileNetV3-Large	MobileNetV4-Large*	ConvNeXt-Base	Swin Transformer Base**
Accuracy	87.41%	90.80%	93.15%	93.38%
Macro Precision	87.16%	90.55%	93.02%	93.27%
Macro Recall	87.25%	90.47%	92.95%	93.23%
Macro F1-Score	87.17%	90.56%	92.98%	93.24%
FN: A. xanthodermus -> A. campestris	102	68	54	44
FP: A. campestris -> A. xanthodermus	70	80	73	68
FN: C. molybdites -> C. rhacodes	75	39	30	30
FP: C. rhacodes -> C. molybdites	53	52	50	50

*Note. *Selected offline model. **Selected online model. FN = false negative (toxic specimen classified as its edible look-alike). FP = false positive (edible specimen classified as its toxic look-alike).*

MobileNetV3-Large reached a macro F1-score of 87.17%, but its recall on the toxic *A. xanthodermus* (83.33%) and *C. molybdites* (83.37%) fell below the study's 85% target, with 102 and 75 false negatives. MobileNetV4-Large raised macro F1 to 90.56% and cleared the 85% recall target on both toxic classes (88.47% and 89.60%), cutting false negatives to 68 and 39 (Figure 3).

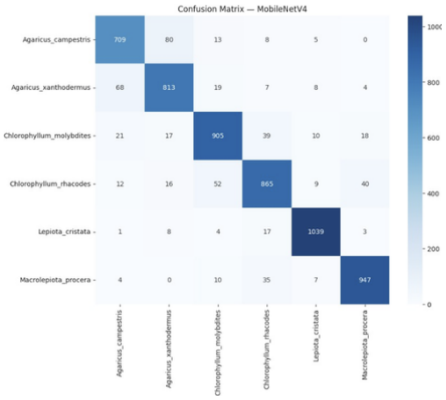


Figure 3: Confusion matrix for the selected offline model, MobileNetV4-Large.

Among the online candidates, ConvNeXt-Base reached 92.98% macro F1, and Swin Transformer Base performed best overall at 93.24% macro F1 and 93.38% accuracy. Swin's recall was 93.46% on *A. xanthodermus* and 93.27% on *C. molybdites*, with the lowest or tied-lowest toxic-to-edible false-negative counts of all models (44 and 30; Figure 4).

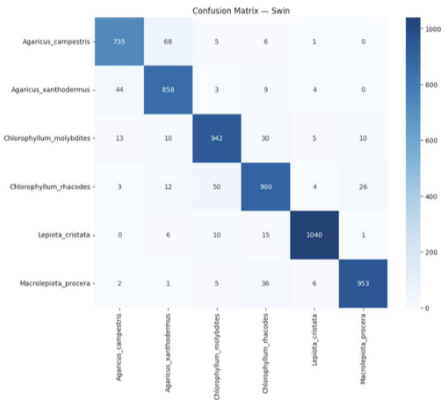


Figure 4: Confusion matrix for the selected online model, Swin Transformer Base.

6.3 System Testing, Multi-View Fusion, and On-Device Latency

All six functional test cases (TC_01 to TC_06) passed their expected outputs. Offline and online three-image classification worked, non-mushroom uploads returned a low-confidence result with an invalid-image message, and incomplete input (two of three views) disabled the submit button.

In the out-of-scope test (TC_06), four untrained species were uploaded in both modes: *Amanita muscaria* (Fly Agaric), *Leucocoprinus birnbaumii*, *Pleurotus ostreatus*, and *Schizophyllum commune* (Splitgill). Of the eight runs, three were withheld as low confidence (Fly Agaric in both modes, and Splitgill online). The other five runs were assigned to one of the six trained classes. Because this was a small, informal test (n = 4), it shows the closed-set behavior of the classifier and should not be read as a validated property of the model.

Controlled testing of three-view averaging on *C. molybdites* showed high agreement across top, bottom, and side views, with a fused confidence of 90.25% (spread 0.7%) on MobileNetV4-Large and 90.80% (spread 2.1%) on Swin Transformer Base.

On the low-end Snapdragon 450 device (2 GB RAM), six end-to-end three-view trials were run per mode (Table 3). Offline mode (MobileNetV4-Large, FP32) averaged 5,050.33 ms, and online mode (Swin Transformer Base) averaged 4,084.50 ms. The online mode was 19.1% faster on average but more variable, consistent with network conditions. Experimental FP16 quantization halved the MobileNetV4-Large file size (about 120 MB to 60 MB) with accuracy preserved within a fraction of a percentage point. However, on-device ExecuTorch inference slowed to about 78.9 seconds because of incomplete native FP16 operator support, so FP32 was retained.

Table 3: End-to-End Three-View Latency on the Selected Low-End Android Device

Mode / Model	Mean (ms)	SD (ms)	Min (ms)	Max (ms)
Offline (MobileNetV4-Large)	5,050.33	522.30	4,414	5,724
Online (Swin Transformer Base)	4,084.50	764.44	3,030	5,033

Note. Six three-image trials on a Qualcomm Snapdragon 450 (2 GB RAM) device.

6.4 Mycological Expert Validation, ML Expert Feedback, and System Usability

Against the 35 independent specimens confirmed by CLSU mycologists, MobileNetV4-Large classified 30 correctly (85.71%) and Swin Transformer Base classified 31 correctly (88.57%) (Table 4). One additional out-of-scope *Marasmius sp.* specimen returned below 50% confidence in both modes and was withheld.

Table 4: External Validation Against CLSU Mycologist-Confirmed Specimens (N = 35)

Species Class	Status	Expert Confirmed	Offline Correct	Online Correct
C. molybdites	Toxic	8	5	6
C. rhacodes	Edible	4	3	3
A. xanthodermus	Toxic	5	4	5
A. campestris	Edible	7	7	6
L. cristata	Toxic*	5	5	5
M. procera	Edible	6	6	6
Total (accuracy)	n/a	35	30 (85.71%)	31 (88.57%)

Note. Presumed toxic on a precautionary basis.

Two independent ML experts reviewed the training pipeline and evaluation. Their main points (split integrity, evaluation consistency, the definition of "low-resource," out-of-distribution handling, and prioritizing recall for the toxic classes) are addressed in the Discussion.

The 20 end users rated usability on the modified 5-point PSSUQ-based instrument (16 rated items). The overall mean was 4.72 out of 5, with Information Quality highest (4.78), then Interface Quality (4.75), and System Usefulness lowest (4.63) (Table 5).

Table 5: Modified PSSUQ Results by Participant Group and Subscale (N = 20)

Subscale	GU	ME	EG	TP	Mean
Interface Quality	4.80	4.78	4.78	4.58	4.75
Information Quality	4.80	4.73	4.87	4.70	4.78
System Usefulness	4.71	4.42	4.88	4.38	4.63
Total	4.76	4.58	4.85	4.52	4.72

Note. GU = General Users (n = 10), ME = Mushroom Enthusiasts (n = 3), EG = Environmental Groups (n = 3), TP = Future Researchers and Technology Practitioners (n = 4). Scores are on a 5-point scale.

ACM Seventh Page

7. DISCUSSION

7.1 Safety-Critical Metric Prioritization and Architecture Trade-offs

A central finding, reinforced during the ML expert consultations, is that overall accuracy is an insufficient primary metric for safety-critical foraging applications. Misclassifying a toxic mushroom as an edible look-alike (false negative) poses a far greater health risk than the reverse (false positive), so model selection prioritized toxic-class recall and the direction of confusion-matrix errors. Upgrading the offline model from MobileNetV3-Large to MobileNetV4-Large reduced *A. xanthodermus* false negatives by 33.3% (102 to 68) and *C. molybdites* false negatives by 48.0% (75 to 39), while keeping inference at about 5 seconds on entry-level hardware. This is consistent with the efficiency-oriented design principles described by Cai et al. (2022) and Qin et al. (2024). In online mode, Swin Transformer Base had the lowest toxic false-negative counts (44 and 30), which is consistent with the ability of shifted-window self-attention (Liu et al., 2021) to separate fine-grained morphological differences.

7.2 Edge Quantization Bottlenecks and Multi-View Fusion

FP16 quantization is expected to halve memory use and speed up inference (Taklai et al., 2025). In this study, however, running the FP16 MobileNetV4-Large model through ExecuTorch on Android took about 80 seconds per identification, which we attribute to upcasting and downcasting overhead for operators without native FP16 support. This may reflect the specific runtime and hardware used rather than a shortcoming of FP16 itself. Retaining FP32 gave a practical latency of about 5.05 seconds for three images. Requiring top, bottom, and side views also addresses the weakness of single-cap photographs, since ambiguous features in one view can be balanced by diagnostic features (such as gill attachment or a stalk ring) in another. Because the models were trained on single images, the fusion step is post-hoc averaging, and its benefit was verified only in a controlled test.

7.3 External Generalization and User Trust

On the 35 CLSU-confirmed specimens, accuracy was about 5 percentage points below test-set accuracy (85.71% offline vs. 90.80%; 88.57% online vs. 93.38%). Possible reasons include the temporal separation from the training data and the use of three-view fused inputs, but with only 35 specimens this gap should be read cautiously.

In the PSSUQ evaluation, overall satisfaction was high (4.72 out of 5). Mushroom Enthusiasts (4.42) and Future Researchers and Technology Practitioners (4.38) rated System Usefulness lower than General Users (4.71) and Environmental Groups (4.88). Because the enthusiast and environmental groups had only three participants each, these differences are descriptive. The item with the most varied responses concerned comfort in relying on the app for toxicity checks (mean 4.20, range 2 to 5). This is consistent with literature linking distrust of "black-box" AI to health-critical settings (Li et al., 2021), and it motivated the morphological explanations added in the third prototype iteration.

8. LIMITATIONS

Several factors limit these findings. First, although all six species are documented in the Philippines, training images came from iNaturalist without a geographic restriction, so regional morphological variation and tropical field lighting were not isolated. Second, the model is a closed-set classifier restricted to six species. In informal testing on four out-of-scope species (*Amanita muscaria*, *Leucocoprinus birnbaumii*, *Pleurotus ostreatus*, and *Schizophyllum commune*), three of eight runs were withheld as unclear, while the other five were assigned to a trained class. Third, the 85% and 50% confidence cutoffs rely on raw, uncalibrated softmax outputs and were set by research judgment, not by empirical calibration. Fourth, the toxic status of *L. cristata* is presumed rather than toxicologically confirmed. Fifth, models were trained on single images without view-angle labels, and the three-view fusion has not been validated beyond a controlled test. Finally, results come from a single fixed data split, a small expert set (35 specimens), a purposive sample of 20 users on a modified PSSUQ instrument, and one low-end Android device. The study did not measure any effect on actual poisoning incidents.

9. CONCLUSION

This study demonstrated the feasibility of a low-resource, hybrid offline-online mobile application for distinguishing toxic wild mushrooms (*C. molybdites*, *A. xanthodermus*, and presumed-toxic *L. cristata*) from their edible look-alikes (*C. rhacodes*, *A. campestris*, and *M. procera*). From 58,184 curated images split by specimen, a unified transfer-learning pipeline produced MobileNetV4-Large for offline inference (90.56% macro F1; 5.05 s on a 2 GB RAM device) and Swin Transformer Base for online inference (93.24% macro F1). Validation with CLSU mycologists (85.71% offline, 88.57% online accuracy), two ML experts, and 20 end users (4.72 out of 5 on the modified PSSUQ) suggests that lightweight deep learning with multi-view fusion and confidence-based warnings can serve as a practical decision-support aid for foraging safety in low-connectivity settings. It is an aid alongside expert mycological consultation, not a replacement for it.

10. RECOMMENDATIONS FOR FUTURE WORK

Future work should expand the dataset to more toxic and edible Philippine macrofungi and prioritize field-verified specimens photographed in Philippine ecosystems. To address the closed-set limitation, researchers should add out-of-distribution detection or open-set recognition and apply post-hoc confidence calibration (for example, temperature scaling) to set empirically validated rejection thresholds. Models should also be trained directly on grouped multi-view inputs rather than relying on post-hoc probability averaging. Visual explanation methods such as Grad-CAM could highlight diagnostic regions on the cap, gills, and stalk and improve user trust. Finally, future development should evaluate mobile runtimes and hardware accelerators with native INT8 and FP16 support, to compress the model without a latency penalty.

ACM Eighth Page

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ACM Ninth Page

ABOUT THE AUTHORS

	Danielle Laurence B. Lenon is a Computer Science student at Holy Angel University with experience in Python, Java, Dart, and Flutter, as well as web technologies and relational databases. She has contributed to student projects including a weather-based packing recommendation app, a quiz management system, and a supply management system. She is a recipient of the President's Lister (2024) and Dean's Lister (2023–present) honors.
	Hans Christian S. Ocampo is a fourth-year BS Computer Science student at Holy Angel University with hands-on experience in web and mobile development, databases, and training computer vision models. He is strong in troubleshooting, debugging, and quality assurance, and he completed on-the-job training at the Occupational Safety and Health Office of the Department of Labor and Employment in 2023. He holds Cisco Networking Academy certificates in Cyber Threat Management, Database Analytics Essentials, and Endpoint Security.
	Bianca Claire L. Ochoa is a Computer Science student at Holy Angel University with skills in full-stack and mobile development, REST API integration, database design, and deep learning for computer vision. Her projects include a full-stack order and payment tracker, a skin lesion classifier with Grad-CAM explainability, and the mobile front end of a weather-based packing app. She has been a Dean's Lister (2023–2024) and a President's Lister (2024–present).
	Bianca Venice L. Ochoa is a Computer Science student at Holy Angel University with skills in computer vision, transfer learning, and full-stack development. She was the project lead of this study's mushroom identification app, managing planning and contributing to model development and back-end services, and she was the machine learning lead of a skin lesion classifier. She is a President's Lister (second semester 2023-24 to second semester 2025-26) and a Dean's Lister (first semester 2023-24), and she holds an AWS Academy Cloud Foundations badge and Cisco Networking Academy and IBM SkillsBuild certificates.

Appendix X

Researchers' Resume

Researcher 1

DANIELLE LAURENCE LENON

Computer Science Undergraduate

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Mansfield Residences, Angeles City, Pampanga, Philippines
https://github.com/danlenoon

I'm currently an undergraduate college student studying BS Computer Science with experience in utilizing technologies. Moreover, I have a consistent academic performance while understanding and applying the concepts, theories, and foundations of algorithms, data structures, and other technologies to the best of my abilities.

PROJECTS

2025 - Present
Weather-Based Packing Recommendation System
I planned the timeline for the fruition of our Weather-Based Packing Recommendation System named PackAssist. Moreover, I also designed the wireframes (Low-Fidelity and High-Fidelity) for a more organized and structured appearance and functionality of our application. I also assisted in testing our application to ensure the operations work as intended.

2025
Quiz Management System
I oversaw the overall vision and development of our Quiz Management System named PyFlyQuiz, which has options to add, edit, take, view, or delete a quiz – providing an opportunity to be a student and an instructor as a use case in one program. Python is the programming language used here.

2024
Supply Management System
I oversaw the overall vision and development of our Supply Management System named DL AutoCare, which has options to supply vehicles as a supplier (their colors, models, variants, and respective prices) and purchase vehicles as a buyer. Java is the programming language used here.

EDUCATION

Holy Angel University
2023 - PRESENT

Bachelor of Science in Computer Science

Relevant Coursework
- Data Structures and Algorithms, Software Engineering, Object-Oriented Programming, Design and Implementation of Programming

Honors & Recognition
- President's Lister, Holy Angel University (2024)
- Dean's Lister, Holy Angel University (2023 - Present)

SKILLS

Technical
Python, Java, Dart, HTML, CSS, JavaScript, PostgreSQL, SQL, Flutter
- Mobile Development
- Web Development
- Python Development
- Java Development

Professional
Software Development
System Design
CRUD Operations
Agile Development
Microsoft Excel
Jamovi
Figma

Researcher 2



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PERSONAL INFORMATION

Age: 21

Born: September 2, 2005

Place of birth: Santo Tomas, San Matias, Pampanga

Civil status: Single

Religion: Catholic

TECHNICAL SKILLS

LANGUAGES

Python · Java · JavaScript · Dart

WEB & MOBILE

React · Flutter · HTTP

DATABASES & CLOUD

AWS · PostgreSQL · Supabase

MACHINE LEARNING

Computer vision · Model training

TOOLS & SYSTEMS

Microsoft Office, Google Workspace, operating systems, hardware and software troubleshooting

References available upon request.

Hans Christian S. Ocampo

COMPUTER SCIENCE STUDENT · SOFTWARE & MACHINE LEARNING

Fourth-year BS Computer Science student at Holy Angel University with hands-on experience in web and mobile development, databases, and training computer vision models. Strong in troubleshooting, debugging, and quality assurance.

EDUCATION

BS Computer Science — 2023 – 2027
Holy Angel University, Angeles City

Basic Education, Secondary Level — 2017 – 2023
New Era University, Pampanga Branch (graduated)

Basic Education, Elementary Level — 2010 – 2017
Infant Jesus Learning Center / New Era University

EXPERIENCE

Trainee, Occupational Safety and Health (OSH) Office, DOLE — 2023
On-the-job training · assigned through New Era University

Designed visuals for Facebook posts and looked up certificates for graduates, among other office support tasks.

THESIS PROJECT

Wild Mushroom Species Classification (computer vision)

Group research project that trains and compares deep learning models to classify six visually similar wild mushroom species, using a standardized training pipeline from dataset preparation to results analysis.

CERTIFICATIONS & TRAINING

Cisco Networking Academy (badges and certificates)
Cyber Threat Management · Database Analytics Essentials · Endpoint Security

Work Ethics Seminar, Parts 1 & 2 · NEU Pampanga HS MPH · Feb–Mar 2023

ABILITIES

Time management
Interpersonal skills
Analytical problem-solving
Project management
Adaptability
Attention to detail

TALENTS

Troubleshooting · Debugging · Quality assurance · ML model training · Digital & traditional drawing · Creativity

Researcher 3

Bianca Claire L. Ochoa
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Education

HOLY ANGEL UNIVERSITY — Angeles, Pampanga
Bachelor of Science in Computer Science — April 2027
Thesis: Enhancing Foraging Safety: A Low-Resource Machine Learning Model for Poisonous Mushroom Identification in the Philippines
Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Object-Oriented Programming, Mobile Development, Automata Theory, Applications and Systems Integration, Application Development and Emerging Technologies
Honors: President's Lister (2024 – Present); Dean's Lister (2023 – 2024)

Projects

ProxyPal – Full-Stack Order & Payment Tracker — 2026 – Present
Solo project • React, Vite, Tailwind CSS, Express, PostgreSQL, Supabase • Live: c1aus-y.github.io/ProxyPal
- Built from scratch a web app that lets proxy-shopping buyers track orders, items, payments, remaining balances, and package status, replacing spreadsheet-based tracking.
- Designed the PostgreSQL schema with row-level security and a database view that computes balance owed (items total minus payments), so balances can never drift out of sync.
- Developed an Express REST API that verifies Supabase Auth tokens and scopes every parameterized query to the logged-in user; integrated the 17TRACK API with a Supabase Edge Function webhook for automatic shipment updates; deployed on GitHub Pages, Render (Docker), and Supabase.

Thesis – SmartSpore: Poisonous Mushroom Identification App — 2025 – 2026
Team thesis • Python, PyTorch, TensorFlow Lite, Flutter, Dart
- Trained and compared four image classifiers (MobileNetV3/V4-Large, ConvNeXt-Base, Swin Transformer) on a curated 58,184-image, six-class dataset of toxic Philippine mushrooms and edible look-alikes; best model reached 93.2% macro F1.
- Prioritized safety by tracking false negatives on toxic species; cut Yellow Stainer misses from 102 to 44 and False Parasol misses from 75 to 30 across model iterations.
- Delivered a hybrid Flutter Android app: offline TFLite MobileNetV4 and cloud Swin Transformer, with confidence-threshold warnings; averaged about 5 seconds per prediction on a 2GB-RAM low-end phone.
- Validated with expert-verified samples (85.7% offline, 88.6% online) and a 20-participant usability survey scoring 4.70 out of 5.

Skin Lesion Classifier with Explainability — March 2026
Team project (4 members) • Python, PyTorch, Albumentations, Grad-CAM, scikit-learn, Tkinter
- Trained an EfficientNet-B4 transfer-learning model for benign vs. malignant classification on 35,000+ dermoscopic images (HAM10000 and ISIC 2019).
- Achieved 96.6% accuracy, 0.995 AUROC, and 98.9% sensitivity at 10% false-positive rate on a 1,924-image held-out test set.
- Improved robustness with Albumentations, 6-view test-time augmentation, and ablations; added Monte Carlo dropout uncertainty and Grad-CAM/saliency explainability.
- Extended the system into a multi-modal pipeline: an NLP module scores clinical notes for malignancy-risk symptoms, and a contextual bandit agent adapts the decision threshold to patient context, prioritizing recall in high-risk cases; all components integrated in a Tkinter GUI.

PackAssist – Weather-Based Packing Recommendation App — 2025
Team project • Flutter, Dart, Supabase
- Developed a cross-platform mobile application that combines real-time weather forecasts with packing assistance to recommend packing lists for travelers.
- Implemented the full mobile front end and the application logic that adapts recommendations to forecast conditions.

Apartment & Hardware Inventory Management Systems — March 2024
- Built an offline apartment management app (room availability, tenants, billing) and a Windows inventory system (stock, suppliers, retailers) using CRUD operations.

Skills & Interests

Languages: Python, Java, Dart, JavaScript, TypeScript, SQL, HTML/CSS, Perl, Bash
Frameworks & Tools: Flutter, React, Express, Tailwind CSS, Tkinter, PyTorch, TensorFlow / TensorFlow Lite, scikit-learn, timm, Albumentations, PostgreSQL, SQLite, Supabase, Docker, Git/GitHub Actions, Figma
Areas: Full-stack and mobile development, REST API integration, database design, deep learning (computer vision), Agile development

Researcher 4

BIANCA VENICE L. OCHOA

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EDUCATION

HOLY ANGEL UNIVERSITY — Angeles City, Philippines
Bachelor of Science in Computer Science — *2023 – Present*
Honors: President's Lister (2nd Sem 2023–24 to 2nd Sem 2025–26), Dean's Lister (1st Sem 2023–24)
Relevant Coursework: Data Structures and Algorithms; Intelligent Systems; Operating Systems; Information Management; Software Engineering; Information Assurance and Security; Cloud Computing; Probability and Statistics

TECHNICAL SKILLS

AI/ML: Computer Vision, NLP, CNNs, Transformers, Transfer Learning, Model Evaluation, Grad-CAM, PyTorch, TensorFlow Lite, scikit-learn
Programming Languages: Python, JavaScript, Dart, Java, SQL, C++, C#, HTML/CSS
Frameworks & Tools: React, Vite, Node.js, Express, Flask, Flutter, Tkinter, PostgreSQL, Supabase, REST APIs, Render, Google Cloud, AWS (Cloud Foundations), GitHub Actions, Git

WORK EXPERIENCE

FREELANCE DIGITAL ARTIST — Pampanga, Philippines
Self-Employed
- Deliver custom digital illustrations on commission, managing project briefs, revisions, and final delivery independently

PROJECTS

SMARTSPORE (THESIS): MUSHROOM IDENTIFICATION APP – PROJECT LEAD — *2025 – 2026*
- Led a student team developing an Android application for poisonous mushroom identification; managed project planning and contributed to model development, back-end services, and interface design
- Evaluated MobileNetV3/V4, ConvNeXt, and Swin Transformer on a 58,184-image, six-class dataset, achieving 93.2% macro F1 with the best-performing model
- Delivered a hybrid application supporting on-device (TFLite) and cloud-based (Swin Transformer) inference, averaging approximately 5 seconds per prediction on a 2GB-RAM device

SKINTECT: SKIN LESION ANALYSIS PLATFORM – FULL-STACK DEVELOPER — *2026*
- Designed and deployed a multi-service web platform: React/Vite client (GitHub Pages), Node.js/Express REST API with PostgreSQL (Render), and Flask/PyTorch inference service (Google Cloud)
- Implemented token-based authentication, HTTP POST integration between the API and inference service, Grad-CAM visual explanations, scan-history tracking, and PDF report generation

SKIN LESION CLASSIFIER – ML LEAD AND BACK-END LEAD (TEAM OF 4) — *Mar 2026*
- Led model development, fine-tuning EfficientNet-B4 on 35,000+ dermoscopic images to achieve 0.995 AUROC and 96.6% accuracy on held-out data
- Developed back-end services, including HTTP communication and POST request handling, integrating the image model with an NLP clinical-note scorer and a contextual bandit that adapts decision thresholds per patient

PACKASSIST: WEATHER-BASED PACKING APP – PROJECT LEAD, BACK-END DEVELOPER — *2025*
- Led a team of four through planning, development, and release of a travel and packing assistant app
- Developed the complete back end, including integrations with Open-Meteo weather and geocoding APIs and the recommendation logic that generates weather-based packing lists
- Implemented HTTPS communication and POST request handling between the Flutter client, external APIs, and the Supabase PostgreSQL database
- Tested back-end functionality and released the application as an Android APK

HARDWARE INVENTORY SYSTEM – PROJECT LEAD, FULL-STACK DEVELOPER — *Mar 2024*
- Led development of a Windows desktop inventory system in Python and Tkinter, backed by a PostgreSQL database, managing stock, supplier, and retailer records through full CRUD operations

APARTMENT MANAGEMENT APP – FULL-STACK DEVELOPER — *Mar 2024*
- Developed an offline apartment management application handling room availability, tenant records, and billing through full CRUD operations

CERTIFICATIONS & TRAINING

AWS Academy Graduate – Cloud Foundations (Training Badge); Cisco Networking Academy: Cyber Threat Management (Sep 2026), AI Fundamentals with IBM SkillsBuild (Mar 2026), Data Analytics Essentials (Mar 2026), Endpoint Security (Aug 2025), JavaScript Essentials 1 (Oct 2024)